



Indian Food Packer

Bi-Monthly Journal

Issue No - 2

March - April 2026



India's Mango Magic

Sustenance Needs Policy & Innovation

Celebrating our
national fruit!



Gits®
Since 1963

Sabka Instant Favorite



**KHAMAN
DHOKLA**

**RICE
IDLI**

Ready in 3 easy steps

EDITORIAL TEAM

Technical Editor

Dr. Prabodh Halde
Head-Technical & Regulatory Affairs, Marico Ltd.
& West Zone Chairman, AIFPA

Advisory Board

Dr. Raghav Jadli
Director, Jadli Foods Pvt. Ltd.
& President, AIFPA

Dr. Subodh Jindal
MD, Excelsior Food & Chemical Industries
& Immediate Past President, AIFPA

Sh. Gokul Patnaik, IAS (Retd.)
Chairman, Global Agrisystem Pvt. Ltd.
& Former President, AIFPA

Sh. P. L. Kaul
MD, Mariental, Mariental India Pvt. Ltd.
& Former President, AIFPA

Sh. M.A. Tejani
MD, Gits Food Products Pvt. Ltd.
& Former President, AIFPA

Technical Board

Dr. V. Prakash
Former Director, Scientist (Retd.), CSIR-CFTRI

Sh. Arun Om Lal
President, Hexagon Nutrition

Dr. A. K. Tyagi
Executive Director, Haldiram Group of Companies
& Vice President, AIFPA

Sh. K. K. Debnath
Chief Regulatory Officer, DFM Foods Pvt. Ltd.
& North Zone Chairman, AIFPA

Dr. V. K. Modi
Professor & Head,
Amity Institute of Food Technology

Sh. K. K. Joshi
Head-Regulatory & Scientific Affairs,
ITC Foods Division

Sh. Subhas Bhattacharjee
Former MD, NERAMAC &
Consultant-NER (CSIR-CFTRI)

Dr. C. Tara Satyavathi
Director, ICAR-IIMR

Dr. Shatadru Sengupta
Sr. Director-Legal, Hardcastle Restaurants Pvt. Ltd.

Prof. Rekha S. Singhal
Head, Dept. of Food Engg. & Tech., ICT Mumbai

Sh. Harsh Gursahani
Partner, PLR Chambers

Dr. Nilesh Amritkar
MD, Envirocare Labs Pvt. Ltd.

Sh. Sanjeev Sharma
Author & R&D Expert
(Course Developer, FoodTech Pathshala)

Ms. Subha Nishtala
Director, ITCFSAN, FSSAI Training Centre

Dr. Sridevi A. Singh
Director, CSIR-CFTRI

Dr. J. V. Parekh
Dairy Consultant

Ms. Shashi Sareen
Former Director, EIC

Sh. Jayan Mehta
MD, AMUL

Dr. H.S. Oberoi
Director, NIFTEM-K

Prof. V. Palanimuthu
Director, NIFTEM-T

Sh. Krishan Gupta
CEO & MD, Organic India

Ms. Preeti Saluja
Cold Chain Expert & Regional Sales Head, va-Q-tec

Contents

Foreword

- Mango: The King of Fruits 03

Government Insights

- MoFPI Article – India Mango: Orchard to Global Market 05
- India's Coordinated Policy Response 08

Mango Production, Trade, GI & Market Landscape

- GI Tagging and Its Role in Mango Value Enhancement 13
- Readers Window – Indian Mango – GI Lineage 15
- Readers Window to the World of Mango - India Horticultural footprints – at a glance 18
- Innovation in mango-based products 27
- Indian Production / Horticulture data 36
- Global Mango Production..... 38

Regulatory & Export Compliance (MRL & Pesticide Management Perspective)

- MRLs & Pesticide Management in the Mango Sector: A Supply Chain Imperative 43
- MRLs & Pesticide Management in the Mango Sector 45

Innovation & Sustainable Processing

- Innovation in Mango Based Products 51
- Next-Generation Mango Processing: Technologies Driving Value Addition.. 53
- Solar Refrigerated Evaporative Cooling (SREC) System : A Sustainable and Cost – Effective Solution for Safe Storage of Mangoes in India 61
- Sustainable Management and Environmental Implications of Mango Processing Waste 66
- Rethinking Mango Waste : Sustainable Utilization of Mango Waste 69
- Valorisation of Mango Processing Waste : Unlocking the Potential of Mango Kernel Fat for Sustainable Lipid Applications 72

Post- Harvest Infrastructure & Logistics

- Cold Chain Readiness and Infrastructure 79

Guest Article

- Misbranded Food : An Avoidable Violation 82

AIFPA Activity 85

Mango News 90

Industry News 91

Product Launches 93

News Round Up 95

Food Safety and Regulatory Issues 98

Upcoming Events 100



OneCert International Pvt Ltd.

"Your One Stop Solution For Certification Services Globally"

Organic	Food Safety	Sustainable
---------	-------------	-------------



OneCert Services

Organic	Food Safety	Sustainable & Other Services
Individual Farmers	Food Safety Management System (FSMS)	Trustea
Grower Groups	Food Safety System Certification -22000 (FSSC-22000)	Sustainable Rice Platform (SRP)
Processing & Trading Units	HACCP	Life Cycle Assessment (LCA)
Dairies, Poultry, Livestock Production	India- HACCP	Good Agricultural and Collection Practices (GACP)
Wild Harvest/Forest Collection	GMP	World Castor Sustainability Forum (WCSF)
Tea Estates	India- GHP	Voluntary Certification Scheme for Medicinal Plant Produce (VSCMPP) [Applied]
Apiculture (Bee Keeping)	Third Party Audits (TPA)	ISO 17029 [Applied]
Aquaculture, Aquatic Plant & Sea Weeds	FoSTaC Trainings	ISO 14044 [Applied]
Mushroom Production & Green House Production	PCQI Trainings	ISO 14064-65 [Applied]
Organic Standard Trainings (ICS/Individual Farmers/Handlers)	Warehousing Audits	
	Supplier Audits	

H-274, RIICO Industrial Area, Sitapura, Jaipur -302022, Rajasthan (India).

Phone: 09414046706 (PBX). Email : info@onecert.net. Website : www.onecertinternational.com

Mango - The King of Fruits

Shailendra Rajan (Ex Director)

ICAR-Central Institute for Subtropical Horticulture, Lucknow
E-mail:- srajanlko@gmail.com



Mango, widely celebrated as the "King of Fruits," is an economically valuable fruit crop in many parts of the world. Besides its appealing tropical flavour, mango offers important nutrients and makes healthy eating an enjoyable sensory experience. Although mango cultivation is reported from more than 120 countries, over 60% of global production is contributed by India, China, Thailand, Indonesia and Mexico. A large share of the crop is consumed fresh, while hundreds of processed mango products are also produced commercially. Mango fruit is naturally low in sodium and provides useful amounts of potassium, phosphorus and calcium. In addition to its nutritional value, mango contains bioactive compounds such as mangiferin and lupeol, which have been associated with several health-promoting properties, including antitumour-promoting activity.

Mango is among the leading fruit crops of the tropical and subtropical world. The tree is respected not only for its sacred and cultural importance, but also for its considerable economic value, as almost every part of the plant is useful in one form or another. In many traditional Asian societies, mango supports food, fuel and fodder needs and provides livelihood opportunities for millions of people. At the same time, this crop faces growing challenges from changing climate conditions and increasing biotic stresses across different agro-ecologies. As mango becomes an export-oriented commodity for many countries, competition in the global market is also intensifying.

Fascinating history of distribution

Mango has a long and interesting history of domestication and dispersal, particularly because sea routes were crucial for crop movement before the early twentieth century. *Mangifera indica* is believed to have originated in the Indo-Burma region. Over several centuries, traders, travellers and rulers were attracted by Indian mangoes, and their efforts helped establish mango cultivation across tropical and subtropical areas. Historical accounts indicate that mango was taken to the Malay Peninsula by the Tamils and also introduced from the Indo-China Peninsula. The crop is now moving into regions where it was not previously grown and is currently cultivated in about 140 countries. In parts of the Western Hemisphere and in some developed countries, modern production practices for mango were adopted, many of them inspired by systems used for temperate fruit crops.

Deeply associated with culture

The importance of mango in Indian mythology, customs

and religious practices reflects its deep connection with the country's long cultural history. Ancient Hindus valued mango not only for religious and emotional reasons, but also for its role in the economic and social life of communities. The Sanskrit word "Amra" refers to mango as a fruit of abundance, while the Hindi word "Aam" is also associated with the common people. Mango inflorescences and leaves are used by Hindus in several rituals and in the worship of Saraswati, the goddess of knowledge and arts.

The influence of mango on Indian culture is widely recognised. In ancient India, both men and women were linked with mango through names and stories; for example, the woman who offered mango to Lord Buddha was known as "Amradarika." Names of musical ragas, such as Amra-takeswara and Amra-panchma, are also connected with this fruit. Mango cultivation in India is nearly as old as Indian civilisation itself. Indian art further records the importance of mango, with evidence seen in the Barhut Stupa of 110 BC and in the sculptures of Ajanta and Ellora.

In Malaysia and Indonesia, the Sanskrit or Tamil-derived names for mango suggest its movement from India to Peninsular Malaysia and eastern Asia. The fruit also received literary admiration, notably when the Turkman saint and poet Amir Khusroo praised mango in his poetry. The *Ain-i-Akbari*, the encyclopaedic work of Mughal emperor Akbar, provides detailed descriptions of mango and its varieties. Akbar is also remembered for establishing the famous "Lakh Bagh," an orchard of one lakh mango trees near Darbhanga, remnants of which are still known.

An economically important crop

Until the middle of the twentieth century, mango was considered an important economic crop mainly in Asian countries. Since then, mango cultivation has expanded and continues to develop in new subtropical and tropical regions. It has become an important export commodity in several non-traditional mango-producing countries, where a sizeable share of fruit production is directed to export markets. FAOSTAT data from 2014 indicate that Mexico was the leading mango exporter, exporting about 17% of its national production.

The development of Floridian mango cultivars helped adapt the crop to a wide range of agro-ecological conditions and encouraged the expansion of production into additional regions. Nevertheless, Asia continues to dominate mango production. The top five mango-producing countries are in Asia, and Bangladesh and the Philippines also appear

among the top ten producers.

Changing varietal preference in international markets

Red-skinned cultivars have long dominated the global export market because of their visual appeal. In Asian countries, however, consumers generally select mangoes on the basis of well-established varietal knowledge, flavour, aroma and local preference. As a result, many premium local cultivars with yellow or greenish-yellow skin fetch high prices in domestic markets but often receive less attention in international trade because they do not develop a red peel even at maturity.

This situation is gradually changing. Yellow-skinned mangoes are now gaining acceptance in the United States because of their eating quality. The yellow cultivar "Ataulfo" has attracted increasing consumer attention in the US market and has helped reduce the exclusive preference for red-skinned varieties. This shift may create opportunities for excellent Indian cultivars, many of which have yellow or greenish peels and have not been strongly favoured in export markets. Mango breeding programmes in India are also including red peel colour among their important objectives, and several newly developed cultivars now show red colouration.

A nutritionally rich fruit

Mango fruit is a good source of carbohydrates, minerals, dietary fibre such as pectin, vitamin C, vitamin A in the form of beta-carotene and several health-supporting phytochemicals. For millions of people living in subtropical, semi-arid and tropical regions, especially in countries where vitamin A deficiency is common, mango can serve as an important dietary source of vitamin A. In many areas, mango is available in abundance during the season, including in rural regions of developing countries.

Mango pulp contains very little lipid and may be regarded as virtually fat-free. The seed, however, is rich in lipids, and its fatty acid composition resembles cocoa butter. Mango fat may therefore have potential value in cosmetic, pharmaceutical and food industries. Mango is also a source of essential minerals such as calcium, iron, magnesium, phosphorus, potassium, zinc, copper, manganese and selenium, while the fruit remains naturally low in sodium.

Storehouse of bioactive compounds

Mango has been associated with a wide range of pharmacological properties, including antidiabetic, antioxidant, antiviral and anti-inflammatory effects. Other reported activities include antibacterial, antifungal, anthelmintic, antiparasitic, anti-HIV, anti-bone-resorption, antispasmodic, antipyretic, antidiarrhoeal, immunomodulatory, hypolipidaemic, antimicrobial, hepatoprotective and gastroprotective effects.

Mangiferin, a major bioactive compound isolated from different parts of the mango tree, has been widely studied

under in vivo and in vitro conditions. Reported effects include antioxidant, antidiabetic, antitumour, lipid-metabolism-regulating, cardioprotective, antihyperuricaemic, neuroprotective, anti-inflammatory, antipyretic, analgesic, antibacterial, antiviral and immunomodulatory activities. Zynamite, a mango leaf extract rich in natural polyphenols and mangiferin, has also been reported to improve sprint exercise performance when administered in combination with luteolin or quercetin. Lupeol, a natural pentacyclic triterpene present in mango fruit, has recently attracted attention because of its anti-inflammatory, anti-arthritic, antimutagenic and antimalarial properties, along with reported antitumour-promoting effects.

Numerous uses

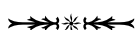
Mango is extensively used in culinary preparations, cosmetics and pharmaceutical applications, and useful products can be obtained from almost every part of the tree. The fruit is utilised at different developmental stages. Immature fruits are used for chutneys, curries, beverages, pickles and similar products, while ripe fruits are processed into puree, squash, nectar, drinks, mango leather, fruit bars, canned and frozen slices and sweets.

Mango stones are used for raising seedlings and can also provide oil, fibre and animal feed. The kernel is a rich source of carbohydrates, calcium, fat and starch, which may be processed for industrial purposes. Goats and cattle readily consume mango leaves, while branches and wood are used as fuel and timber. The bark is also used for extracting tannins and gums.

Invaluable processed products

Mango is a distinctive crop because it can be processed at both raw and mature stages. Technologies have been developed for many products, including canned and frozen slices, pulp, jelly, squash, tea, nectar and ready-to-serve drinks. A large number of traditional mango products are also prepared at the household or cottage-industry level. Mango is used in several hundred dishes and is combined with pulses, buttermilk, yoghurt, soups, meat, fish curry, spices and many other ingredients.

Mango pulp and pickles are among the most important mango products exported from India. In recent years, research has focused on innovative processing methods that help retain nutritional and sensory quality. The large diversity of mango pickles reflects regional recipes and culinary traditions across the world. In India, several varieties are specifically known for pickle making in particular regions. The famous Appimedi pickle from the Western Ghats of Karnataka is prepared from tender mangoes collected mainly from forests and riverbanks. Along with pulp, mango pickles continue to be major processed export products from India.





Indian Mango: From Orchard to Global Markets

Strengthening Value Chains, Processing Ecosystems and Global Competitiveness

By PA Division, Ministry of Food Processing Industries

India's mango sector stands as a symbol of agricultural richness, economic

“India accounts for nearly 40% of global mango production, yet unlocking value lies in strengthening processing, infrastructure, and market linkages.”

Oppportunity, and cultural heritage. As the world's largest producer, India contributes nearly 40% of global mango production, with annual output exceeding 20 million tonnes. The sector sustains millions of farmers across key producing states and showcases a diverse range of premium varieties, including GI-tagged products.

With evolving consumption patterns and rising global demand, the mango ecosystem is transitioning from a fresh-fruit dominated system to a diversified, value-added, and export-oriented sector.

Mango-Based Products: Unlocking Value Addition

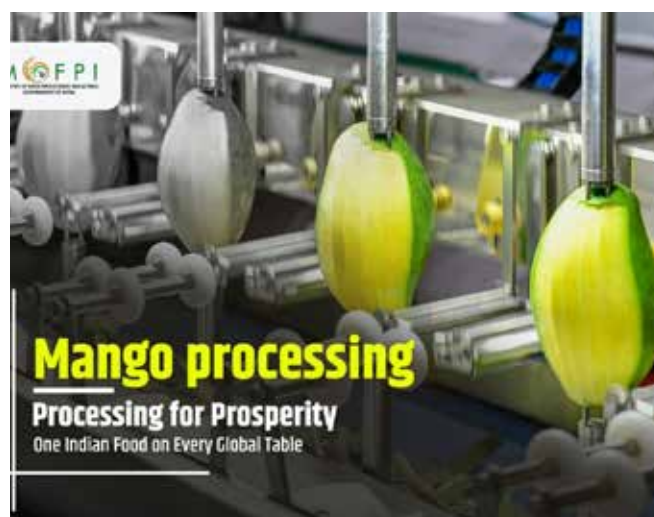
India is a global leader in mango processing, particularly in pulp, puree, concentrates, beverages, and traditional products like pickles and chutneys. The country contributes over 55% of global mango pulp production, underlining its strong processing capabilities.

Emerging segments such as freeze-dried mango, nutraceuticals, and clean-label products are expanding the scope of value addition. These innovations are not only enhancing shelf life but also enabling better utilization of lower-grade produce, thereby improving farmer incomes.

Value Chain & Supply Chain: Bridging the Gaps

The mango value chain—from farm to consumer—remains fragmented, with inefficiencies in aggregation, logistics, and market connectivity. Post-harvest losses are estimated at 20–30%, reflecting critical infrastructure gaps.

Strengthening Farmer Producer Organizations (FPOs), cluster-based approaches, and integrated packhouses is essential to improve efficiency, reduce losses, and enhance price realization.



Processing Ecosystem: Expanding Infrastructure

India has a well-established base of mango pulp processing units, particularly in southern and western regions. However, the need for decentralized primary processing infrastructure—such as washing, grading, and pulping at the farm gate—remains critical.

Through schemes like PMKSY and PMFME, the Ministry is promoting micro-enterprises, cluster-based processing, and value chain development, enabling inclusive growth and rural industrialization.

Under the PMKSY scheme Ministry has provided financial assistance to three projects under Operations Green scheme at Chittoor district to create more than 100000 MT processing capacity of mango processing and 30000 MT cold storage capacity for mango under the cold chain scheme.

“Reducing post-harvest losses by even 10% can significantly enhance farmer incomes and national output.”

The One District One Product (ODOP) approach under the PMFME Scheme plays a pivotal role in promoting localized value addition by identifying district-specific strengths and aligning them with targeted support for micro food processing enterprises. Under the ODOP framework, mango-based products have been identified and approved across 35 districts in 16 states, namely Bihar (Bhagalpur, Madhepura); Chhattisgarh (Dantewada); Gujarat (Gir Somnath, Navsari); Himachal Pradesh (Kangra); Jharkhand (Pakur, Sahebganj, Godda, Simdega); Karnataka (Dharwad, Haveri); Madhya Pradesh (Betul, Sidhi, Umaria); Maharashtra (Ratnagiri, Sindhudurg); Mizoram (Lawngtlai); Odisha (Angul, Subarnapur); Punjab (Roopnagar); Rajasthan (Banswara, Dungarpur); Tamil Nadu (Krishnagiri); Telangana (Jagitial, Mancherial, Nagarkurnool); Uttar Pradesh (Amroha, Lucknow, Mau, Sitapur, Unnao); and Uttarakhand (Udham Singh Nagar). This widespread identification underscores the geographical diversity of mango cultivation and highlights how the PMFME Scheme, through ODOP, is enabling micro-enterprises in these districts to develop specialized processing capabilities, strengthen value chains, and enhance market access while reducing post-harvest losses.

The impact of the PMFME Scheme is increasingly visible across states, with more than 550 loans sanctioned for micro enterprises engaged in mango processing with significant participation from states like Maharashtra, Andhra Pradesh, Jharkhand, and Telangana, reflecting both the availability of raw material and the growing uptake of value addition of Mango at the grassroots level. This transformation is also evident in the expanding range of products being developed by PMFME Scheme beneficiaries. Beyond conventional offerings, scheme beneficiaries are creating a diverse portfolio that includes mango pickle, aam panna, mango chutney, raw mango powder, mango jam, alongside other products such as dried mango slices, mango bars, candies, mango katli, and regional specialties like mango gorkeri. Notably, products like the Mango Gutli Mix—made from mango seed kernels blended with spices and natural ingredients—highlight how even by-products of Mango are being repurposed into nutritious, value-added foods. Together, these innovations reflect a shift towards minimizing waste, diversifying income streams, and aligning traditional processing practices with contemporary market demand, reinforcing the role of micro-enterprises as key drivers of India's mango processing ecosystem.

“Under PMKSY scheme 1 lakh MT processing capacity and 30000 MT cold storage capacity created for mangoes. 35 districts in 16 states have been identified for mango based products under PMFME scheme”

Efficient post-harvest management is vital for a perishable commodity like mango. Scientific ripening using ethylene chambers is replacing unsafe practices, ensuring better quality and consumer safety.

Cold chain infrastructure—including pre-cooling, reefer transport, and storage—remains a key enabler, though gaps persist in production clusters. Ongoing investments are addressing these challenges through integrated cold chain projects.

Packaging: Innovation Driving Quality

Packaging plays a critical role in preserving quality during transit. Traditional wooden boxes are being replaced by corrugated fibreboard boxes, ventilated crates, and advanced packaging solutions.

Innovations such as biodegradable and smart packaging are gaining momentum, enhancing sustainability, traceability, and brand value in global markets.

Exports & Logistics: Expanding Global Reach

India exports mangoes to over 60 countries, with major markets including the Middle East, Europe and North America. Fresh mango exports exceed 100,000 tonnes annually, alongside a robust processed product segment.

However, high logistics costs, freight constraints, and supply chain disruptions remain key challenges. Strengthening air cargo infrastructure and reducing transit time are critical to improving competitiveness.

Barriers to Trade: Navigating Global Standards

Indian mango exports face stringent phyto-sanitary requirements, irradiation protocols and documentation standards. Compliance gaps can lead to shipment rejections, affecting exporter confidence.

“Modern cold chain, irradiation and scientific ripening are transforming mango quality, safety and export readiness.”

Addressing these barriers requires capacity building, improved traceability, and stronger institutional support to meet global regulatory expectations.

Quality & Certification: Building Trust

Certifications such as Global G.A.P., HACCP, ISO, and organic standards are increasingly important for accessing premium markets. GI tagging further enhances authenticity and traceability.

Strengthening quality assurance across the value chain is essential to maintain consumer trust and ensure compliance with food safety regulations

Emerging Technologies: Driving Transformation

The mango sector is witnessing adoption of cutting-edge

technologies such as:

- AI-based grading and sorting
- Blockchain-enabled traceability
- IoT-based cold chain monitoring
- Advanced processing technologies like freeze-drying and aseptic packaging

These innovations are enhancing efficiency, reducing losses, and expanding product diversity.

Scientific & Technical Advancements

Scientific orchard management practices—including high-density planting, integrated pest management, and climate-resilient varieties—are improving productivity and quality.

With increasing climate variability, there is a growing emphasis on climate-smart agriculture and research-driven interventions to ensure sustainability of the mango sector.

“Quality certification and traceability are no longer optional—they are essential for global market access.”

The Way Forward

India's mango sector is poised for significant transformation. While production leadership is well established, the future lies in value addition, infrastructure development, quality assurance and global integration.

The Ministry of Food Processing Industries remains committed to strengthening the mango value chain through targeted interventions and collaborative partnerships. With a focused and integrated approach, India can emerge not only as the largest producer but also as a global leader in high-value processed mango products.



India's Coordinated Policy Response to Sustain Agricultural Exports During the Persian Gulf Crisis

In a volatile global trade environment, recent disruptions in the Persian Gulf, a key route for India's agricultural and processed food exports have posed significant challenges. For a sector reliant on timely delivery and product integrity, the impact extends beyond delays, affecting supply chain reliability and buyer confidence.

In response, the Ministry of Commerce and Industry, in coordination with DGFT and APEDA, has initiated structured and timely intervention. Led by the Department of Commerce, with DGFT's regulatory support and APEDA's operational role, the approach combined policy measures, logistical adjustments and stakeholder engagement to sustain export flows, minimize disruptions especially for perishable goods and reinforce confidence in India's export ecosystem.

1. Dedicated Helpdesk for Real-Time Grievance Redressal

A key early step was the launch of a dedicated Helpdesk on 6th March 2026. Acting as a centralized support system, it has enabled exporters to report issues and seek timely assistance. Concerns are being addressed on a case-by-case basis, with APEDA coordinating directly with shipping lines, port authorities and other stakeholders to resolve bottlenecks efficiently.

2. Strategic Diversification of Logistics Routes

To reduce dependence on disrupted Gulf ports, alternative logistics pathways are actively being explored. In coordination with Indian and UAE authorities, APEDA facilitated engagement with Khorfakkan Port, leading to its emergence as a viable transit hub. Priority berthing for food shipments and a bonded corridor linking Khorfakkan to Jebel Ali have ensured continuity of exports to GCC markets.

3. Leveraging Regional Corridors: Oman as an Alternative Gateway

Efforts have also been extended to develop Oman as an additional access point. APEDA engaged with Oman Port Authorities and exporters to operationalize these routes, enabling the use of regional transit corridors for smoother cargo movement. This has broadened routing options and reduced reliance on congested channels.

4. Regulatory Facilitation and Export Flexibility Measures

To ease operational challenges, DGFT introduced key relaxations in coordination with APEDA. These included

extending the validity of RCAC for basmati rice exports, allowing changes in destination ports and providing case-specific support for non-basmati shipments. Such measures have given exporters the flexibility to adapt to shifting logistics conditions.

5. Port-Level Interventions and Cost Mitigation Efforts

APEDA has engaged with major ports such as JNPA, Kandla, Mundra, Cochin and Chennai to address cost-related concerns, including demurrage and detention charges. Discussions around force majeure have also been taken up with DGFT. Additionally, efforts were made to secure dedicated storage space for perishable goods and facilitate off-port warehousing solutions for delayed consignments.

6. Strengthening Coordination with Shipping and Logistics Operators

Regular interaction with shipping lines and logistics providers has been central to maintaining cargo flow. APEDA has been working closely with operators such as Transworld, DP World, Maersk, Evergreen and Unifeeder to prioritize sensitive consignments, ensure vessel space and arrange alternative movement for stranded containers.

7. Air Cargo Facilitation for Time-Sensitive Shipments

For urgent shipments, APEDA has coordinated with airlines including Air India, SpiceJet, and Emirates, along with airport authorities, to streamline air cargo operations. These efforts have been particularly important for exports to Middle Eastern markets.

8. Focused Stakeholder Engagement and Sectoral Consultations

Regular consultations have been held with exporters across sectors such as rice, meat, poultry, eggs and fresh produce. These interactions, supported by the Department of Commerce and DGFT, is a helping factor in identifying sector-specific challenges and enabled more targeted responses.

9. Digital Communication Channels for Enhanced Coordination

To improve real-time coordination, APEDA has established dedicated communication channels,

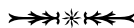
including WhatsApp groups for sharing updates, advisories and resolving issues quickly. This has strengthened responsiveness across stakeholders.

10. Proactive Representation and Inter-Ministerial Coordination

The Department of Commerce, along with DGFT and APEDA, has actively taken up exporter concerns with DG Shipping, port authorities and other ministries. This

continued coordination has been essential in addressing systemic issues and ensuring a unified response.

Overall, the measures reflect a balanced and coordinated approach, combining regulatory flexibility, logistical innovation and sustained engagement. Together, these efforts have helped stabilize export operations while strengthening the resilience of India's agricultural trade ecosystem.





Goglio Flex Pack System India Pvt. Ltd.

Goglio: a global player in Integrated Packaging Systems



Founded in 1850 in Rho, near Milan, Goglio is a key player in the packaging industry: over 176 years, it has evolved from a local manufacturer into a global leader, known for its innovative, integrated packaging solutions, establishing production facilities in Italy, the Netherlands, the United States, Brazil and China and offices in France, Spain, Poland, India and Japan.

One of the company's most transformative/disruptive innovations is the development of the Fres co System®. Introduced in the 1960s, it is the core of Goglio's philosophy, integrating proprietary filling machinery, high-performance flexible laminates and specialized spouts degassing valves to create complete solutions tailored to specific product needs - including those for the fruit and vegetable processing industry.

Goglio's mission focuses on innovation and sustainability, delivering solutions to protect product integrity, reduce environmental impact and optimize efficiency across the supply chain. By investing heavily in R&D and embracing digital transformation, Goglio ensures its clients benefit from the latest technological advancements - such as Industry 4.0 tools for real-time production monitoring through its proprietary MIND platform.

Sustainability is a driving force in Goglio's development of new materials and formats featuring monomaterial Design for Recycling laminates for pouches, and eco-efficient design.

Multilayer laminates and high-barrier structures are produced for applications such as coffee, powdered and liquid food and

are available in reels, pre-made pouches or large aseptic bag formats.

The in-house machine plant designs and builds highly customized packaging lines, for powders, solids, or liquids.

A standout product family is the GNova series - modular, horizontal filling systems supporting multiple processes such as hot fill, retort, aseptic and ambient. The GNova lines are suitable to manage pre-formed pouches ranging from 500 ml to 10l majorly required for HoReCa. Dedicated to this application is also the Pillow UP® pouch: a flat, flexible pouch that stands upright after filling, combining space-saving transport with strong shelf appeal, and ideal for both retail and foodservice use, for which it ensures sustainability, efficiency, and convenience.

Expanding its range of packaging lines for liquid products, Goglio has launched also the G18 W, a Vertical Form Fill & Seal (VFFS) line specifically designed for liquids. Starting from a reel, the line enables liquid products to be packaged in a three-seal pouch, offering a highly efficient and cost-effective solution for high-volume production.

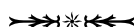
A cornerstone of Goglio's liquid packaging capabilities is its Fres co Aseptic System®, which provides sterile, barrier-protected packaging for liquid foods, produced in controlled ISO-6/8 environments and available in a wide selection of barrier films as well as in mono-PE/EVOH films that are fully designed for circular economy applications.

For companies in the fruit and vegetable industry seeking scalable, sustainable and reliable packaging solutions, Goglio represents a trusted partner: from aseptic bulk bags to high-speed horizontal fillers, its systems ensure your products are packaged safely, efficiently and in line with evolving global standards for food safety and sustainability.

From its historical roots in 19th century to its modern-day leadership in integrated packaging, Goglio exemplifies innovation, reliability and sustainability. Its deep expertise in liquid packaging, especially through its Fres co Aseptic System® and GNova filling lines, makes it a preferred partner for companies seeking flexible packaging solutions ready for the future.

Whether you're exporting processed fruit or vegetables, or launching a new retail-ready product, Goglio delivers performance, flexibility and innovation needed in today's competitive marketplace

To know more, Visit our website www.goglio.it or get in touch with Goglio Flexpacks India Pvt. Ltd. (Bangalore) – Mr. Vinay Birari on 9900079098.





The Comprehensive Packaging Solution



PRODUCTS CAN BE PACKED

POUCH STYLES

HANDLE



HANGING HOLE

SHAPES



- Packaging Lines
- Flexible Packaging High Barrier Laminates
- Advanced Service And Technical Assistance
- Clean Pouch Technology



OUR PRODUCTS

Goglio Flex Pack System India Pvt. Ltd.

www.goglio.it

Contact here for more details
Mr. Vinay Birari – 9900079098
Email – vinay.birari@gmail.com

Indian Mango - GI lineage

GI mangoes from India

Alphonso	Appemidi
Banaganapalle	Bhagalpuri Zardalu
Laxman Bhog	Gir Kesar
Malda Khirsapati (Himsagar)	Marathwada Kesar
Rataul	Kuttiattoor
Mallihabadi Dusseheri	Fazli



Geographical Indication (GI) Tagging and Its Role in Mango Value Enhancement

K. Prasad, PhD (Horticulture-Postharvest)

Assistant Professor (Horticulture)

Dr. Rajendra Prasad Central Agricultural University (RPCAU), Bihar

E-mail:- kprasad.tca@rpcau.ac.in



India contributes approximately 40–45% of global mango (*Mangifera indica* L.), production and bear the highest diversity of this crop. Yet it historically accounts for only 5–6% of international mango trade, indicating a gap between production and value realization. Geographical Indication (GI) tagging has emerged as a strategic instrument for enhancing the economic value and global competitiveness of region-specific distinct mango varieties of India. GI certification addresses this gap by linking product quality, reputation, and unique agro-ecological characteristics to defined geographical origins, thereby enabling market differentiation and premium pricing. Recent trade data indicate a 19% increase in India's mango export value, reaching about US\$ 47.98 million within a five-month period (FY 2023–24), reflecting improved market access and quality assurance mechanisms. GI-tagged mangoes, particularly premium varieties such as Alphonso, Himsagar, and Jardalu, have contributed to this growth through enhanced traceability, branding, and compliance with international standards. Technological interventions linked with GI systems—such as QR-based traceability—have enabled coverage of up to 100,000 metric tonnes of mangoes, directly benefiting over 1,000 farmers in organized value chains. Case-based evidence further demonstrates that GI tagging has facilitated export diversification, including the first commercial export of GI-certified Jardalu mango to the United Kingdom, marking the integration of non-traditional regions into global supply chains. At the regional level, export-oriented states such as Gujarat have reported cumulative mango exports exceeding 3,000 metric tonnes over five years, driven partly by GI-linked branding and infrastructure improvements. Empirical observations indicate that GI-tagged mangoes can generate 20–30% higher farm-gate returns due to price premiums and reduced market asymmetry.

G.I.: A tool for mango Value Enhancement

India produces 24–25 million metric tonnes (MMT) of mango annually, representing ~40–45% of global output, yet contributes <5% of global mango exports by value. This discrepancy highlights structural inefficiencies in value realization. GI tagging, by linking product quality to geographical origin, is hypothesized to reduce information asymmetry and enable premium market positioning.

GI Ecosystem in India: Latest Quantitative Status

(2025–26)

Indicator	Value
Total GI-registered products	651
Agricultural GI products	163
GI-tagged mango varieties	12
Year of GI Act enforcement	2003
Major GI-tagged mangoes	Alphonso, Kesar, Dashehari, Langra, Himsagar, Jardalu
GI Tagging growth trajectory	>300% increase in GI registrations since 2010

Mango Export Dynamics of India

Year	Export Volume (MT)	Export Value (USD million)
2021-22	27,000	~50
2022-23	22,964	48.5
2023-24	32,104	60.1
2024-25 (YTD)	29,938	56.5
Destination countries	UAE, USA, UK, and Qatar	
Remarks	Export growth rate of 6–8% annually (volume) while value growth slightly higher due to premium segments	

Source: PIB & APEDA (2026)

Market Distribution (%) and Role of GI:

Region	Share percentage
Middle East (UAE, Saudi Arabia, Qatar)	55–60
European Union	15–18
USA	10–12
Others	10–15
Remarks	GI-tagged mangoes dominate exports to EU and USA, where traceability and origin labelling are critical.

Data Source: APEDA (2026)

Price Differential Analysis

Mango Type	Price Range (USD/kg)	Premium (%)
Non-GI mango	2–3	-
GI Alphonso	5–8	100 to 160%
GI Kesar	4–6	60 to +120%
Remark	GI tagging significantly shifts mangoes into the high-value export segment.	
Data: Ranges derived from published studies of ICAR, NABARD & APEDA (2023-25)		

Supply Chain Value Distribution

Supply chain member	Non-GI Share (%)	GI Share (%)
Farmer	25–30	35–45
Intermediaries	40–50	30–35
Retail/export margin	20–30	25–30
Remark	GI systems improve farmer share in consumer price by ~10–15 percentage points, particularly when mediated through FPOs.	
Data source:	NABARD & ICAR Value and Supply Chain Studies (2023-25)	

Regional Case Evidence:

Following are the case studies where GI tagging led mangoes variety trade to significant value enhancement.

Alphonso (Maharashtra)

- Accounts for ~70–80% of premium mango exports
- Export prices consistently 2–3× higher than non-GI varieties
- Major markets: UAE, UK, EU

Kesar (Gujarat)

- GI tagging improved export branding
- Integrated packhouse systems led to ~25% reduction in post-harvest losses

Jardalu (Bihar)

- Post-GI export revival to Gulf countries
- Niche positioning as “heritage mango” increased visibility despite low volume

Impact on Post-Harvest Losses and Quality

Parameter	Non-GI System	GI-Oriented System
Post-harvest loss	25–30%	15–20%
Export rejection rate	High	Reduced by ~10–15%
Traceability	Minimal	Increasing (QR/blockchain pilots)
Remarks	GI oriented systematic mango trade lowers loss and rejection rate	
Data Source	ICAR e-Publication (2023), APEDA & Business Standard (2025)	

Constraints for GI Effectiveness and potential interventions

Various reports of WIPO, NCAER, APEDA, suggest that despite measurable gains, there is still scope of improvement in the quantitative gaps such as efficient use of GI Labelling (<30% of GI farmers actively use GI labelling), traceability adoption (<20% in mango supply chains) and Export-grade handling infrastructure covers (<25% of production zones bear this). Apart from this infrastructure, enforcement, and farmer awareness continue to limit its full economic potential utilisation.

The Data-Backed Policy and Technological Interventions which include cold chain expansion, GI branding campaigns, Blockchain traceability, FPO integration and such other measures are utilised by governmental and non-governmental organisations for enhancing the efficacy of mango trade and its value enhancement through GI.

Conclusion

GI tagging contributes to value enhancement through improved price realization, export expansion, supply chain transparency, and rural income generation. The way India shifting towards origin-certified, quality-assured mango trade, it’s on the path to become mango value leader in the global mango economy.



READERS WINDOW TO THE WORLD OF MANGO*

Indian Mango - GI lineage



**GI of
Kerala**

KUTTIATTOOR MANGO (KERALA)

Popular in Kannur district of Kerala, It is famous for its orange yellow colour of flesh, taste and flavor. The flesh is neither very soft nor too firm and has medium juiciness.

RATAUL MANGO (UTTAR PRADESH)

It is medium-sized fruit with distinct aroma and taste. The fruit is very sweet and delicious.

**GI of
Uttar
Pradesh**



**GI of
Gujrat**

GIR KESAR MANGO (GUJRAT)

Grown in Junagadh region, Gir Kesar mangoes have distinctive and rare characteristics like saffron colour of the pulp hence the name kesar mango. It also has distinct taste, aroma, pulp colour and sweetness.

MARATHWADA KESAR MANGO (MAHARASHTRA)

Grown in Marathwada region, it contains highest amount of total soluble solids among all mango varieties grown in India. It is also called Kesar Mango due to taste and colour similar to saffron.

**GI of
Maharashtra**



*Advice & disclaimer: Reader's discretion may be required on the pictorial/ graphical/ tabulated info depicted in this section which is intended only for general information. Reader may do a thorough check for detailed & authentic information on respective source site/ data source for further / detailed information.

READERS WINDOW TO THE WORLD OF MANGO*

Indian Mango - GI lineage

LAXMAN BHOG MANGO (WEST BENGAL)

It is thick-skinned, yellow green in colour with orange yellow flesh. Grown in Malda district of West Bengal. It is very sweet in taste and has characteristically pleasant flavour.

**GI of
West
Bengal**



FAZLI MANGO (WEST BENGAL)

It is a variety of mango grown in the district of Malda, West Bengal. It is big in size with medium thickness. Flesh is light yellow, firm and juicy with pleasant flavour and sweet taste.

**GI of
West
Bengal**



MALDA KHIRSAPATI (HIMSAGAR) MANGO (WEST BENGAL)

Grown in Malda region, it is oval in shape with yellowish-green colour. The taste is very sweet and the flavor is characteristically pleasant.

**GI of
West
Bengal**



APPEMIDI MANGO (KARNATAKA)

Mainly grown in Shimoga district, the fruits are fragile with fewer fiber content and sour in taste. The unripe fruit are used for pickle. The latex has a characteristic aroma by which the sub-varieties are differentiated.

**GI of
Karnataka**



READERS WINDOW TO THE WORLD OF MANGO*

Indian Mango - GI lineage

BANAGANAPALLE MANGOES (ANDHRA PRADESH)

Oblique to obliquely oval, shining golden yellow colour with sweet, meaty and fibreless flesh, it is known for its pleasant and delightful flavor.

GI
of Andhra Pradesh



GI
of Bihar

BHAGALPURI ZARDALU (BIHAR)

It is creamy yellow coloured, obliquely shaped fruit, with enticing aroma. It has a pleasant flavor, sweet taste and has moderately abundant juice.

GI mangoes from India

ALPHONSO (MAHARASHTRA)

Nicknamed as "King of Mangoes" it is loved for its colour, wholesome fragrance and irresistible taste. It is grown in Konkan region of western ghats. The pulp of the fruit is firm and fibreless with good sugar acid blend which gives distinct flavour and aroma.

GI
of Maharashtra



GI
of Uttar Pradesh

MANGO MALIHABADI DUSSEHERI (UTTAR PRADESH)

Grown on the banks of river Gomti, it has fibreless flesh with a pleasant flavor and taste. The unique characteristics of Malihabadi Dusseheri is attributed to the unique soil of the region.

Source: APEDA publication (21 Apr 2022); Indian GI Mangoes, Unique & bright it's flavourous delight

READERS WINDOW TO THE WORLD OF MANGO*

Mango – India Horticultural foot prints – at a glance

2025-26 (First Advance Estimates): Area and Production of Horticulture Crops

Category-wise : All India

Area(A) in '000 Ha

Production(P) in '000 MT

Sl.No.	Crops	2023-24		2024-25		2025-26	
		(Final)		(Final)		(First Advance Estimates)	
	Fruits	Area	Production	Area	Production	Area	Production
20	Mango	2396	22398	2418	23143	2417	23445
Total Fruits		7131	112978	7288	117649	7278	118680

Source: Ministry of Agriculture & Farmer's Welfare; 18-03-2026

All India Area, Production and Productivity of Guava, Litchi and Mango

Area in '000 Ha

Production in '000 MT

Productivity in MT/Ha

Year	Guava			Litchi			Mango		
	Area	Production	Productivity	Area	Production	Productivity	Area	Production	Productivity
1991-92	94.0	1095.1	11.7	49.3	243.8	4.9	1077.6	8715.6	8.1
2001-02	154.6	1715.5	11.1	58.1	355.9	6.1	1575.8	10020.2	6.4
2002-03	154.6	1793.0	11.6	54.1	476.4	8.8	1623.4	12733.2	7.8
2003-04	166.4	1830.7	11.0	53.7	478.5	8.9	1906.7	11490.0	6.0
2004-05	161.0	1682.8	10.5	60.0	368.6	6.1	1970.4	11829.7	6.0
2005-06	166.5	1736.6	10.4	63.2	392.1	6.2	2080.7	12663.1	6.1
2006-07	176.0	1831.0	10.4	65.0	403.0	6.2	2154.0	13734.0	6.4
2007-08	179.0	1981.0	11.1	69.0	418.0	6.1	2201.0	13997.0	6.4
2008-09	204.0	2270.0	11.1	72.0	423.0	5.9	2309.0	12750.0	5.5
2009-10	219.7	2571.5	11.7	74.4	483.0	6.5	2312.3	15026.7	6.5
2010-11	205.0	2462.0	12.0	78.0	497.0	6.4	2297.0	15188.0	6.6
2011-12	219.9	2510.4	11.4	80.4	538.1	6.7	2378.1	16196.4	6.8
2012-13	235.6	3198.3	13.6	82.7	580.1	7.0	2500.0	18002.4	7.2
2013-14	268.2	3667.9	13.7	84.2	585.3	7.0	2516.0	18431.3	7.3
2014-15	246.2	3993.5	16.2	85.0	528.3	6.2	2163.5	18527.0	8.5
2015-16	254.9	4047.8	15.9	90.1	558.8	6.2	2208.6	18642.5	8.4
2016-17	260.1	3826.4	14.7	93.3	568.2	6.1	2212.2	19506.2	8.8
2017-18	264.9	4053.5	15.3	92.3	686.4	7.4	2258.1	21822.3	9.7
2018-19	276.2	4252.6	15.4	95.5	721.4	7.6	2295.8	21378.1	9.3
2019-20	292.4	4361.2	14.9	96.6	726.2	7.5	2281.1	20264.9	8.9
2020-21	308.1	4582.3	14.9	97.9	720.1	7.4	2316.8	20386.0	8.8
2021-22	353.8	5526.6	15.6	99.3	737.2	7.4	2350.3	20772.3	8.8
2022-23	351.9	5364.5	15.2	96.2	746.0	7.8	2346.1	20872.2	8.9
2023-24	361.4	5368.1	14.9	99.0	577.1	5.8	2396.2	22398.2	9.3

Source: Horticulture Statistics Unit, D/o Agriculture & Farmers Welfare

Source: Ministry of Agriculture & Farmer's Welfare; Govt. of India

*Advice & disclaimer: Reader's discretion may be required on the pictorial/ graphical/ tabulated info depicted in this section which is intended only for general information. Reader may do a thorough check for detailed & authentic information on respective source site/ data source for further / detailed information.

READERS WINDOW TO THE WORLD OF MANGO*

Mango – India Horticultural foot prints – at a glance

Crop-wise Area and Production of Horticulture Crops for 2019-20 to 2023-24

Sl.No.	Crops	Area in '000 Ha Production in '000 MT									
		2019-20		2020-21		2021-22		2022-23		2023-24	
		Area	Production	Area	Production	Area	Production	Area	Production	Area	Production
	Fruits										
1	Almond	9	11	10	11	10	12	10	12	10	12
2	Aonla/Gooseberry	97	1164	100	1197	105	1278	106	1318	107	1378
3	Apple	310	2814	313	2276	315	2589	304	2876	304	2626
4	Banana	897	32597	924	33062	963	34528	994	36614	948	37614
5	Ber	54	606	53	580	52	547	52	531	52	526
	Citrus										
6	(i) Lime/Lemon	322	3687	327	3548	313	3776	312	3787	319	3836
7	(ii) Mandarin	454	6136	477	6219	454	6303	460	6178	474	6076
8	(iii) Sweet Orange (Mosambi)	217	4040	217	3988	228	3895	231	3927	240	3852
9	(iv) Other citrus	81	704	77	489	103	782	97	713	93	789
	Total Citrus (i to iv)	1075	14568	1097	14245	1098	14757	1100	14605	1126	14553
10	Custardapple	50	395	47	407	48	418	52	460	62	829
11	Dragon Fruit (Kamalam)							5	21	15	54
12	Grapes	150	3181	155	3358	163	3401	169	3740	180	3904
13	Guava	292	4361	308	4582	354	5527	352	5364	361	5368
14	Jackfruit	187	1739	188	1893	195	3301	190	3260	193	3392
15	Kiwi	5	14	5	16	5	16	5	18	4	17
16	Litchi	97	726	98	720	99	737	96	746	99	577
17	Mango	2294	20317	2317	20386	2350	20772	2346	20872	2396	22398
18	Muskmelon	61	1368	75	1478	68	1498	67	1498	67	1541
19	Papaya	142	5780	146	5540	150	5341	148	5240	149	5292
20	Passion Fruit	12	56	11	70	11	56	11	57	11	48
21	Peach	19	119	18	111	19	121	16	102	17	109
22	Pear	43	290	42	276	44	309	35	271	35	274
23	Picanut	3	0	1	0	1	0	1	0	1	0
24	Pineapple	106	1732	106	1799	108	1851	108	1828	106	1733
25	Plum	23	88	24	84	24	86	18	65	18	68
26	Pomegranate	283	3186	288	3271	270	3086	258	3098	224	2842
27	Sapota	84	906	79	822	78	941	73	875	72	887
28	Strawberry	1	9	3	20	3	11	2	19	3	21
29	Walnut	108	286	109	281	104	289	97	316	97	318
30	Watermelon	116	3157	119	3254	120	3308	126	3626	128	3721
31	Other fruits	258	2610	294	2741	306	2726	281	2776	346	2874
	Total Fruits	6774	102080	6930	102481	7064	107507	7025	110207	7131	112978

Source: Ministry of Agriculture & Farmer's Welfare; Govt. of India

Estimated Irrigated and Unirrigated Area by size of Operational Holdings under Mango and Banana

Sl. No.	Size Class (in Ha.)	MANGOES (601)				Banana (606)			
		No. of Holdings	Irrigated Area	Unirrigated Area	Total	No. of Holdings	Irrigated Area	Unirrigated Area	Total
1	Below 0.5	4313549	43807	91267	135074	1097874	49441	15199	64640
2	(0.5-1.0)	599141	55678	106025	161703	345825	59035	17880	76915
	Marginal	4912690	99485	197292	296777	1443699	108476	33078	141554
3	(1.0-2.0)	473748	105596	167997	273592	320722	106583	26654	133237
	Small	473748	105596	167997	273592	320722	106583	26654	133237
4	(2.0-3.0)	175976	58640	103633	162273	130498	60703	16015	76718
5	(3.0-4.0)	77580	37595	59961	97557	59981	31129	9831	40961
	Semi Medium	253556	96235	163595	259830	190479	91833	25846	117678
6	(4.0-5.0)	45377	27971	41977	69948	36582	19990	7650	27640
7	(5.0-7.5)	44200	36003	52561	88564	34170	21673	7509	29182
8	(7.5-10.0)	15019	18719	25917	44636	11075	11889	1834	13723
	Medium	104596	82693	120455	203148	81827	53551	16994	70545
9	(10.0-20.0)	11013	18673	25826	44499	8293	8098	3417	11515
10	20 & Above	2495	7986	10915	18901	1918	2849	1636	4485
	Large	13508	26659	36742	63401	10211	10947	5053	16000
11	All Classes	5758098	410667	686080	1096748	2046938	371390	107625	479015

Source: Agriculture Census 2015-16

Source: Ministry of Agriculture & Farmer's Welfare; Govt. of India

READERS WINDOW TO THE WORLD OF MANGO*

Mango – India Horticultural foot prints – at a glance

Sl.No.	STATES/UTS	2019-20			2020-21			2021-22			2022-23			2023-24		
		Area	Production	Productivity	Area	Production	Productivity	Area	Production	Productivity	Area	Production	Productivity	Area	Production	Productivity
1	ANDHRA PRADESH	376.49	4688.10	12.45	376.49	4517.93	12.00	393.94	4727.32	12.00	398.82	4985.28	12.50	398.82	4985.28	12.50
2	ARUNACHAL PRADESH													0.00	0.01	5.50
3	ASSAM	4.78	62.34	13.04	4.80	63.18	13.15	4.83	63.63	13.17	4.86	63.98	13.18	5.68	69.14	12.17
4	BIHAR	160.29	1543.48	9.63	160.24	1549.97	9.67	160.24	1549.97	9.67	162.99	1576.06	9.67	164.63	1580.43	9.60
5	CHHATTISGARH	75.63	463.57	6.13	76.12	459.13	6.03	77.44	465.45	6.01	78.24	471.15	6.02	77.45	471.50	6.09
6	GUJARAT	166.36	1222.29	7.35	163.78	997.83	6.09	166.33	917.20	5.51	173.52	960.17	5.53	177.51	1115.14	6.28
7	HARYANA	9.62	116.60	12.12	9.68	113.42	11.72	9.65	108.77	11.28	7.77	87.22	11.23	7.80	98.34	12.60
8	HIMACHAL PRADESH	42.39	31.88	0.75	42.42	51.55	1.22	42.38	39.11	0.92	42.48	45.50	1.07	42.42	36.57	0.86
9	JAMMU & KASHMIR	13.69	28.67	2.09	14.00	29.14	2.08	13.96	29.41	2.11	14.36	31.36	2.18	14.24	30.81	2.16
10	JHARKHAND	57.16	437.37	7.65	55.40	452.70	8.17	55.93	459.38	8.21	60.97	522.75	8.57	61.81	530.17	8.58
11	KARNATAKA	176.28	1700.97	9.65	172.79	1646.76	9.53	166.71	1620.21	9.72	148.38	1462.25	9.85	137.50	1457.03	10.60
12	KERALA	78.55	488.07	6.21	76.57	475.22	6.21	75.84	470.04	6.20	76.69	475.59	6.20	80.33	495.54	6.17
13	MADHYA PRADESH	52.89	747.16	14.13	56.79	798.99	14.07	60.33	842.33	13.96	62.94	885.49	14.07	66.28	965.84	14.57
14	MAHARASHTRA	155.53	450.59	2.90	162.76	442.35	2.72	164.40	459.15	2.79	169.76	390.94	2.30	170.28	405.92	2.85
15	MANIPUR	0.09	0.92	10.38	0.09	0.89	9.88	0.09	0.89	10.13	0.09	0.89	10.13	0.13	1.39	11.07
16	MIZORAM	0.96	4.27	4.46	0.96	4.27	4.46	0.96	4.27	4.46	0.96	4.27	4.46	0.96	4.27	4.46
17	NAGALAND	0.55	3.66	6.66	0.54	3.56	6.60	0.54	3.70	6.84	0.56	3.94	7.09	0.56	3.98	7.15
18	ODISHA	189.94	517.52	2.72	218.46	847.72	3.88	218.59	850.30	3.89	219.32	850.36	3.88	229.36	888.34	3.87
19	PUNJAB	7.01	118.78	16.94	7.99	137.28	17.19	8.90	156.20	17.56	9.00	164.58	18.28	9.13	173.95	19.06
20	RAJASTHAN	4.50	86.96	19.32	5.44	93.15	17.14	5.67	70.85	12.51	5.14	70.65	13.74	4.97	63.99	12.88
21	TAMIL NADU	158.20	546.17	3.45	146.07	639.64	4.38	147.98	943.37	6.37	145.89	974.85	6.34	145.90	949.45	6.51
22	TELANGANA	128.58	1270.36	9.88	124.20	1149.00	9.25	129.82	1165.13	8.98	131.23	1024.46	7.81	124.05	939.23	7.57
23	TRIPURA	10.27	52.28	5.09	10.29	52.37	5.09	10.36	52.72	5.09	10.16	51.14	5.04	10.19	51.37	5.04
24	UTTAR PRADESH	275.17	4734.50	17.21	279.25	4806.65	17.21	279.31	4662.45	16.69	279.31	4662.45	16.69	323.29	5990.53	18.53
25	UTTARAKHAND	36.91	156.79	4.25	36.60	150.67	4.12	36.60	150.67	4.12	21.05	113.41	5.39	21.14	112.09	5.30
26	WEST BENGAL	106.59	825.83	7.75	109.52	886.17	8.09	113.90	942.99	8.28	116.01	1027.58	8.86	116.16	882.27	7.60
27	OTHERS	5.58	17.46	3.13	5.61	16.47	2.94	5.61	16.85	3.00	5.62	15.92	2.83	5.64	15.65	2.78
	TOTAL	2294.01	20316.60	8.86	2316.81	20385.99	8.80	2350.30	20772.33	8.84	2346.09	20872.22	8.90	2396.22	22398.21	9.35

Source : Horticulture Statistics Unit, Department of Agriculture & Farmers Welfare

Source: Ministry of Agriculture & Farmer's Welfare; Govt. of India

READERS WINDOW TO THE WORLD OF MANGO*

Mango – India Horticultural foot prints – at a glance

State-wise Value of Output of Mango (at Current and Constant Prices)

(Rs. in Lakhs)													
Sl. No.	States/UTs	(AT CURRENT PRICES)						(AT 2011-12 PRICES)					
		2011-12	2019-20	2020-21	2021-22	2022-23	2023-24	2011-12	2019-20	2020-21	2021-22	2022-23	2023-24
1	Andhra Pradesh	265437	1836959	1882516	1969763	1984087	1599389	265437	593742	572191	598709	631380	631380
2	Arumachal Pradesh	221	37	37	37	0	16	221	12	12	12	0	5
3	Assam	16836	23320	25286	26704	27477	29694	16836	20458	20736	20883	20998	22692
4	Bihar	417208	1026706	1199548	1199548	1110345	1190951	417208	518562	520744	520744	529510	530977
5	Chhattisgarh	78701	171197	217988	198658	272190	219579	78701	134378	133089	134922	136573	136676
6	Goa	8778	24773	25705	25400	25470	32760	8778	11588	10822	10693	10723	10609
7	Gujarat	171079	295196	472041	512593	664702	739369	171079	216468	176716	162435	170046	197492
8	Haryana	19517	43441	42933	41595	35286	40228	19517	29401	28599	27427	21993	24797
9	Himachal Pradesh	7523	11241	23330	25287	21997	21546	7523	8269	13371	10145	11803	9486
10	Jammu & Kashmir	9597	7167	12765	12881	9427	11828	9597	6020	6120	6176	6585	6470
11	Jharkhand	96417	226338	234499	344533	394674	400805	96417	81788	84655	85903	97754	99141
12	Karnataka	223710	409084	354696	394911	404137	810907	223710	203675	197183	194004	175090	174464
13	Kerala	54562	144618	134645	137603	172144	151546	54562	71355	69478	68720	69531	72449
14	Madhya Pradesh	38790	253421	306254	289760	416709	366395	38790	165517	176999	186599	196160	213960
15	Maharashtra	234373	398869	432709	460385	440670	499000	234373	209954	206112	213940	182156	226413
16	Manipur	0	887	853	751	1000	2367	0	480	462	463	463	721
17	Mizoram	248	921	939	971	919	939	248	341	341	341	341	341
18	Nagaland	120	1523	1530	1589	1771	1791	120	1098	1067	1109	1181	1194
19	Odisha	177348	215388	301289	351089	373750	348709	177348	128330	210209	210849	210864	220282
20	Punjab	27731	40948	47476	56651	60533	64059	27731	31672	36605	41650	43884	46382
21	Rajasthan	22569	40561	41221	29937	38216	52253	22569	21807	23359	17768	17717	16046
22	Tamil Nadu	202064	85916	182214	274114	968531	2089446	202064	112316	145288	214277	210070	215657
23	Telangana	198646	543388	495704	525355	465617	420759	198646	177851	160859	163118	143424	131492
24	Tripura	6996	38340	37808	38378	51625	54984	6996	15306	15330	15432	14972	15037
25	Uttar Pradesh	749079	1512589	1353615	1482795	1531123	2348355	749079	923379	937451	909328	909327	1168346
26	Uttarakhand	36929	77725	86886	86886	27573	25813	36929	39176	37646	37646	28337	28006
27	West Bengal	302453	536904	554400	595784	733528	507686	302453	377588	405175	431154	469835	403395
28	A & N Islands	672	1293	1370	1454	1629	1313	672	722	658	778	654	654
29	Chandigarh	128	52	54	56	57	0	128	43	43	44	44	0
30	Dadra & Nagar Haveli	2214	5268	6129	5501	7325	0	2214	3390	3390	3390	3390	0
31	Daman & Diu	0	206	239	215	286	0	0	132	132	132	132	0
32	Lakshadweep	5	0	0	0	0	0	5	0	0	0	0	0
33	Puducherry	279	1721	1725	1548	1795	1701	279	534	535	535	466	441
All India		3370228	7975994	8478403	9092732	10244595	12034190	3370228	4105352	4195375	4289325	4315402	4605005

Source: State-wise estimates of Value of Output from Agriculture & Allied Activities, National Accounts Division (NAD), National Statistical Office, M/o Statistics & PI

Source: Ministry of Agriculture & Farmer's Welfare; Govt. of India



Source: APEDA AgriExchange

READERS WINDOW TO THE WORLD OF MANGO*

View India's Area: Horticulture Crops: Fruits: Mango: Andhra Pradesh from 2012 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: Telangana from 2014 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: Karnataka from 2012 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: Gujarat from 2012 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: Madhya Pradesh from 2012 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: Chhattisgarh from 2012 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: Odisha from 2012 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: West Bengal from 2012 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: Bihar from 2012 to 2024 in the chart:



View India's Area: Horticulture Crops: Fruits: Mango: Uttar Pradesh from 2012 to 2024 in the chart:



Source: <https://www.ceicdata.com/en/india/area-of-horticulture-crops-in-major-states-fruits-mango/area-horticulture-crops-fruits-mango>



PRAYAGH
CONSUMER CARE (P) LTD.



CHOCOLATES | CONFECTIONERY | CAKES



Let's Sweeten the World!

www.prayagh.com

www.lavian.in

www.lavianexotique.com

Indian Mango Pulp Industry: Challenges, Policy Imperatives and Emerging Opportunities



By Dr. Raghav Jadli

Business Managing Director, Jadli Foods
President, All India Food Processors' Association (AIFPA)

Introduction & Industry Overview

Mango pulp, a processed product derived from ripe mangoes by removing the peel and seed followed by pulping and preservation (aseptic or canned), is one of the most important value-added tropical fruit products globally. It serves as a key input for the food and beverage industry, particularly in juices, dairy applications, and processed foods. With increasing consumer preference for natural, fruit-based, and clean-label products, the global demand for mango pulp continues to grow steadily.

Nutritionally, mango pulp is rich in Vitamins A, C, and E, along with essential minerals such as potassium and magnesium, dietary fiber, and antioxidants. It supports immunity, digestion, and overall wellness, making it highly relevant in today's health-conscious consumption landscape. India processes multiple varieties such as Totapuri, Alphonso, Kesar, and Banganapalli, offering products in natural pulp, concentrate, and sweetened formats.

Globally, the mango pulp market is valued at approximately USD 1.61 billion (2025) and is projected to reach USD 3.01 billion by 2034, growing at a CAGR of 7–8%. India remains the largest exporter, with nearly 650,000 MT exported annually across key markets including the Middle East, Europe, and emerging Asia-Pacific regions.

Industry Leadership and Role of Jadli Foods

India's leadership in the mango pulp sector is strongly supported by established processors and exporters who have built global trust through quality, scale, and reliability. Among them, Jadli Foods stands as one of the leading manufacturers and exporters of mango pulp and concentrate, including Totapuri, Alphonso, Kesar, and Neelam varieties.

Incepted in 1999, Jadli Foods operates state-of-the-art manufacturing facilities in Chittoor, one of India's प्रमुख mango-growing regions. Over the years, the company has developed a strong backward linkage with more than 2,500 farmers, ensuring consistent quality sourcing and supporting rural livelihoods.

With exports spanning over 55 countries and enabling more than USD 10 billion in business value for its global partners, Jadli Foods has established itself as a trusted name in the industry. Over its 27-year journey, the company has earned

recognition for its commitment to quality, reliability, and long-term partnerships, reflecting the broader strengths of India's mango processing ecosystem.

Key Highlights

Strategic Importance to India

- Processes 10 lakh+ MT annually
- Supports 180+ lakh livelihoods
- Ensures farmer market linkage & income stability
- Generates significant foreign exchange earnings
- Strengthens rural economies, especially regions like Chittoor

Current Challenges

- Global disruptions: COVID aftermath, Russia-Ukraine war, Middle East crisis
- Logistics issues: Suez Canal disruptions, high freight costs
- Trade barriers: Import duties in key markets
- Demand slowdown: Shift toward synthetic & energy beverages
- Inventory stress: ~1.5 lakh MT pulp unsold/unexported
- Farmer impact: Reduced procurement, price volatility, distress sales

Policy Solutions Required

- ECLGS-type credit guarantee support
- MSME liquidity & SME Growth Fund access
- 3–6 month loan moratoriums
- Force majeure recognition
- RELIEF scheme inclusion for export facilitation

GST Rationalization – A Game Changer

- Current GST: ~40% on carbonated fruit beverages (sin category)

Recommendation:

- Reduce to 5%–12% GST



MANGO PULP, GUAVA PULP, TOMATO PASTE AND OTHER FRUIT PULPS & CONCENTRATES FOR INDUSTRIAL EXCELLENCE



Consistent
Brix & Quality



Aseptic &
Canned Packing



Bulk Supply
Capability



Export-Ready
Standards



Strict Quality
Control

Ideal for
Beverages | Ice Cream
Dairy | Confectionery
Bakery | And More

Your Trusted Partner in Premium Fruit Pulps Solutions



JADLI FOODS (INDIA) PVT. LTD.
An ISO 22000 : 2018 Certified Company

📍 219, R.G. Complex - II, Plot No. 5, Sector-14, Rohini, Delhi - 110085, India
☎ 0091-11- 27555459, 27555460, 27555461
✉ info@jadlifoods.net | 🌐 www.jadlifoods.com



- Ensure 7–10% minimum fruit pulp content

Impact:

- Boosts mango pulp demand
- Creates “Fun + Health” beverage category
- Attracts Gen Z consumers
- Promotes healthier alternatives
- Enhances farmer income

Opportunities Ahead

- **Domestic consumption:** Defence, police, school nutrition programs
- **Product innovation:** Sauces, RTD beverages, functional drinks
- **HORECA integration:** Hotels, restaurants, catering

- **Global expansion:** New markets beyond traditional regions

- **Branding:** Trade exhibitions and global positioning

Conclusion

The Indian mango pulp industry stands at a critical juncture. While global disruptions have created short-term pressures, the long-term outlook remains strong. With progressive policy support, particularly GST rationalization, along with innovation, diversification, and stronger industry–government collaboration, India can further strengthen its leadership in the global mango economy.

At its core, this sector represents far more than processing—it is about securing farmer livelihoods, promoting healthier consumption, strengthening rural economies, and enhancing India’s export competitiveness.



Innovation in Mango-Based Products

Ch. Pallavi Sri Padma*, Shruti Sethi and Sindhu PM

Division of Food Science and Postharvest Technology, ICAR-Indian Agricultural Research Institute, New Delhi, India

(*corresponding author email: pallavisripadma@gmail.com)



Ch. Pallavi Sri Padma

INTRODUCTION

Mango (*Mangifera indica* L.), a member of the Anacardiaceae family, is widely consumed worldwide for its rich aroma and balanced sweet-acidic taste. Mango production in India is estimated to be 228.37 lakh metric tonnes in 2024–25, of which approximately 7% is utilized for processing. Major mango producing states include Uttar Pradesh, Andhra Pradesh, Bihar, Karnataka, Gujarat, Madhya Pradesh and Tamil Nadu, with Uttar Pradesh leading at 26.75% share and highest productivity (2023–24). India is also a key exporter, with major markets in the UAE, USA, UK, Kuwait and Qatar. In FY25, India exported 29,938.40 metric tonnes of mangoes, generating a total export value of USD 56.50 million.

Mango processing and its importance

Mangoes are processed at two maturity stages: unripe fruits are used for chutneys, pickles, curries and dried products, while ripe fruits are processed into canned or frozen slices, purée, juices, nectar and other products for domestic and small-scale industries. Only fresh, undamaged fruits are suitable for processing. However, industrialization faces challenges such as alternate bearing, short shelf life and wide varietal diversity, which affect consistency and continuous processing. The absence of reliable maturity indices and mechanized peeling systems further limits efficiency, especially given variations in fruit size and shape. High production costs also restrict affordability in producing regions. Despite these issues, there is strong export potential, particularly to developed markets, although competition with established processed fruits remains high. Rising demand for natural mango purée is driving market growth, with the Middle East leading in imports and consumption, followed by Europe (over 20%). Mango purée is widely used in beverages, as well as in ice cream and baby food industries.

Primary mango products include puree, pulp and concentrate, which support a secondary processing sector accounting for over 60% of the market. Key secondary products include juices, beverages, fruit bars, jellies, candies, jams and fruit-based cosmetics, along with squashes and powdered drink mixes. Rising use of mango puree in confectionery and dairy products such as puddings, bakery fillings, ice creams and baby foods is driving demand. Additionally, pulp is widely used in beverage production, while concentrates enhance sweetness and flavor, further supporting market growth.

Mango Varieties Used for Processing

In India, Alphonso, Totapuri and Kesar are the primary processing varieties, while Sindura, Raspuri, and Neelam are used in smaller amounts. Processing mainly uses southern and western varieties, with northern types largely consumed fresh. Totapuri (76%) and Alphonso (14%) dominate (90% total) in processing sector. These varieties have strong domestic and export demand. The season runs from April to July, with peak processing during May–July.

Urbanization and higher incomes are boosting demand for ready-to-eat foods and ready-to-drink beverages globally. In response, mango-processing industries are shifting toward convenient products, with manufacturers favoring mango puree over artificial flavors to retain natural taste and quality. Changing lifestyles and wider use of puree are expected to further drive growth in processed mango products.

A. RAW MANGO PRODUCTS

In India, green mangoes are widely utilized for pickles and chutneys, with pickles prepared both domestically and commercially.

1. Mango Pickle

Mango pickles are prepared from green, hard mangoes (2–3% acidity, 10–15°Brix) as salt or oil pickles using whole or sliced fruits, typically with added salt. They are valued for their tangy taste, long shelf life and nutritional contribution. Rich in vitamin C, they support immunity and iron absorption. Preservation using sugar, salt, chilli powder, citric acid and potassium metabisulphite (250–1000 ppm) improves shelf life compared to dry salting (Nupur, 2020). Typically, raw mango pieces are deseeded, sun-dried, mixed with spices and mustard oil and cured in sunlight for several days to develop flavour (Deeksha & Sunita, 2020).

Instant mango pickle technology developed at CFTRI involves washing and cutting mature mangoes, brine curing and drying (sun or mechanical), followed by mixing with ground spices (chilli, turmeric, mustard), packaging and sealing. The product is reconstituted in water overnight to obtain ready-to-eat



pickle. FSSAI specifications for pickles include a minimum drained weight of 60%, sodium chloride content of 12%, acidity of 1.2%, and a preservative limit of 100 ppm of SO₂.

2. Mango Chutney

Mango chutney is a sweet-sour condiment prepared from peeled, sliced, or grated unripe/semi-ripe mango varieties like Totapuri and Fazali cooked with salt, sugar/jaggery, vinegar and spices to a thick purée, then cooled and stored in sterilized jars. Additional ingredients such as onion, garlic, dried fruits and nuts may be included. FSSAI standards specify a total soluble solids content of 50%, with preservative limits of either 100 ppm of SO₂ or 250 ppm of benzoic acid for chutney.



3. Amchur (Green Mango Powder)

Amchur (mango powder) is a tangy souring agent made from dried unripe mangoes, widely used in curries, chutneys and snacks in Indian cuisine. It is rich in minerals such as iron, calcium, potassium and magnesium, supporting overall health. Traditionally, unripe mango slices are sun-dried (16 h at 35–40°C) before grinding into powder.



4. Beverages: Green Mango RTS Beverage

Green mango beverage is a tangy, refreshing drink made from unripe mango pulp blended with spices and sweeteners. The pulp is obtained by grading, washing, boiling (20 min), peeling, slicing, pulping and homogenizing raw mangoes, and is typically used at 10% in RTS drinks, with acidity 4%, TSS 11–13°B and low pH.



5. Raw Mango Candy

Hard boiled raw mango candy is preferred for its strong tangy-sweet taste that closely resembles raw mango (kaccha aam). FSSAI standards specify fruit pulp -55%, TSS -75% and citric acid -1-1.5% for candy.



B. RIPE MANGO PRODUCTS

1. Fresh-cut Pieces/ Slices

Minimal processing involves basic steps like cleaning, peeling and slicing to preserve fresh quality and nutrition, using non-thermal methods to extend shelf life with minimal additives. The fresh-cut mango pieces are prepared by sorting and washing ripe, firm mangoes in sanitized water to remove contaminants, then peeled and sliced under hygienic conditions. Slices are treated with antioxidant and calcium solutions to prevent browning and



maintain firmness and finally packed, often using modified atmosphere packaging to extend shelf life.

2. Canned Mango Slices

Canning preserves food by heating in sealed containers to destroy microbes and enzymes, forming a vacuum seal. It extends shelf life up to several years, using water bath canning for high-acid foods and pressure canning for low-acid foods. Canning mango slices involve placing peeled, firm-ripe slices in jars, adding hot sugar syrup (or water) and heat-processing in a boiling water bath for 15–20 minutes.



3. Mango pulp or puree

India is a leading exporter of mango pulp (98,369.74 MT worth Rs. 714.41 crores/96.43 million USD in (2020–21), with major markets including Saudi Arabia, Yemen, the Netherlands, Kuwait and Oman (Ministry of Food Processing Industries Report).



The Middle East dominates imports of processed mango products, followed by Europe (>20% share). Mature mangoes are harvested, transported, inspected, washed and graded; high-quality fruits are ripened under controlled ethylene conditions and processed into pulp. The pulp is preserved by freezing, canning, or barrel storage to ensure year-round availability. For peeling, fruits are steam-treated (2 min) to soften skins, then mechanically peeled, pulp is separated by centrifugation and refined through screening to remove fibres and skin. Pasteurization (85°C/30 min) is followed by addition of citric acid (0.3%) and KMS (0.5–0.7 g/kg). Pulp/concentrate from Alphonso, Totapuri and Kesar are widely used in beverages, jams, bakery, dairy, and confectionery products.

4. Beverages

a) Mango Wine

Due to its high sugar content (TSS >16°), mango pulp is well suited for fermentation (Ogodo et al., 2018). Mango wine is a tropical, semi-sweet, full-bodied fruit wine with a golden-amber color and rich fruity aroma. Made from ripe mangoes, sugar and yeast, it undergoes 7–10 days of fermentation followed by aging. Moreover, the choice of yeast strain significantly influences aroma, flavor, and ethanol levels (Patel et al., 2021).



b) Ready to serve beverages (RTS)

RTS is a diluted beverage containing at least 10% fruit juice/pulp, 10% total soluble solids (sugar), water, and added

acid/preservatives. FSSAI specifications include juice content -10%, acidity – 0.2-0.3%, TSS-10 % and preservative -70 ppm of SO₂ for RTS.



c) Nectar

Nectar is a non-carbonated beverage prepared from fruit pulp or juice, usually diluted with water and sweetened. They contain at least 20% fruit content and include natural fibers, pectin and added sugar or acid. Mango nectar contains about 20% pulp with 15°B TSS and 0.3% acidity and is heat processed at 85°C before canning.



d) Squash

Squash is a concentrated fruit drink made from juice or pulp, sugar and water, diluted before use and often containing added acids and preservatives. It typically has 25% juice, 45% TSS and 1.2–1.5% acidity, preserved using sulphur dioxide (350 ppm) or sodium benzoate (1000 ppm). FSSAI specifications include juice content of 25%, acidity of 1%, TSS of 45 %, preservative of 350 ppm of SO₂ (or) 600 ppm of benzoic acid.



e) Carbonated mango beverage

A carbonated mango beverage is a fizzy drink made from mango pulp or juice, blended with sugar, water, acidity and carbonated with about 3% CO₂ to create a refreshing, sweet and effervescent beverage.



f) Mango sago

Mango sago is a chilled dessert made with mango puree, coconut milk, and small chewy sago (tapioca) pearls, often sweetened and served cold for a creamy, refreshing taste. It is prepared by boiling tapioca pearls for about 10 min until they turn mostly translucent. After draining and rinsing with cold water, they are set aside. Mangoes are blended with sweetened condensed milk, adjusting sweetness as needed. The cooked pearls are then mixed with diced mango and nata de coco, followed by the addition of coconut milk, and served chilled, with ice if desired.



g) Blended mango beverages

Mango aloe vera juice

Mango-aloe vera juice is a refreshing functional beverage made by blending mango pulp with aloe vera gel or extract. It combines the natural sweetness



and flavor of mango with the hydrating and digestive benefits of aloe vera, making it popular as a health drink.

Mango flavoured milk shake and soymilk

Mango flavoured milk shake is a popular summer beverage prepared by adding mango powder to sweetened, boiled and cooled milk whereas mango soy milk is prepared by blending ripe mango with soy milk and sweetener until smooth. Optionally vanilla or cardamom maybe added for flavor.



Mango lassi

Mango lassi is a traditional yogurt-based drink made by blending ripe mango pulp with curd, sugar and sometimes cardamom, resulting in a thick, creamy and refreshing beverage.



Mango bubble tea

Mango bubble tea is a vibrant, tropical beverage that combines the rich sweetness of ripe mango with the playful texture of chewy tapioca pearls or bursting fruit beads. Typically prepared with a tea base such as black or green tea or as a tea-free smoothie, it may include milk or plant-based alternatives for a creamy consistency. The drink is served chilled, offering a refreshing balance of fruity flavor, mild tanginess and subtle sweetness. Its distinctive appeal lies in the contrast between the smooth liquid and the soft, chewy or juicy pearls, making mango bubble tea both a flavorful and texturally engaging drink.



Mango black iced tea

Mango black iced tea is made by steeping a tea bag in 100 ml of hot water for about 5 min to obtain mild astringency without bitterness. The tea is blended with mango puree, strained over ice, and topped with soda..



Mango energy drink

Mango energy drink is commercially available in the market which is a functional beverage that blends mango flavor or pulp with energy-boosting ingredients like caffeine, taurine, vitamins and sugars. It provides a refreshing taste along with enhanced alertness and energy.



Mango probiotic drink

This is a functional beverage made by combining mango pulp in a probiotic



drink containing beneficial live microorganisms. It offers a pleasant fruity taste while supporting gut health, digestion and overall wellness.

Mango smoothie

A smoothie is a thick, creamy beverage made by blending fruits with liquids such as milk, yogurt, or juice, often enhanced with sweeteners, ice, or nutritional ingredients. A mango smoothie is made by blending mango, banana, coconut yogurt, coconut milk, maple syrup and vanilla until smooth, adjusting the milk as needed.



5. Intermediate Moisture Foods

a) Mango Jam

Mango jam is a semi-solid product prepared by cooking mango pulp with sugar, acid and pectin in suitable proportions. (Krishna et al., 2020). Jam preparation typically involves heating pulp with almost equal sugar to 65–68°C and adjusting acidity to 0.6–0.7% with citric acid for better colour and flavour. FSSAI specifications include fruit pulp of 45% and, minimum total soluble solids content of 65% alongwith 40 ppm of SO₂ for jam.



b) Mango murabba

Mango murabba is a traditional Indian sweet-tangy preserve made by cooking raw mangoes in sugar or jaggery syrup, often flavored with cardamom, cloves or saffron and served as a condiment or spread. FSSAI specifications are fruit pulp -55%, TSS -70% and citric acid - 1-1.5%.



c) Mango jelly

Fruit jelly is a transparent, semi-solid fruit preserve made by cooking fruit juice with sugar and pectin. It has a gel-like consistency that holds its shape and is valued for its clear appearance and pure fruit flavor, commonly used as a spread or pastry filling. Mango jelly is prepared by blending mango with sugar without adding water and alongwith addition of agar-agar solution. The mixture is strained, poured into molds and refrigerated. FSSAI specifications include 45% fruit pulp, 65% TSS and 40 ppm of SO₂ as preservative.



d) Mango Leather and mango energy bar

Mango leather (*Aam papad*) is an intermediate moisture product made by blending mango pulp with sugar, acids and preservatives, then drying into sheet form. A mango energy bar is a nutrient-dense, ready-to-eat snack made from dried mango or pulp combined with cereals, nuts and natural sweeteners. It has a soft, chewy texture and a pleasant tropical flavor,



with mango providing natural sweetness, vitamins (A and C), antioxidants, and fiber. Rich in energy, it is suitable for athletes, children and busy individuals, and serves as a healthier alternative to confectionery due to its low refined sugar and minimal additives. It is a convenient and portable functional food for on-the-go consumption.



a) Mango toffee and hard confectionery

Mango toffee can be prepared from mango pulp by mixing mango pulp with sugar, glucose and skim milk powder alongwith some hydrogenated fat and suitable amounts of flavouring essence and colour. Hard confectionery are sugar-based candies heated to the "hard crack" stage, resulting in a brittle, glassy texture. Primarily made of sucrose and glucose syrup, these candies are often flavored, colored and produced in various shapes through molding or cutting.

6. Mango Ice Cream

Mango ice cream is a creamy frozen dessert made from mango pulp, milk or cream, sugar, and sometimes flavorings, offering a smooth texture and rich, fruity taste. It can be prepared by blending mango powder with milk in a 3:10 ratio and 17% sugar. Peres et al. (2024) developed a sugar-free mango ice cream using 82% of pasteurized mango pulp, 7% of pasteurized passion fruit, 6% of whey protein and 5% of coconut oil, which showed good acceptance among cancer patients, providing nutritional benefits and helping ease chemotherapy-related effects.

7. Drying or dehydration

Fruit juice powder production has grown in importance due to its advantages. Dried mango powder is widely used in ice cream, bakery, and confectionery products and is prepared by concentrating mango pulp or juice, mixing with sugar and drying.

a) Foam mat dried mango powder

Foam mat drying is an alternative to techniques such as spray drying and freeze drying. In this method, a liquid is whipped into a stable foam and then dehydrated using heat. The increased surface area of the foam enhances moisture removal, making drying faster and more efficient for high-moisture foods. Essentially, the process involves converting a liquid into a stable foam, followed by air drying.



For mango powder preparation, mango juice is whipped with liquid milk along with non-fat condensed milk, fat, minerals and proteins. The mixture is blended using a hand blender until a uniform foam is formed. This foam is then spread in a steel tray at a thickness of 3 mm and dried at 65°C. Once dried, the material is scraped off and ground into powder.

b) Vacuum dried mango pulp

Vacuum drying is a low-temperature, low-pressure moisture

removal process that utilizes a sealed chamber to evaporate water at temperatures below 40–50°C. Jaya et al. (2003) standardized vacuum drying of mango pulp using 2.6 mm thickness and plate temperatures of 72.3°C under 30–50 mmHg.

c) Vacuum-fried mango chips

Vacuum-fried mango chips are prepared by vacuum-frying of mango slices (4 mm) at 10 kPa using an industrial fryer with fresh high-oleic sunflower oil. A high oil-to-fruit ratio minimizes temperature drop upon immersion. The process ensures reduced fat uptake and crunchy texture (Ayustaningwarno et al., 2020).



c) Freeze dried mango

Freeze drying is a preservation method that removes moisture from food by freezing it and then sublimating the ice under vacuum, helping retain flavor, nutrients and structure while extending shelf life. Freeze-dried mango is prepared by thawing frozen mango puree at 4°C for 12 h followed by rapid freezing to –50°C until the core reaches –20°C, followed by two-stage drying at 40 Pa. Primary drying continues until the core temperature reaches –10 to 10°C, after which secondary drying is carried out at 40°C (Mawilai et al., 2019).



d) Ohmic-heated mango pulp

Ohmic (Joule) heating is a method in which electrically conductive foods are heated by passing alternating current through them, generating internal heat due to electrical resistance. It provides rapid, uniform heating, making it suitable for pasteurization and sterilization while minimizing nutrient loss and preserving quality. In mango pulp processing, ohmic heating is an advanced volumetric technique that ensures better retention of nutrients such as vitamin C, phenols and color, while effectively inactivating enzymes. Typically, about 500 mL of mango pulp (at set TSS) is placed between stainless steel electrodes in an ohmic chamber and heated at 15 V/cm. The temperature is raised to 70–80°C and held for 2 minutes to achieve pasteurization (Priyadarshini et al., 2023).

e) Fruit drink mix

Fruit drink mix is a powdered or concentrated product that produces a refreshing beverage when mixed with water. It is made from fruit powders or concentrates, sweeteners, acids and sometimes added nutrients. These mixes are convenient, shelf-stable and easy to store, dissolving quickly to provide a balanced fruit flavor. They are cost-effective, portable and ideal for instant drink preparation. Pour one sachet into 200 ml water, mix for 30 seconds, and serve.



f) Mango yogurt cubes

Mango yogurt cubes are bite-sized frozen snacks made by mixing mango puree with yogurt and sweetener, then freezing

the mixture into small cubes for a creamy, tangy treat. Freeze-dried mango yogurt bites are made by combining equal parts thick yogurt and diced mango, adding sweetener if desired, then portioning the mixture onto a lined tray. It is frozen until solid and subsequently freeze-dried to obtain a light, airy texture.

8) MANGO BASED COMPOSITE FOODS

- a) **Mango yoghurt:** Mango yogurt is a creamy dairy product made by blending yogurt with mango pulp or puree, often sweetened to create a smooth, fruity and refreshing snack or dessert. It can be served as a suitable alternative to high-calorie dessert puddings containing cream and curd. Moreover, yoghurt prepared with mango powder and milk in a 3:10 ratio.



- b) **Mango soy yogurt:** Mango soy yogurt is a plant-based yogurt made by blending soy yogurt with mango pulp or puree, resulting in a creamy, fruity and dairy-free product. It is prepared by blending 1 cup of ripe mango with ½ cup soy yogurt and 1–2 tablespoons of sweetener until smooth to obtain a creamy mango soy yogurt.



- c) **Mango peanut butter:** Mango peanut butter is a flavored spread made by blending roasted peanuts with mango pulp or powder, giving a sweet, nutty, and slightly tangy taste with a creamy texture. It typically contains ground peanuts, mango, sweeteners, and sometimes oils or stabilizers. Mango adds natural sweetness, color and nutrients like vitamins A and C. The product can be used as a spread, dip, or in smoothies and desserts, offering a nutritious and convenient alternative to regular peanut butter.



9. OTHER MANGO PRODUCTS

- a) **Mango custard powder:** Mango custard powder is a ready-to-use dessert mix made from cornflour, sugar, milk solids and mango flavoring. When cooked with milk, it forms a smooth, creamy custard with a rich mango taste and yellow color. It can be served hot or chilled and is often enhanced with fruits or nuts. Widely used in homes and food service, it is ideal for fruit custards, puddings, trifles and bakery fillings, offering convenience, consistency and long shelf life.



- b) **Mango cereal flakes, vermicelli and powder:** A process for producing mango cereal flakes was

developed at CFTRI, Mysore, involving pH adjustment of pulp (5.4) using sodium bicarbonate, blending with precooked wheat flour and sugar and drying on a double drum dryer. The product was golden yellow, flavorful and hygroscopic, requiring moisture-proof packaging.



- c) **Strained baby food:** Mango-based strained pulps and custards were developed and evaluated for their quality and consumer acceptability. The pulp was filtered through a 60-mesh sieve to eliminate fibrous material. Sugar was then incorporated to achieve the desired formulation, followed by homogenization of the mixture and subsequent canning.



C. MANGO BY-PRODUCT UTILIZATION

Mango processing generates about 40–50% by-products, including culled fruits, seeds (20–60% of fruit), kernels (45–75% of seed), de-oiled kernel meal, and peels (7–24%). These residues are valuable for livestock feed and as sources of pectin and phenolics. Mango kernels contain 7–12% oil rich in stearic (24–57%) and oleic (34–56%) acids, suitable as a cocoa butter substitute in chocolate, as well as for soap, fat-based products, and soil enrichment. Globally, 150,000–400,000 tons of mango waste are generated, posing environmental concerns; however, their utilization in feed and value-added products helps mitigate these issues.

Mango peel powder, owing to its high pectin content, functions as an effective emulsifier and fat replacer, enabling up to 50% oil substitution in products like chicken patties while improving texture and color. Its incorporation in muffins (75% replacement of wheat flour and sugar) enhances physicochemical, sensory, and functional qualities (Ramirez-Maganda et al., 2015). Peel extracts also improve packaging films by increasing antioxidant activity and reducing solubility and permeability (Adilah et al., 2018). Mango seed constitutes about 10–25% of the fruit and contains a kernel rich in protein (6–8%), fat (11–13%), and starch (8–10%), along with phenolics and phytosterols. The fat, high in oleic and stearic acids (“mango butter”), is used in low-fat cookies (Kumar et al., 2023) and as a cocoa butter substitute in chocolate (Kaur et al., 2022). Kernels exhibit high phenolic content (~112 mg/100 g) and strong antioxidant activity (Soong et al., 2004). The extracted fat has edible and industrial uses, while kernel meal can be utilized for starch production, animal feed, and manure. Pectin and vinegar are also derived from peel and kernel waste. Mango leaves contain bioactives such as mangiferin, flavonoids, phenolic acids, and xanthenes, showing strong antimicrobial activity against pathogens like *Escherichia coli* and *Staphylococcus aureus*. These properties support their use in edible or biodegradable films for extending food shelf life (Fang et al., 2025).

D. OTHER INNOVATIVE MANGO PRODUCTS

- a) **Mango halwa:** Mango halwa is a rich dessert prepared by cooking mango pulp with sugar, ghee, and a thickening agent like semolina or flour. It has a soft, smooth texture, bright golden color and strong mango flavor. The mixture is cooked and stirred until thick, with nuts and cardamom added for taste. It is a sweet, energy-rich delicacy and a popular way to use ripe mangoes in a traditional dessert.
- b) **Mango noodles:** Mango noodles are a unique value-added product made by incorporating mango pulp or powder into a wheat- or millet-based dough. They offer a mild tangy-sweet flavor, a natural yellow color, and enhanced nutritional value due to the presence of vitamins, fiber, and antioxidants from mango. Typically processed through mixing, kneading, extrusion, steaming, and drying, these noodles are designed as a clean-label, kid-friendly alternative to conventional noodles, combining taste with improved health appeal.



E. BAKED PRODUCTS

Mango cookies are prepared by first creaming butter, sugar and vanilla until smooth. An egg and milk are then added and mixed, followed by incorporating flour and baking powder to form a thick dough. Fresh mango chunks are gently folded in and the dough is chilled for 30–60 min. Portions are scooped onto lined trays and baked at 180°C for 14–16 min. After cooling, the cookies are finished with a light glaze made from powdered sugar, milk and vanilla.

Mango cheesecake is prepared by combining graham cracker crumbs, brown sugar and melted butter, pressing the mixture into a pan and baking to form the base. The filling is made by blending cream cheese, sugar, eggs, cream, sour cream (or yogurt), vanilla, cornstarch, and salt, then folding in mango. The cheesecake is baked in a water bath and allowed to cool gradually in the oven for a smooth, creamy texture.



Mango muffins are prepared by preheating the oven to 200°C and lining a muffin tray. Fresh or thawed mango is pureed and dry ingredients such as flour, sugar, baking powder and salt are mixed separately. In another bowl, mango puree is combined with eggs, melted butter and vanilla, then gently folded into the dry mixture. The batter is poured into molds up to three-quarters full and baked for about 15 minutes until done. After brief

cooling, the muffins are transferred to a rack.

Mango-incorporated bread is prepared by mixing flour with strained mango puree followed by baking.

Mango doughnuts are prepared by combining mango pulp with flour (wheat, maize, or rice) and eggs. The dough is kneaded for 10 min, shaped and cooked by frying

F. MANGO FLAVOUR NUTRACEUTICALS

- a) **Mango sticky rice protein powder:** Mango sticky rice protein powder is a premium supplement delivering 25 g of protein per serving with only 125 calories and 5 g of carbohydrates. It features a mango sticky rice inspired flavor, combining ripe mango with creamy coconut and is suitable for both intra and post-workout consumption.



- b) **Mango protein powder:** Mango protein powder is a mango-flavored, plant-based supplement designed to support muscle growth, energy and overall fitness. It typically blends proteins such as pea and rice to provide a complete amino acid profile for effective muscle repair and development.



- c) **Mango flavoured kids multivitamin gummies:** Mango flavoured kids multivitamin gummies are supplements formulated to support overall growth and development. It includes a blend of essential nutrients such as curcumin, DHA, vitamins A, B-complex (B3, B6, B7, B9, B12), C, D3, E and zinc to help meet daily nutritional requirements.



- d) **A To ZNS Mango Flavour Syrup:** A To ZNS Mango Flavour Syrup supplies essential multivitamins and minerals to support overall health, including astaxanthin, B-complex vitamins, vitamins A and C, biotin, and trace elements. It helps address nutrient deficiencies, strengthen immunity, improve energy levels and protect cells from oxidative damage caused by factors such as pollution and UV exposure.



- e) **Mango flavoured vitamin C dietary supplement:** Mango flavoured vitamin C supplement is enriched with zinc and natural antioxidants to promote overall health. It is available as chewable tablets, offering a convenient and enjoyable way to consume.



Reference

- i. Adilah Z.M., Jamilah B., Hanani Z.N. Functional and antioxidant properties of protein-based films incorporated with mango kernel extract for active packaging. *Food Hydrocolloids*. 2018;74:207–218.
- ii. Ayustaningwarno, F., van Ginkel, E., Vitorino, J., Dekker, M., Fogliano, V., & Verkerk, R. (2020). Nutritional and physicochemical quality of vacuum-fried mango chips is affected by ripening stage, frying temperature, and time. *Frontiers in Nutrition*, 7, 95.
- iii. Dars, A. G., Hu, K., Liu, Q., Abbas, A., Xie, B., & Sun, Z. (2019). Effect of thermo-sonication and ultra-high pressure on the quality and phenolic profile of mango juice. *Foods*, 8(8), 298.
- iv. Deeksha, K., & Sunita, M. (2020). Utilization of mango and its by-products by different processing methods. *Asian J. Sci. Technol*, 9, 8896–8910. Green mango products
- v. Fang, G., Liu, B., Guo, J., Goksen, G., Tosif, M. M., Jafarzade, S., ... & Pandi, A. (2025). Valorization of mango byproducts for sustainable active packaging: Advances in functionalized biopolymer films. *Food Chemistry*: X, 102961.
- vi. Kaur G., Kaur D., Kansal S.K., Garg M., Krishania M. Potential cocoa butter substitute derived from mango seed kernel. *Food Chemistry*. 2022;372 doi: 10.1016/j.foodchem.2021.131244.
- vii. Krishna, B., Banik, A. K., & Das, S. (2020). Assessment of quality attributes and storage behavior of jam prepared from different mango varieties grown at Malda district. *International Journal of Chemical Studies*, 8(2), 443–448.
- viii. Kumar S., Seluriyal P., Sharma S., Kumar V., Bisht B., Joshi S., Saikumar A. Functional and nutritional perspectives of low-fat cookies fortified with jamun pulp, jamun seed, mango kernel powder. *Applied Food Research*. 2023;3(2)
- ix. Mawilai, P., Chaloeichitratham, N., & Pornchaloempong, P. (2019, August). Processing feasibility and qualities of freeze dried mango powder for SME scale. In *IOP Conference Series: Earth and Environmental Science* (Vol. 301, No. 1, p. 012059). IOP Publishing.
- x. Nupur, A., Hossain, M. A. M., Uddin, M., & Sultana, R. (2020). Effect of potassium meta-bisulphite on quality and acceptability of formulated green mango pulp during freezing. *Journal Agricultural University, of Bangladesh* 18, 734–741. <https://doi.org/10.5455/JBAU.105300>
- xi. Ogodo, A. C., Ugbogu, O. C., Agwaranze, D. I., & Ezeonu, N. G. (2018). Production and evaluation of fruit wine from *Mangifera indica* (cv. Peter). *Applied Microbiology*, 4, 144.
- xii. Patel, V., Tripathi, A. D., Adhikari, K. S., & Srivastava, A. (2021). Screening of physicochemical and functional attributes of fermented beverage (wine) produced from local mango (*Mangifera indica*) varieties of Uttar Pradesh using novel *saccharomyces* strain. *Journal of Food Science and Technology*, 58, 2206–2215.
- xiii. Peres, A. P., Santos, E. B., Morzelle, M. C., Siqueira, P. B., Cunha, J. S., Cerzosimo, A. M. A., & Kapitango-a-Samba, W. K. K. (2024). Sensory acceptance of mango ice cream added with whey protein by consumers affected and not affected by cancer. *Brazilian Technology, Journal* 27, of Food e2024008. <https://doi.org/10.1590/1981-6723.00824>
- xiv. Priyadarshini, A., Rayaguru, K., Routray, W., Biswal, A. K., & Misra, P. K. (2023). Ascertaining optimal ohmic-heating characteristics for preserving mango (*Mangifera indica* L.) pulp through analysis of physicochemical properties and hurdles effect. *Food Chemistry Advances*, 3, 100458.
- xv. Ramirez-Maganda J., Blancas-Benitez F.J., Zamora-Gasga V.M., García-Magaña M.D.L., Bello-Perez L.A., Tovar J., Sayago-Ayerdi S.G. Nutritional properties and phenolic content of a bakery product substituted with a mango (*Mangifera indica*) 'Ataulfo' processing by-product. *Food Research International*. 2015;73:117–123.
- xvi. Ministry of food processing industries report: assessment studies for identifying gaps in infrastructure and processing facilities for development for potential value chains for perishable products under operation green scheme by Ministry of food processing industries.
- xvii. Soong YY, Barlow PJ and Perera CO (2004). A cocktail of phytonutrients: identification of polyphenols, phytosterols and tocopherols from mango (*Mangifera indica* L.) seed kernel. In IFT annual meeting, July 12–16, Las Vegas.



Growing With India: The Making of Parle Agro's Fruit Beverage Leadership

Indian Food Packer Special Edition: Indian Mango

When Parle Agro launched Frooti in 1985, packaged mango drinks did not exist in India. What followed was not just the birth of a brand, but the creation of an entirely new category, one that the company has led, for four decades. That origin story reflects something fundamental about how Parle Agro operates: it does not enter markets, it builds them.

Today, the company stands among India's leading beverage players, with a diversified portfolio spanning fruit drinks, sparkling beverages, flavoured milk and packaged water through brands including Frooti, Appy Fizz, B Fizz, Smoodh, Bailey and Bombay 99. With a strong workforce and a distribution network that reaches every tier of the Indian market, the company that began with a single mango drink now serves multiple beverage occasions across every consumer's day.

Frooti remains the clearest expression of what Parle Agro does best. Four decades after its launch, it continues to be one of India's largest-selling mango drink; a position that reflects continuous reinvention across packaging, communication and distribution. The mango, after all, is not simply a flavour choice; for Parle Agro, it is a strategic commitment. India is the world's largest mango producer, and the company has built its manufacturing infrastructure, processing, and multi-format packaging for maximising of this advantage.

Rooted Where It Matters

A company whose growth is closely linked to India's agricultural ecosystem has reason to re-invest in it and Parle Agro, through the Prakash Jayantilal Chauhan Foundation (PJCF), does exactly that.

Established in 2019, PJCF works across mango-growing regions including Maharashtra, Andhra Pradesh, Uttar Pradesh, Gujarat and Karnataka. Its focus lies on addressing two critical gaps in rural community infrastructure: early childhood development and youth employability.

The Anganwadi programme addresses the first. These government-run centres form the backbone of early childhood education and nutrition for children aged 0–6 in rural India. However, many continue to operate with limited infrastructure. PJCF collaborates with district and state authorities to identify high-need locations, assess each site individually, and rebuild them to meaningful, functional standards. To date, 120 Anganwadis have been rebuilt across mango-growing regions, positively impacting over 6,000 children—transforming deteriorating spaces into safe, child-friendly learning environments.

The skilling programme addresses the second. Through PJCF's Skill Development Centre, youth from underserved communities receive training in sectors with clear employment demand, including beauty and wellness, IT, and management. This is complemented by structured modules in financial literacy, professional conduct, and workplace readiness. Among the 1,600+ students who have completed the programme, a significant majority have secured employment, often representing a first step into the formal workforce.

What Parle Agro has built over four decades is, at its core, a business deeply connected to the Indian landscape—its fruits, its farms, and its people. Its impact can be seen not only in category creation and market leadership, but also in the communities it continues to grow alongside.



MANGO BOLE TOH FROOTI



READERS WINDOW TO THE WORLD OF MANGO*

Mango – GLOBAL Horticultural Profile - at a glance

Major Mango, Mangosteen, Guava Producing Countries in the World

Area in Million Ha,

Production in Million Tonne,

Yield in Tonne/ Ha

SLNo.	Country	2019			2020			2021			2022			2023		
		Area	Production	Yield	Area	Production	Yield	Area	Production	Yield	Area	Production	Yield	Area	Production	Yield
1	India	2.57	25.63	9.97	2.59	24.68	9.54	2.63	24.97	9.51	2.70	26.30	9.73	2.70	26.24	9.72
2	Indonesia	0.25	3.29	13.38	0.27	3.62	13.56	0.27	3.56	13.42	0.30	4.13	13.58	0.30	4.10	13.74
3	China	0.36	3.45	9.70	0.38	3.97	10.56	0.39	3.96	10.26	0.40	3.91	9.85	0.41	3.99	9.86
4	Mexico	0.22	2.40	11.10	0.21	2.37	11.11	0.22	2.44	11.32	0.22	2.50	11.23	0.24	2.71	11.45
5	Pakistan	0.21	2.27	10.57	0.21	2.34	10.94	0.21	2.68	12.54	0.21	2.78	13.03	0.21	2.59	12.15
6	Brazil	0.09	2.00	22.40	0.10	2.11	21.89	0.10	2.06	20.96	0.10	2.19	21.79	0.10	2.34	22.74
7	Malawi	0.05	1.67	32.53	0.06	1.70	30.19	0.05	1.73	32.21	0.06	2.02	33.84	0.06	2.13	35.47
8	Egypt	0.13	1.40	11.00	0.15	1.54	10.60	0.15	1.12	7.43	0.15	1.65	10.62	0.16	1.72	10.59
9	Thailand	0.21	1.64	7.80	0.14	1.32	9.27	0.13	1.64	12.44	0.13	1.65	12.52	0.13	1.62	12.29
10	Bangladesh	0.13	1.46	10.84	0.12	1.45	11.82	0.14	1.46	10.43	0.14	1.45	10.44	0.14	1.51	10.60
11	Others	1.56	11.36	7.29	1.64	11.37	6.95	1.64	11.85	7.21	1.65	11.93	7.22	1.71	12.16	7.11
World		5.78	56.58	9.79	5.86	56.48	9.65	5.92	57.47	9.70	6.08	60.50	9.95	6.16	61.11	9.92

Source: FAOSTAT (<https://www.fao.org/faostat/en/#data/QCL>)

Mango crop calendar of major producing countries

Countries	Jan	Feb	March	April	May	June	July	August	September	October	November	December
India												
Indonesia												
China												
Pakistan												
Mexico												
Brazil												
Peru												
Egypt												
Thailand												

Lean season Peak season

Source: CRISIL APEDA; Monthly dashboard –Mango; Nov-2025

Crop/ Production data (Global) : MANGO

Country	2024		2023		2022		2021		2020	
	Production	Share (%)	Production	Share (%)	Production	Share (%)	Production	Share (%)	Production	Share (%)
India	27,766.26	44.52	26,236.00	42.84	26,299.00	44.46	24,968.00	43.79	24,748.00	45.13
China	4,242.14	6.8	3,994.81	6.52	3,800.00	6.42	3,790.00	6.65	2,368.18	4.32
Indonesia	4,136.62	6.63	4,104.45	6.7	4,125.24	6.97	3,561.87	6.25	3,617.27	6.6
Mexico	2,650.03	4.25	2,706.11	4.42	2,485.55	4.2	2,441.50	4.28	2,373.11	4.33
Brazil	2,395.93	3.84	2,340.95	3.82	2,111.14	3.57	2,057.77	3.61	2,135.30	3.89
Pakistan	2,146.17	3.44	2,594.33	4.24	2,782.81	4.7	2,677.02	4.7	2,344.65	4.28
Bangladesh	2,100.64	3.37	1,508.55	2.46	1,452.30	2.46	1,458.55	2.56	1,448.40	2.64
Egypt	1,863.40	2.99	1,715.37	2.8	890.34	1.51	1,327.87	2.33	1,395.24	2.54
Vietnam	1,552.91	2.49	1,498.21	2.45	1,443.57	2.44	1,439.27	2.52	1,224.58	2.23
Thailand	1,486.72	2.38	1,618.79	2.64	1,398.90	2.36	1,635.23	2.87	1,657.59	3.02
Total Production	50,340.82		48,317.57		46,788.85		45,357.08		43,312.32	

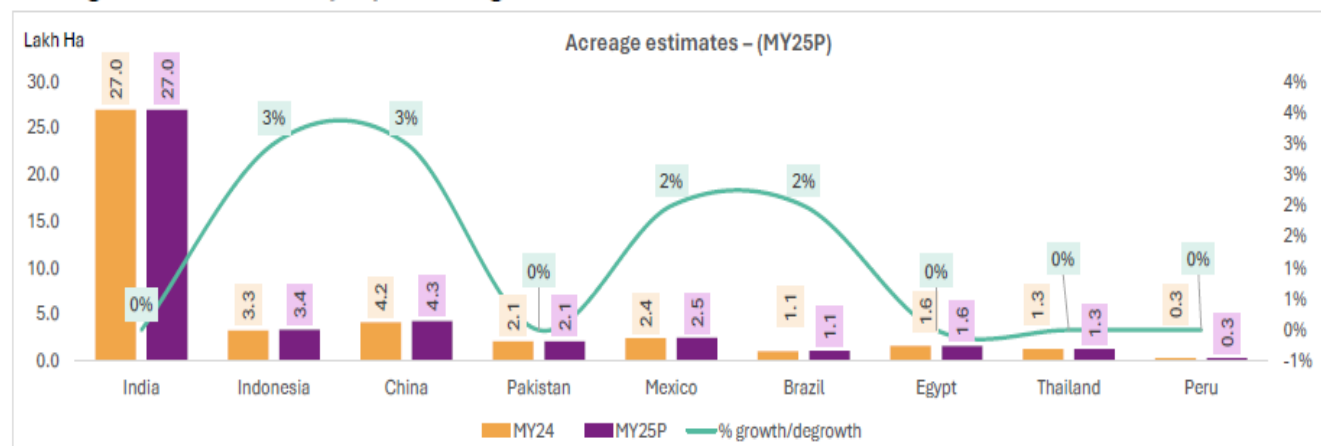
Source: Food & Agricultural Organisation (FAO)

*Advice & disclaimer: Reader's discretion may be required on the pictorial/ graphical/ tabulated info depicted in this section which is intended only for general information. Reader may do a thorough check for detailed & authentic information on respective source site/ data source for further / detailed information.

READERS WINDOW TO THE WORLD OF MANGO*

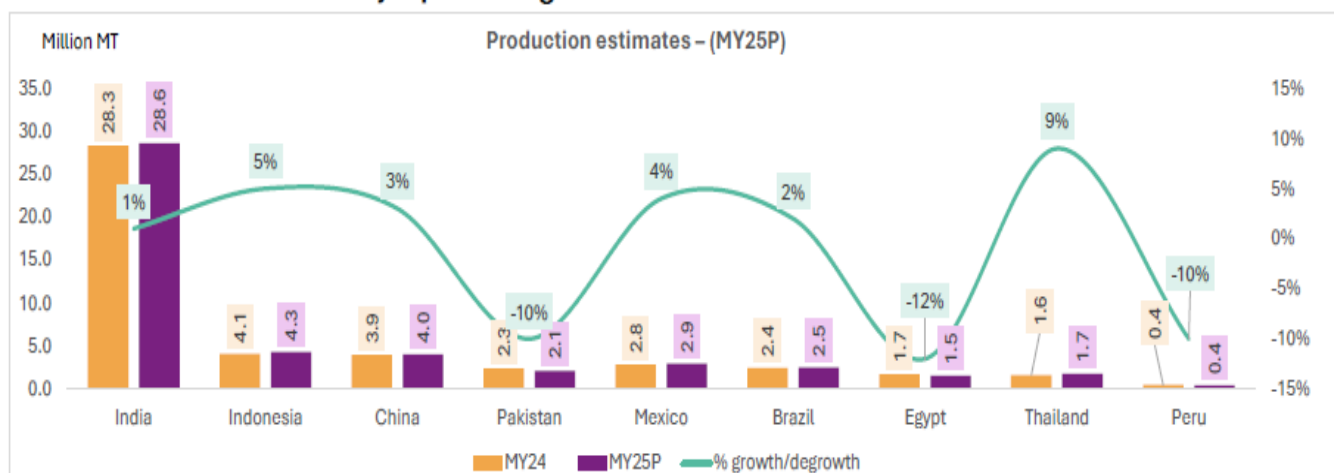
Mango – GLOBAL Horticultural Profile - at a glance

Acreage estimates of major producing countries

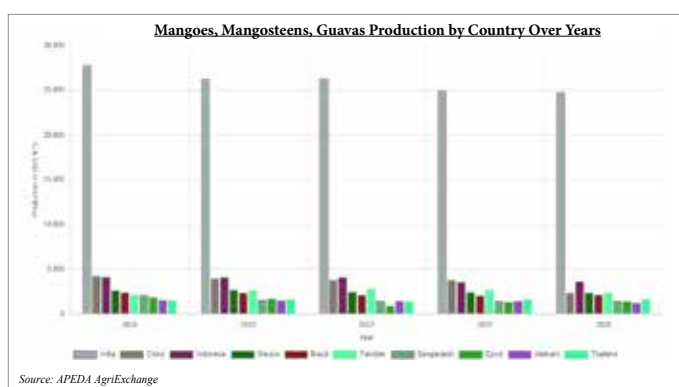
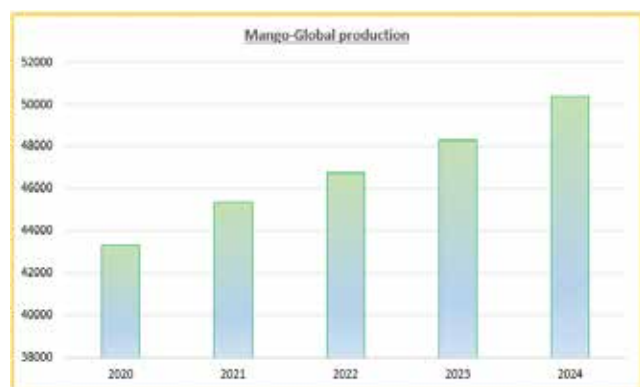


Source: CRISIL APEDA; Monthly dashboard –Mango; Nov-2025

Production estimates of major producing countries



Source: CRISIL APEDA; Monthly dashboard –Mango; Nov-2025

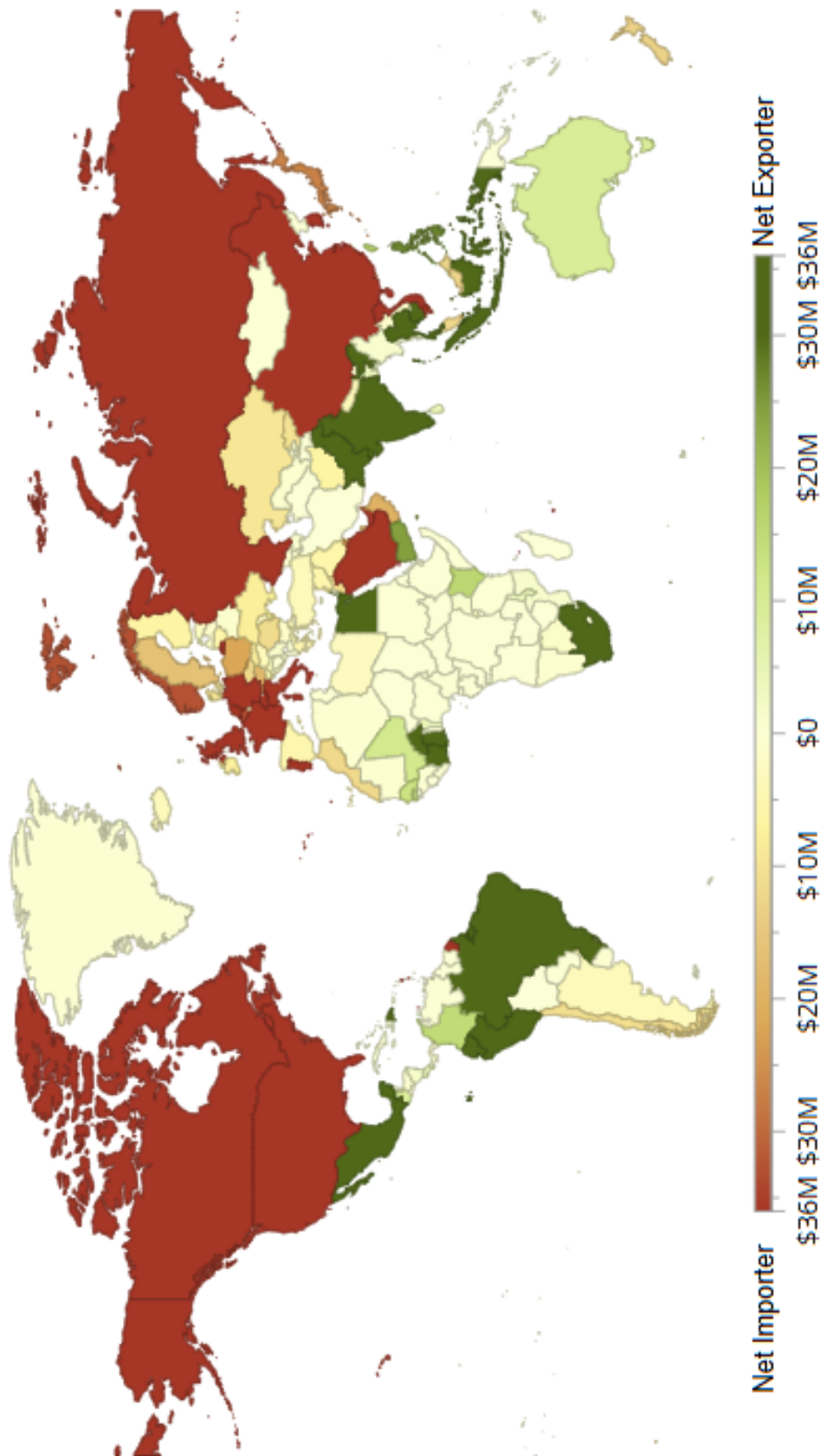


Source: CRISIL APEDA; Monthly dashboard –Mango; Nov-2025

Mango – GLOBAL Trade Profile - at a glance

Global Trade Balance of Guavas, Mangoes & Mangosteens (Fresh/Dried)

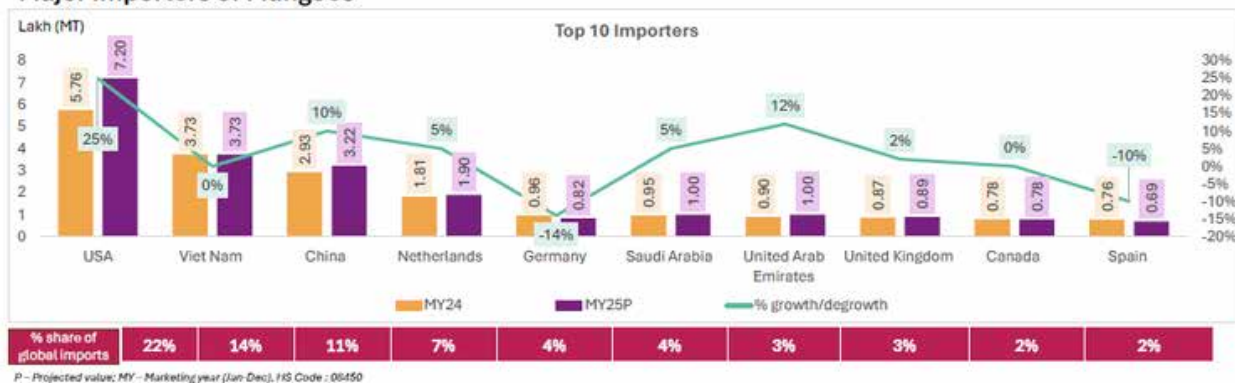
(2024)



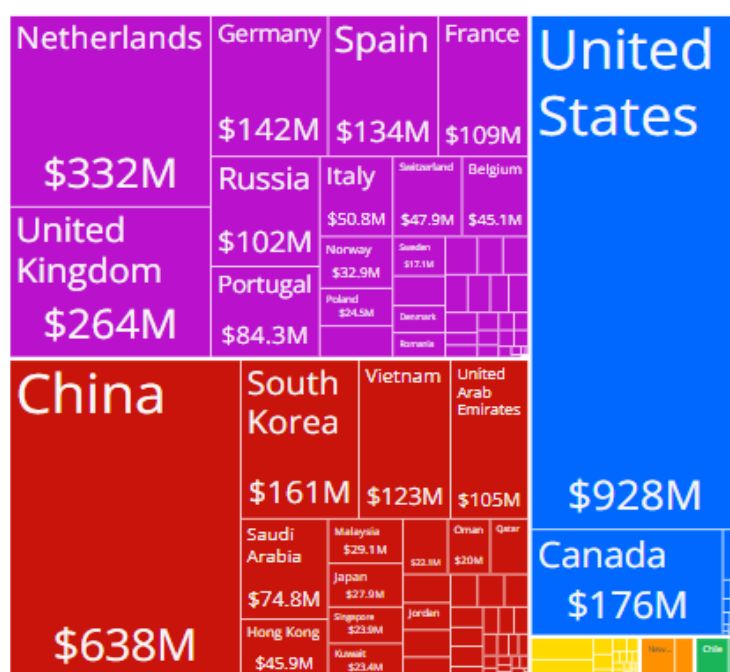
Source: OECD-FAO Agriculture Output 2019-2028

READERS WINDOW TO THE WORLD OF MANGO*

Major importers of Mangoes



Importers of Guavas, mangoes & Mangosteens (Fresh/Dried) in 2024



Source: OEC website

Import Tariffs : Other Fresh Fruits, Processed Fruits, Juices & Nuts

HS Code: 80450 | Exporting Country: all WTO Members (incl. India)

Product Description: Guavas, mangoes & mangosteens, fresh/dried

Sl. No.	Importing Country	Import Duty (%)	Non-AV Duties	FTA
1	U S A	0	[6.6 cents/kg] [6.6 cents/kg] [1.5 cents/kg]	MFN
2	United Arab Emirates	0	N/A	MFN
3	Saudi Arab	0	N/A	MFN
4	Netherland	0	N/A	MFN
5	U K	0	N/A	MFN
6	France	0	N/A	MFN
7	Japan	0	N/A	Free-trade agreement duty rate for India
8	Russia	4	N/A	MFN
9	Canada	0	N/A	MFN
10	Germany	0	N/A	MFN
11	Kuwait	0	N/A	MFN
12	Yemen Arab Republic	25	N/A	MFN
13	Qatar	0	N/A	MFN
14	Oman	0	N/A	MFN
15	China P Republic	15	N/A	MFN

Source: APEDA - Agri Exchange; 15-Apr-26;

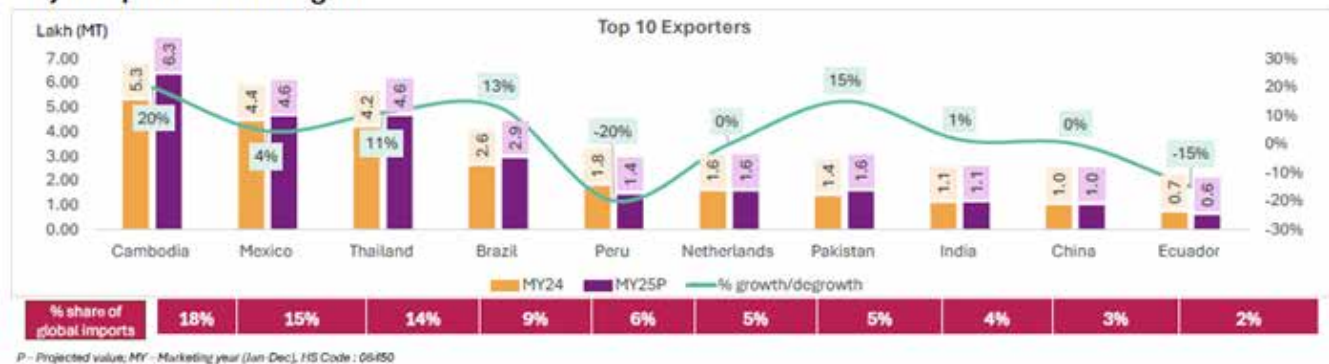
<https://agriexchange.apeda.gov.in/Reports/Marketreport/ImportTariffs>

Price trends of key importing nations

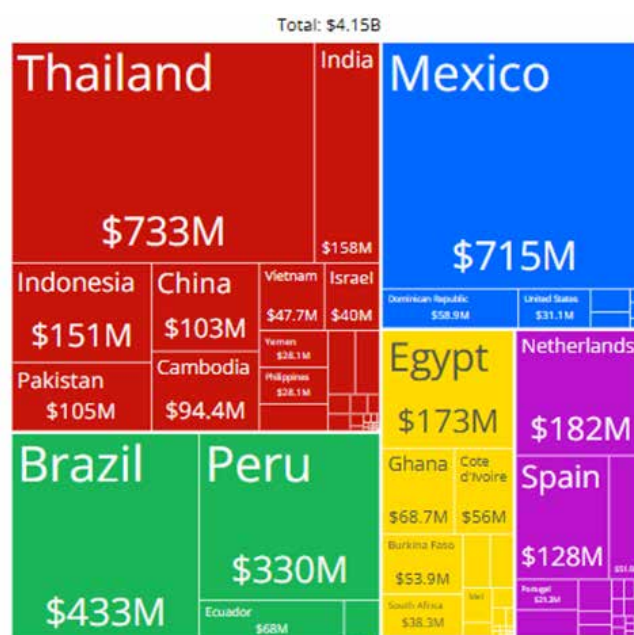


READERS WINDOW TO THE WORLD OF MANGO®

Major exporters of Mangoes

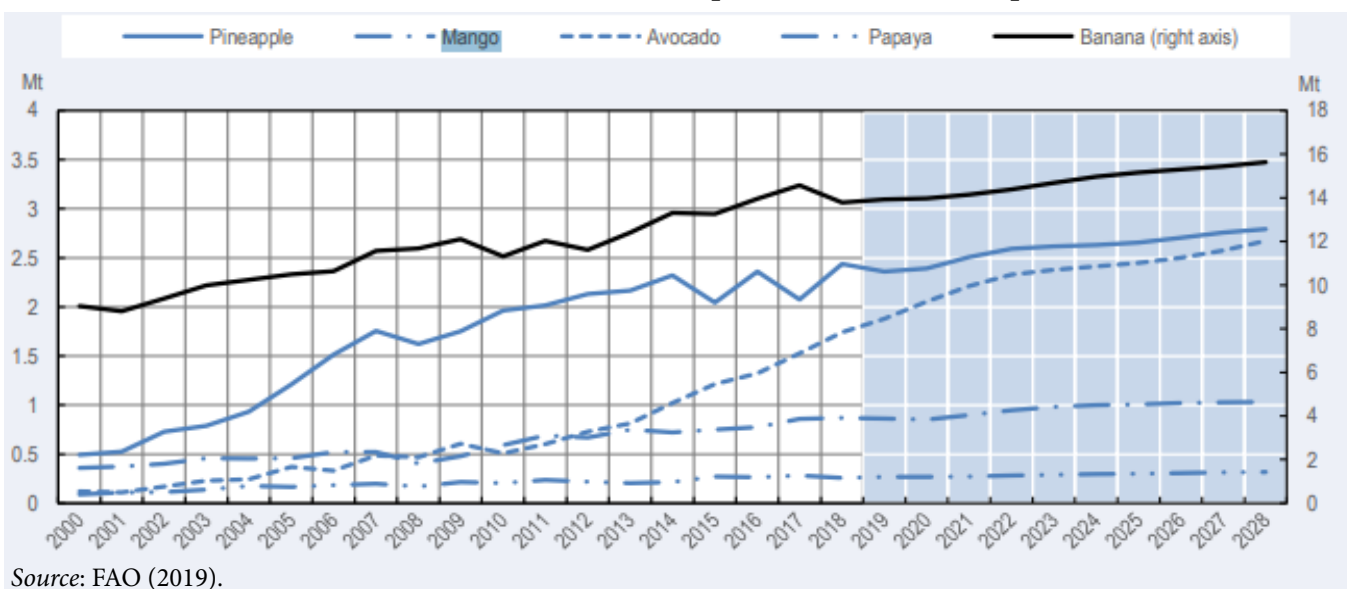


Exporters of Guavas, Mangoes & Mangosteens (Fresh / Dried) in 2024

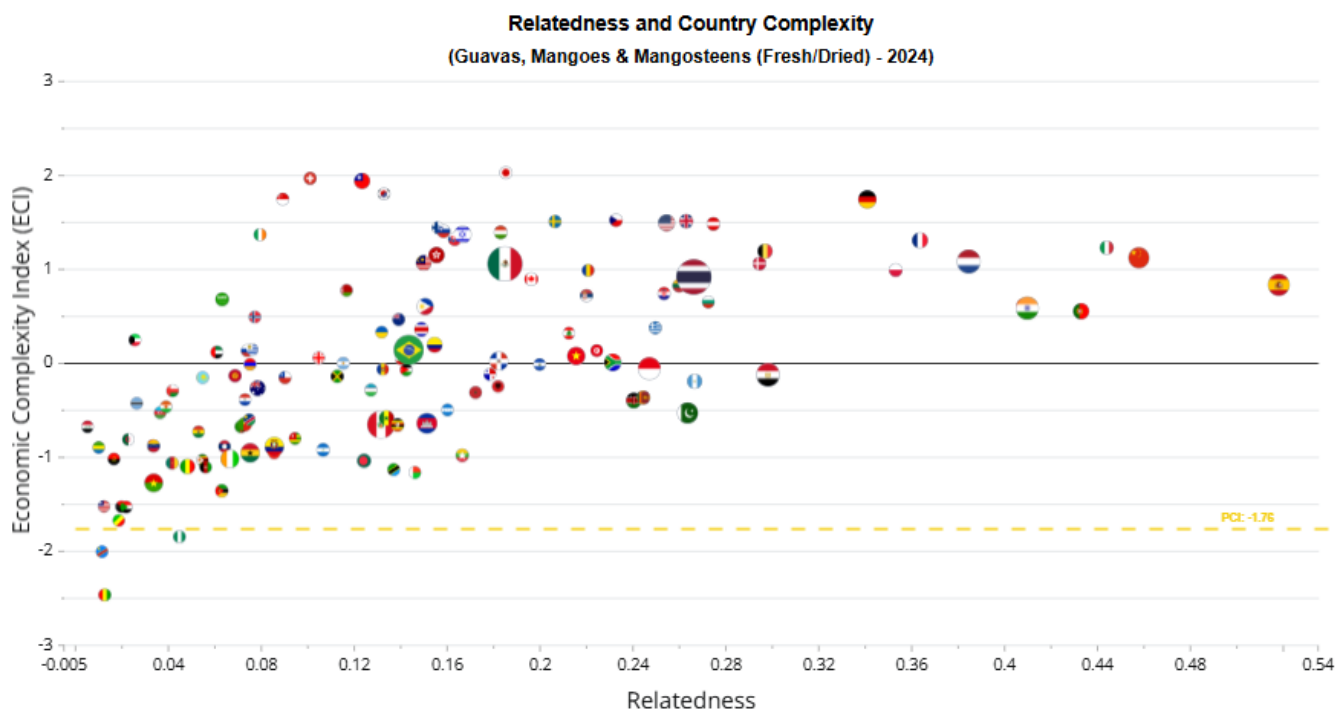


Source: OEC website

Latin America and the Caribbean net exports of bananas and tropical fruit



READERS WINDOW TO THE WORLD OF MANGO*



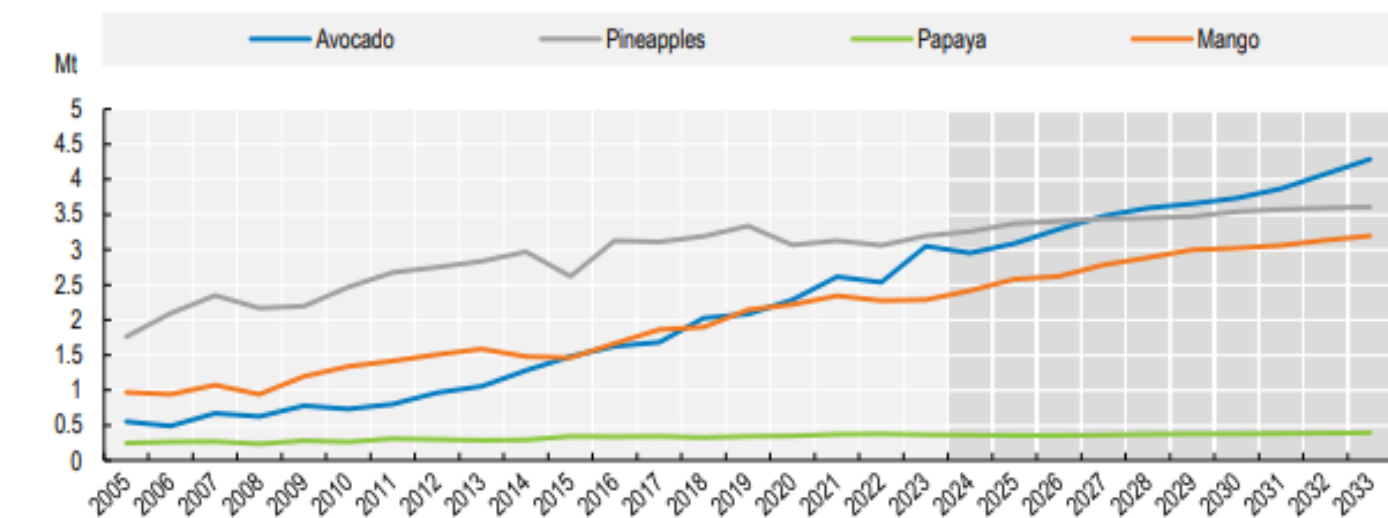
Global Economic Complexity*: The Complexity-Relatedness diagram compares the risk and strategic value of potential export opportunities for a product. Relatedness measures the likelihood that an economy will expand its exports to a specific product, while complexity is associated with higher income levels, economic growth, reduced inequality, and lower emissions. In the chart, a rightward movement reflects a higher probability of exporting a new product, and an upward movement indicates an increase in the complexity of the territory's productive matrix.

*In the chart, the size of the circles is proportional to the volume of exports of the product in each country.

*Relative relatedness captures variation within each country, while absolute relatedness compares across countries.

Source*: The Observatory of Economic Complexity (OEC)

Global exports of the four major tropical fruits



Source: FAO data.

READERS WINDOW TO THE WORLD OF MANGO*

India Export Statistics -All Agri HSCode (8 Digit)

Country: All

Product: 08045022(Banganapalli), 08045021(Alphonso (Hapus)), 08045029(Other Mangoes), 08045027(Totapuri), 08045023(Chausa), 08045025(Langda), 08045026(Kesar), 08045024(Dasheri), 08045030(Mangoes, Sliced Dried), 08045040(Mango Pulp), 20079910(Jams Jellies Mrrmlds Etc. Of Mangoes), 20089911(Mango Squash), 20098910(Mango Juice)

Sl. No.	Value In USD Million				Qty In MT		
	Country	2022-23	2023-24	2024-25	2022-23	2023-24	2024-25
	Total from India	418.14	299.03	262.64	280,428.97	219,347.52	182,512.91
1	U S A	33.04	33.72	39.38	15,425.31	15,063.26	18,043.82
2	U Arab Emts	52.63	44.29	31.58	32,256.28	29,993.98	22,025.72
3	Saudi Arab	55.83	28.98	31.43	44,756.64	31,345.09	32,106.98
4	Netherland	69.18	40.26	29.56	46,569.92	28,664.66	19,908.00
5	U K	27.46	23.85	20.53	14,857.05	13,462.62	11,503.95
6	France	16.78	9.88	13.22	8,366.40	5,140.40	6,205.45
7	Japan	9.94	9.68	8.97	4,585.94	4,393.68	4,112.92
8	Russia	21.93	9	8.33	11,258.67	6,771.00	6,632.09
9	Canada	7.32	9.08	8.12	4,547.89	5,129.39	4,428.07
10	Germany	8.67	8.17	7.75	5,521.92	5,120.95	4,634.06
11	Kuwait	13.21	7.44	6.06	10,729.22	5,657.60	4,552.14
12	Yemen Republc	7.5	6.79	4.67	7,978.93	8,287.02	5,032.77
13	Qatar	6.96	4.25	4.37	4,438.68	3,264.88	3,091.36
14	Oman	6.95	6.19	4.19	5,251.17	5,734.12	3,250.82
15	China P Rp	8.08	8.29	4.03	5,607.69	5,826.71	2,954.29
% of India's export of this Product		82.6	83.6	84.6	79.2	79.3	81.4

Source: DGCIS ; APEDA- Agri Exchange; 15-Apr-26;

India Export Statistics -All Agri HSCode (8 Digit)						
Region: All						
Product: 08045022(Banganapalli), 08045021(Alphonso (Hapus)), 08045029(Other Mangoes), 08045027(Totapuri), 08045023(Chausa), 08045025(Langda), 08045026(Kesar), 08045024(Dasheri), 08045030(Mangoes, Sliced Dried), 08045040(Mango Pulp), 20079910(Jams Jellies Mrrmlds Etc. Of Mangoes), 20089911(Mango Squash), 20098910(Mango Juice)						
Country	Value In USD Million			Qty In MT		
	2022-23	2023-24	2024-25	2022-23	2023-24	2024-25
Total	418.24	299.05	262.59	280,428.92	219,347.49	182,512.84
West Asia- Gcc	136.91	92.44	78.84	98,318.08	76,980.73	65,809.48
Eu Countries	113.92	72.62	63.46	74,169.03	49,883.56	40,121.37
North America	40.74	42.8	47.52	20,020.19	20,192.65	22,496.04
Other European Countries	28.05	24.39	20.77	15,198.86	13,744.84	11,680.22
Ne Asia	23.03	21.92	15.84	13,211.12	12,862.36	8,874.80
Other Cis Countries	22.82	9.49	8.73	11,994.96	7,196.10	6,948.50
Other West Asia	12.95	10.97	7.49	11,868.76	11,678.00	7,484.23
East Asia (Oceania)	6.01	5.6	5.33	3,501.81	3,693.45	2,649.72
North Africa	7.04	4.51	3.95	4,988.35	5,964.00	3,987.94
South Asia	10.85	6.44	3.92	15,776.32	10,936.73	6,855.15
Asean	6.5	4.05	2.82	4,650.69	2,939.08	1,925.43
Other South African Countries	4.38	0.45	1.12	2,743.09	658.48	1,360.92
East Africa	1.67	1.08	1.05	1,099.18	945.9	956.01
European Free Trade Associatipn (Efta)	0.78	0.97	0.66	741.17	661.47	291.08
West Africa	0.96	0.25	0.57	828.17	264.57	685.43
Southern African Customs Union (Sacu)	0.31	0.11	0.23	329.87	48.22	196.94
Latin America	0.6	0.5	0.19	506.31	311.31	113.05
Unspecified	0.42	0.32	0.08	292.8	246.4	69.2
Central Africa	0.3	0.12	0.02	190.06	121.84	6.46
Cars Countries	0	0.02	0	0.1	17.8	0.87

Source: DGCIS

MRLs & Pesticide Management in the Mango Sector: A Supply Chain Imperative



Aleen Mukherjee

Director- Farm Sector

ACCESS Development Services; 22, Hauz Khas Village, Ground Floor, New Delhi - 16

The King of Fruit – The Mango, has no shortage of scale, diversity, or reputation in India. We are home to about 1,000 delicious mango varieties and many more local varieties, though only a limited number are commercially cultivated and exported. APEDA's current mango profile confirms that the major mango-growing states are Uttar Pradesh, Andhra Pradesh, Bihar, Karnataka, Gujarat, Madhya Pradesh, and Tamil Nadu, with Uttar Pradesh ranking first in production with a 26.75% share in 2023–24.¹

But scale alone is not the issue anymore. The real challenge is whether India can consistently translate that scale into market confidence, particularly in export markets. Including my work with international agencies where I was closely involved in building mango value chain linkages with farmers, exporters, and modern retail, one lesson has remained clear, mango quality is never created at the end of the chain.. It is created in the orchard, protected at the farm gate, and either strengthened or weakened by the discipline of the supply system.

That is why MRLs and pesticide management must be seen as a value-chain issue, not merely a compliance issue. India's mango production is projected by APEDA's latest dashboard at around 28.6 million tonnes in Marketing Year'25 (Provisional) up from 28.3 million tonnes in Marketing Year'24 (Estimated)², which shows the sheer size of the crop and the importance of managing it in a more integrated manner. The same dashboard notes that India's production is expected to improve because of productivity gains, especially in southern states such as Karnataka and Andhra Pradesh.

On the trade side, the gap between production and export remains striking. APEDA's current mango page reports that India exported 29,938.40 MT of fresh mangoes worth USD 56.50 million in FY25, with the main destinations being the UAE, USA, UK, Kuwait, and Qatar. That tells us two things at once: first, India has a real export presence; second, the export basket is still tiny compared with the country's overall production base. In practical terms, the domestic market continues to absorb the overwhelming bulk of mango output, while this ensures strong domestic availability, it also raises an important question, why cannot we export more and what needs to be done to bring in quality discipline beginning from farm/crop management till the produce gets loaded for export.

For processors and trade bodies such as AIFPA, this point is especially relevant. Processing is not a downstream repair mechanism for poor orchard management. It is a serious business segment in its own right, and it too depends on predictable raw material. APEDA's mango pulp page shows that India exported 63,253.19 MT of mango pulp worth USD 80.34 million in FY25³, with major destinations including Saudi Arabia, the USA, the UK, Germany, and Canada. That is a strong reminder that the mango economy is not only about fresh fruit. It is also about pulp, value addition, and the ability of the supply chain to convert fruit into reliable industrial and consumer products. The varietal side of the mango business is changing too. APEDA's dashboard identifies key globally traded varieties from India as Alphonso, Kesar, Dasher, and Langra and for pulp, Totapuri and Banganapalli.

The Sept'25 report of monthly dashboard for Mango published by APEDA⁴ makes two important statements – *"Indian exports are facing rising quality concerns for alphonso variety in key importing countries like middle east countries which is expected to impact the demand."*

"India's mango export realization for pulp has declined at a CAGR of -6% in the past five years due to lower product quality."

These statements signal that the problem is structural in nature and not one of case; not necessarily represent market fluctuations but reflect inconsistencies in production practices, weak adherence to residue protocols, and gaps in post-harvest handling. Our approach to orchard management is still largely reactive. We rely heavily on unplanned spraying, informal advice, and input decisions that are not always aligned with market requirements. That approach may reduce visible pressure in the short term, but it creates risks, residue rejection risks, and reputational damage in the longer run. In a market where buyers are increasingly asking for traceability, assurance, and proof of disciplined production, we need to align more closely with what the market demands and shift towards a more proactive approach to orchard and production management. APEDA to that extent is playing its part itself and created Hortinet⁵, a traceability-compliant product identification system linked to a Residue Monitoring Plan for consumer safety and recall

¹ Mango | APEDA

² https://apeda.gov.in/sites/default/files/2025-12/MIC_Monthly_dashboard_Mango_15122025.pdf

³ <https://apeda.gov.in/MangoPulp>

⁴ https://apeda.gov.in/sites/default/files/2025-10/MIC_Monthly_dashboard_Mango_30102025.pdf

readiness, which shows how central this issue has become to the sector.

The real answer is not more paperwork. It is better operating discipline. Standard Operating Procedures, or SOPs, must be treated as the backbone of production and aggregation. A good SOP should define spray windows, approved molecules, pre-harvest intervals, harvest maturity, picking methods, post-harvest management⁶ lot segregation, transport hygiene, and record keeping. SOPs are not administrative documents. They are the practical language through which quality is standardized across a fragmented production system. Without SOPs, every farmer and handler tend to improvise. With SOPs, quality becomes repeatable and far less dependent on individual interpretation.

Farm-gate infrastructure is equally important, and in many places, it is still the weakest link. Too much attention is given to packhouses, cold rooms, and export facilities, while the first point of aggregation remains under-served. Basic infrastructure at the farm gate, viz., shade, clean loading surfaces, field crates, primary sorting space, simple lot identification, hygienic handling, and orderly aggregation can reduce injury, contamination, and traceability loss far more effectively than many of us realize. If the fruit is damaged or mixed at the source, no downstream system can fully recover the value. For processors, this means more sorting loss, lower usable recovery, and less consistent input quality.

Policy has to catch up with this reality. At present, the sector often talks about quality in the language of export rejection and food safety alerts, but far less in the language of orchard management. These needs immediate attention. Mango-growing clusters can be managed as compliance clusters, not just production clusters. Extension support should be crop-specific and season-specific. Input dealers should be drawn into a more accountable advisory framework. FPOs should be empowered to aggregate not only produce but also practices. And processors should increasingly reward compliance at the sourcing stage rather than only penalize failure at the buying stage. That is how the market begins to shape behaviour in a

durable way.

Integrated Pest Management also needs to move from presentation decks into actual field practice. Orchard sanitation, canopy management, pest-scouting and monitoring, biological solutions, and disciplined spraying should be seen as business tools, not as idealistic concepts. Mango is a perennial crop, which means the cost of poor decisions is cumulative. A season of bad management can affect not only the current crop but also the reputation of the orchard and the confidence of buyers in the next season. This is why residue management should be tied to orchard health, not treated as a post-harvest correction exercise and true for all orchard and plantation crops.

For AIFPA and the wider processing community, the message is straightforward. Mango's future will not be determined only by how much we grow. It will be determined by how reliably we can produce, aggregate, trace, and process fruit that meets modern expectations. India already has the varietal depth, the production scale, and the market recognition. What it still needs is convergence: between farm practices and market standards, between policy and execution, and between processors and growers. When that happens, MRL compliance stops being a fear-driven requirement and becomes a business advantage. That is the direction the sector must move in. It is not just about more mangoes. It is about trusted mangoes, responsibly produced and consistently delivered not only to the export market but equally for the domestic consumer!!

At ACCESS, our farm sector initiatives have consistently focused on bridging the gap between policy frameworks, technical knowledge, and field-level adoption. Through our work with farmer collectives and individual farmers and relevant stakeholders in a collaborative approach, we focus on capacity building, and development of market-linked models, the effort has been to translate complex advisories into practical, implementable solutions. In my experience, this last-mile translation is critical because across crops and geographies, the real challenge is not availability of knowledge, but its consistent and disciplined application on the ground.

⁵<https://apeda.gov.in/hortinet-static>

⁶https://www.fao.org/fileadmin/user_upload/inpho/docs/Post_Harvest_Compendium_-_Mango.pdf



Make Your Brand The Next Big Talk !
Promote . Connect . Grow

To book this space
Contact :
Mr. Vishal Sarin
Mobile : +91-7838012842 Email : vishal@aifpa.net

MRLs & Pesticide Management in the Mango Sector

Dr. K. Rama Krishna

Assistant Professor, Department of Horticulture, School of Life Sciences,
Central University of Tamil Nadu, Thiruvaur, TN, India
E-mail:- ramakrishna.karri001@gmail.com



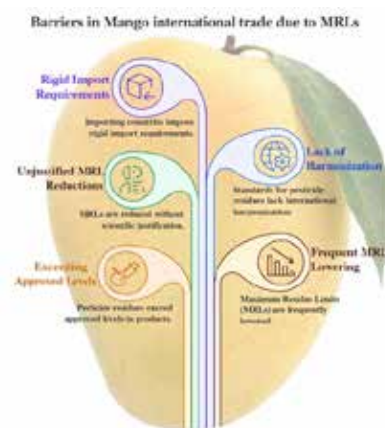
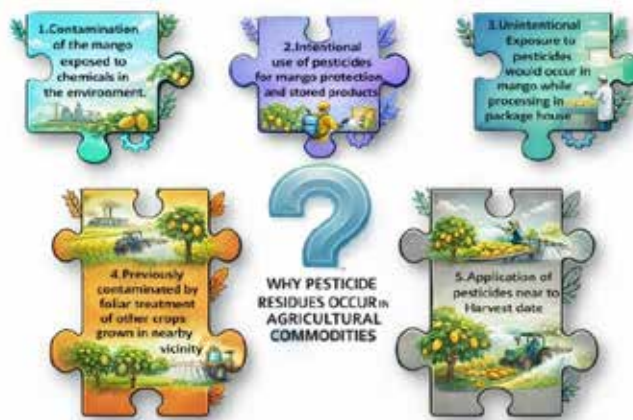
Introduction

The WTO agreements on Sanitary and Phytosanitary Measures (SPS) set out standard rules and regulations to safeguard the food and plant/animal health that are used in trade between countries. The SPS agreement allows the countries to set their own standards, which should be based on scientific data and protocols, not discriminatory between countries with similar conditions and should not be used as a tool for protectionism. Among the several SPS regulations led by the SPS wing of the WTO for the protection of animal and plant health, a particular type of SPS regulation, called Maximum Residue Limits (MRLs), is designed to safeguard human health.

Maximum Residue Limits (MRLs) are the maximum legal level of concentrations of pesticides or feed additives that an importing country will accept on the surface of food products during trade. As projected by the WTO, 60% of non-tariff barriers to trade (NTBs) are related to SPS; of these, 36% are due to MRLs. Nowadays, MRLs for pesticide residue interfere with agricultural trade and can become a serious non-tariff barrier in trade. MRLs for pesticides have been elevated to become the first action growers and exporters should consider in their pest management decisions.

Pesticides, an unavoidable evil of the present modern time, play a significant role in ensuring global food security by increasing crop yield, allowing farmers to grow more food on less land. Around the globe, one-third of the agricultural production depends on pesticides, without which the productivity would be severely hampered due to improper pest/disease control. This is because of the impact of climate change, a recurring issue of the 21st century, where the heat stress and unpredictable rainfall patterns make the crops more vulnerable to pests. Also, the drift in the high and low temperature and moisture levels makes the pest have a higher reproductive rate, shifting their geographical range, making crops susceptible to pest/disease. This has led to indiscriminate use of pesticide application on crops. The toxicity of pesticides depends on the type of function, doses applied, exposure to the chemical, the route by which the exposure occurs, etc. Although banning of several older, less costly pesticides such as dichlorodiphenyltrichloroethane (DDT), lindane, calcium carbide etc., their usage is reported in developing and underdeveloped countries. Adverse effects from these pesticides occur only above a certain safe level of exposure, i.e., large quantities and prolonged exposure may

result in acute poisoning or long-term health issues that may include cancer and loss of reproduction.



As the temperature rises across the equatorial line of the globe during summer, it brings a sweet delicacy 'Mango', the King of the Fruits, to the platter around the world. Mango is the choicest fruit due to its unique taste, flavor, and health

benefits. India is the leading producer of mango in the world, but not the leading exporter, falling behind countries like Mexico, Thailand, Peru, Netherlands and Brazil. Major importing countries from India are UAE, USA, UK, Kuwait, Japan, New Zealand, and Qatar. India was able to trade \$57.98 million worth of mangoes in year 2025 with an export fall of 6% (\$61.68 million, 2024) from the previous year. This impact was linked to mango perishability and logistical problems, including MRLs. However, the mango export from India is estimated to grow to \$77.94 million this year, representing 6.3% growth rate and is predicted to reach \$3.97 billion by 2030 with a CAGR of 6.5%, with the increase in domestic consumption and global demand.

This growth projected in mango might be possible when proper pest management measures are in place, where pesticide application becomes inevitable. The pesticide applied to the mango crop generally dissipated through



various chemical, physical and environmental factors of the plant. Each pesticide has a different Pre-Harvest Interval (PHI) to dissipate and it is also based on the stage of growth of the fruit when applied. Initially, the pesticides applied are removed through the growth dilution effect. As the mango fruit takes

considerable time to reach full maturity, the application of pesticides at the beginning of fruit growth limits the residue deposition on the fruit. While application of pesticides towards the maturity of the mango fruit increases the probability of leaving pesticide residues above MRL at harvest. It is not advisable to apply pesticides where the mango has reached its full maturity (Full size). As a thumb rule NO pesticide application should be given to the mango crop within 1 month of the harvest date, which otherwise will lead to high MRL in mango.

To ensure the food is safe for consumption, adhering to global standards and certifications is essential. Among several standards, the GLOBALGAP standards are internationally recognised and dedicated to Good Agricultural Practices (GAP), ensuring safe food for human consumption. These standards are applied voluntarily in agricultural production to ensure the safety of the food for consumption. It works on the principle of sustainable agriculture development worldwide. GLOADGAP increases the acceptability of the produce in foreign markets, as well as leads to a reduction of cost of production due to lesser use of pesticides, chemicals, etc. The other means of reduction of pesticide residues in the mango fruit would be through certification, i.e., Organic certification (National Programme for Organic Production, Jaivik Bharat, EU Organic certification, PGS-India) if the produce is grown organically; ZERYA certification, Zero/Controlled Pesticide Residue certification, SCS Global Service Pesticide Free Certification and Rainforest Alliance Certification.

Pesticide management for Mango stakeholders

For Farmer/Farm Level



- Strictly adhering to GlobalGap Certification Standards for the production of mangoes.
- Application of Integrated Pest Management (IPM) strategies for protection of crop health.
- Use of labelled, certified, and registered pesticides.
- Application of pesticides as per the limits set by WHO/FAO or other National Agencies.
- Regular training and knowledge sharing with the farmers will help in reducing the pesticide residual load in the long run.

For Package House handling Unit

- For Latex/ Sap removal from the mango surface, use specialised food-grade detergents.
- For pesticide residue removal, use plain water washing through water blasting / high-pressure washing methods.
- Chlorinated water should be used at a concentration of 50 – 150 ppm for the removal of microbial and residual load.
- Use of green cleaning techniques such as ozonated water (0.5 – 2.0 ppm), electrolysed water will help in residue-free cleaning.
- Use of UV – C radiation for photo degradation of pesticide molecules can be adapted.
- For fungicidal application, fumigation of mangoes under controlled conditions using registered and approved fumigants should be done.
- Use of registered ripening methods such as application ethrel or ethephon only instead of calcium carbide.
- Storage of mangoes should be done in a controlled atmosphere (CA) or cold storage that is clean and free of other contaminants or earlier used chemical residues.
- Always use food-grade, residue-free packaging material adhering to packaging standards.
- The mango processing lines should adapt Hazard Analysis Critical Control Point (HACCP) based residue management strategy.

For Export/Logistics Unit

- Map all target-market MRLs (EU, US, Codex, etc.) for your export crops; treat MRLs as the starting point for pest management decisions.
- Implement pre-export residue testing using accredited labs and multi-residue methods (e.g., GC/LC MS/MS, QuEChERS-Quick, Easy, Effective, Rugged, and Safe); apply conservative internal action limits.
- Build a residue database linking farm, pesticide use, residue results, and lot destination to document

compliance and support negotiations with buyers and regulators.

- Provide continuous training for farmers, agronomists and exporters on MRLs, safe use, record keeping and updates in importing country rules.
- Coordinate with national authorities and industry groups on monitoring programs and harmonized MRLs work (e.g., crop grouping, bilateral MRL alignment) to reduce trade disruptions.

For Consumer

- Washing and cleaning: Wash mango fruits thoroughly under running water for 2-3 minutes. Soaking in salt water (2%) or vinegar solution (1:3 ratio) for 10-15 minutes helps remove surface residues.
- Peeling and Trimming: Peel mango fruits to reduce the presence pesticide residues on the surface.
- Cooking and processing: Boiling, steaming and frying can reduce pesticide residues. Blanching before cooking.
- Selection of food: Prefer fresh, seasonal and locally grown produce. Choose certified organic products when possible. Avoid damaged, overly shiny or unsavory large fruit.
- Proper storage: Keep raw and cooked food separate. Refrigeration slows down chemical changes. Store food properly.



For Policy Makers



Reference Links and Sources

- Kurbis. G. 2019. Update on International MRL Coalitions. Crop Protection, Canada Grains Council, May 29, 2019.
- https://apeda.gov.in/sites/default/files/2025-09/MIC_July_Monthly_dashboard_Mango.pdf
- <https://iihr.res.in/sites/default/files/Pesticide%20Residue%20management%20in%20Mango63.pdf>
- <https://openknowledge.fao.org/server/api/core/bitstreams/4aa015cc-f644-4a1a-88dc-d0c2272fca05/content>
- <https://www.who.int/news-room/fact-sheets/detail/pesticide-residues-in-food>
- https://specialtycrops.org/wp-content/uploads/2024/06/18_CanadaGrains_2019.pdf
- https://www.fao.org/fao-who-codexalimentarius/codex-texts/dbs/pestres/commodities-detail/es/?c_id=134#:~:text=Table_title:%20FI%200345%20%2D%20Mango%20Table_content:%20header:mg/Kg%20%7C%20Year%20of%20Adoption:%202021%20%7C

SPREADING LOVE
for the last 65 years...

Fruitomans®
Since 1957

Nature's real taste for Generations...

Ernakulam Bypass, Cochin-682 028
Tel: 0484-2808233, 2808273
E-mail: marketing@fruitomans.com
www.fruitomans.com

[@fruitomans](https://www.fruitomans.com)

Scan here:

www.fruitomans.com Scan QR code for catalogue



IFDC PROFILE

Preserving Nature's Best: From India's Mango Heartland to the World

When you bite into a perfectly ripe Alphonso or Kesar mango, there's a burst of flavour that's unmistakably Indian. It's a taste the world craves, and one that, until recently, was nearly impossible to capture beyond the short summer season.

At Indian Freeze Drying Corp (IFDC), we've made it our mission to change that.

Who We Are

IFDC is a venture of the Hingorani Group of Companies, a conglomerate with over four decades of industrial expertise, established in 1978. Headquartered in Vadodara, Gujarat, we operate a state-of-the-art manufacturing facility dedicated to producing premium freeze-dried and air-dried food products.

Our story is rooted in a simple conviction: India grows some of the world's best fruits, vegetables, and spices, and they deserve technology that does justice to their quality. Freeze drying, we believe, is that technology.

Why Freeze-Dried Mango? Why Now?



India is the world's largest producer of mangoes, yet a significant share of this incredible produce is lost to perishability, seasonal constraints, and limited processing infrastructure. The global demand for clean-label, minimally processed, shelf-stable fruit ingredients is growing rapidly, across snacking, cereals, nutraceuticals, desserts, beverages, and ready-to-eat meals.

Freeze-dried mango sits at the intersection of all these trends. Through our sublimation-based freeze-drying process, we preserve up to 97% of the fruit's natural nutrients, colour, aroma, and flavour, with absolutely no preservatives, additives,

or artificial colouring. The result is a product that looks, tastes, and nourishes almost exactly like fresh mango, but with a shelf life of 18 to 25 years.

Whether it's diced mango for breakfast cereals, mango powder for smoothie blends, or whole slices for premium snacking, our freeze-dried mango products deliver unmatched quality and consistency, season after season.

Our Mango Processing Strengths

Being based in Gujarat gives us a natural advantage. We have direct relationships with certified farms across India's top mango-producing belts, including Gujarat, Maharashtra, Uttar Pradesh, and Andhra Pradesh. Our dedicated sourcing teams work with seasonal procurement protocols to secure the best varieties, Alphonso, Kesar, Totapuri, Dasher, and more, at peak ripeness.

Here's what sets our mango processing apart:

- **Farm-to-Facility Traceability:** We source from Global GAP-certified farms, ensuring complete transparency from orchard to finished product.
- **Individual Fruit Inspection:** Every mango is individually inspected at intake. No shortcuts, no compromises.
- **In-House Quality Lab:** Raw materials are rigorously tested before production begins, for Brix levels, maturity, chemical residues, and microbial safety.
- **Flexible Formats:** Whole slices, dices, cubes, powder, or custom blends, we adapt to your product requirements.
- **Year-Round Availability:** Through strategic procurement and inventory planning, we ensure a consistent supply well beyond the mango season.

Our Full Product Portfolio

While mango is one of our flagship offerings, our capabilities extend across a wide range of produce:

Freeze-Dried Fruits

Mango, Raspberry, Banana, Apple, Peach, Pineapple, Cranberry, Strawberry, Mixed Fruits & more

Freeze-Dried Vegetables & Spices

A comprehensive range of vegetables, herbs, and premium Indian spices

Ingredients & Ready-to-Eat

Custom ingredient blends, instant meals, and convenience food formats

From Indian Mangoes to Global Food Products

Transform your mango processing with IFDC's state-of-the-art freeze-drying technology.



Our mango offerings are widely used in:

- Bakery & Confectionery
- Breakfast Cereals & Snacks
- Dairy & Desserts
- HoReCa & Food Service Applications

Let's build better ingredients together.

📞 95747 93979
🌐 ifdc.co.in

Manufacturing Capabilities

Our facility is purpose-built to meet international food safety standards, and designed to serve partners ranging from early-stage startups to established global brands.

- **Freeze Drying:** Sublimation-based technology preserving nutrition, flavour, and colour with exceptional shelf life.
- **Air Drying:** A cost-effective dehydration option for high-volume, price-sensitive applications.
- **In-House Quality Lab:** Batch-level testing and inspection at every stage of production.
- **Flexible Batch Sizes:** Scalable production that grows with your business.
- **Export-Grade Packaging:** Including custom labelling and branding support to make your product shelf-ready for any market.

B2B & Contract Manufacturing — Built Around Your Needs

IFDC is structured as a dedicated B2B partner. We work with FMCG brands, ingredient companies, HoReCa suppliers, nutraceutical firms, and food exporters, offering tailored, end-to-end manufacturing solutions.

OEM / White-Label Manufacturing

We handle everything, from custom formulations to branding and packaging. Your intellectual property stays yours. Confidentiality is guaranteed, always.

Custom Freeze-Drying

Have your own sourced raw materials? Bring them to our facility. We'll process them to your specifications while you retain full ownership of the finished product.

Bulk Ingredient Supply

Need large, consistent volumes of freeze-dried mango or other ingredients? We deliver — reliably and at scale.

Industries We Serve

- **Food Processing & Packaged Foods:** Cereals, snack bars, trail mixes, confectionery, bakery inclusions
- **Health, Wellness & Nutraceuticals:** Superfood blends, supplement ingredients, functional foods
- **HoReCa & Institutional Catering:** Premium garnishes, dessert ingredients, instant meal components
- **Gourmet, Defence & Aerospace:** Long shelf-life meal solutions for specialised applications

Quality Assurance: It's in Everything We Do

Quality at IFDC isn't a final checkpoint. It's embedded in our culture and our processes, from the farm gate to the shipping dock.

- Farm-to-facility traceability with Global GAP-certified sourcing
- In-house laboratory testing of all raw materials before production
- Individual fruit inspection at intake, because every piece matters
- Compliance with FSSAI, HACCP, and applicable export certifications
- Clean-label commitment, no artificial preservatives, additives, or colouring in any product, ever

Why Partner With IFDC?

Heritage & Credibility

Backed by the Hingorani Group's 45+ years of industrial operations, we bring stability, experience, and a long-term perspective to every partnership.

Advanced Technology

Cutting-edge freeze-drying lines delivering superior product integrity, batch after batch.

Farm-Strong Sourcing

Direct relationships with certified Indian farms, ensuring a stable, traceable, and high-quality supply chain.

End-to-End Partnership

From raw material procurement to shelf-ready product, we handle it all, with complete confidentiality.

Global Compliance

Certifications and quality systems are aligned with what international buyers expect and require.

Customisation

Flexible batch sizes, private labelling, bespoke product development, we shape our capabilities around your goals.

Let's Build Something Together

Whether you're looking for a reliable freeze-dried mango supplier, a contract manufacturing partner for your next product launch, or simply want to explore what's possible — we'd love to hear from you.



Innovation in Mango-Based Products

Abha Singh, Alok Kumar Gupta and Ravi S.C

Division of Post-harvest Management,
ICAR-Central Institute for Subtropical Horticulture,
Lucknow (UP) - 226101, India

Mango is one of the most important tropical fruits with immense potential for value addition and development of innovative processed products. Due to its rich nutritional composition, attractive flavor, and high consumer acceptance, mango is widely utilized in the preparation of functional beverages, dehydrated products, nutraceuticals, fruit bars, and processed pulp products. Recent innovations in mango processing technologies have focused on improving shelf life, reducing post-harvest losses, enhancing nutritional quality, and promoting sustainable utilization of mango waste. Functional mango beverages such as probiotic drinks, fermented beverages, mango kombucha, and fortified smoothies are gaining popularity among health-conscious consumers due to their health benefits and antioxidant properties. Advanced dehydration and freeze-drying technologies have enabled the production of high-quality mango snacks with improved storage stability and export potential. Institutions such as ICAR-CISH have significantly contributed to the development of innovative mango products, pulp processing technologies, export-oriented systems, and waste valorization approaches.

Introduction

Mango is one of the most important tropical fruits in the world and is widely appreciated for its taste, aroma, nutritional value, and economic significance. India is the largest producer of mangoes globally, contributing significantly to world production. However, due to the highly perishable nature of mango fruits, substantial post-harvest losses occur every year. To overcome these challenges and enhance profitability, innovations in mango-based products have emerged as a promising area in food processing, value addition, sustainability, and functional food development.

Recent advancements in food technology, packaging, waste utilization, and nutraceutical research have transformed mango processing industries from traditional pulp and pickle production to high-value functional foods, biodegradable materials, and health-oriented products. Mango peel, kernel, and seed waste are now being recognized as valuable resources for bioactive compounds, dietary fiber, antioxidants, and biodegradable packaging materials.

1. Mango Nutri Bars and Fruit Bars

ICAR-CISH has developed innovative mango-based nutraceutical snack products such as mango nutri bars, fruit leather, and fortified fruit bars to utilize surplus and low-grade mango fruits that are otherwise wasted during peak harvesting season. These products are prepared using ripe mango pulp, raw mango pulp, and blended fruit ingredients with natural sweeteners and nutritional additives. The products are rich in carbohydrates, dietary fiber, vitamins, antioxidants, and minerals, making them suitable as healthy ready-to-eat snacks for children, youth, and health-conscious consumers.

The preparation process involves pulp extraction, formulation standardization, concentration, drying, sheet formation, and hygienic packaging. Advanced dehydration techniques help maintain natural color, aroma, flavor, and nutritional quality. Mango fruit bars developed by ICAR-CISH possess good shelf stability and can be stored for several months without major



quality deterioration. The institute also developed low-sugar and functional variants enriched with herbal extracts and micronutrients to meet the growing demand for healthy convenience foods.

2. Dehydrated and Freeze-Dried Mango Products

ICAR-CISH has standardized modern dehydration technologies for preparation of high-quality mango products with extended shelf life and export potential. Conventional drying methods often lead to loss of flavor, nutrients, and color therefore, the institute adopted advanced processing technologies such as freeze drying, osmotic dehydration, infrared drying, microwave-assisted drying, and vacuum drying.

Using these technologies, ICAR-CISH developed several value-added products including:

1. Dried mango slices
2. Dehydrated mango cubes
3. Mango chips



4. Freeze-dried mango snacks
5. Mango powder

Freeze drying is considered one of the most advanced technologies because it removes moisture at low temperatures under vacuum conditions, preserving natural taste, texture, nutrients, and aroma. The resulting products are lightweight, crispy, and highly stable during storage. Such products are increasingly popular in international markets due to growing demand for healthy fruit snacks. Osmotic dehydration technology developed by ICAR-CISH involves partial removal of moisture using sugar or salt solutions before final drying. This method improves texture, flavour retention, and product stability while reducing processing damage. The institute has also optimized packaging systems using moisture-proof materials to improve shelf life and maintain product quality.

3. Functional Mango Beverages

Modern consumers are increasingly demanding healthy, natural, and functional beverages enriched with vitamins, antioxidants, probiotics, and bioactive compounds. Mango has emerged as an important ingredient in the development of innovative functional drinks such as probiotic mango beverages, mango smoothies, mango sports drinks, fermented mango beverages, and mango kombucha. These products not only provide excellent flavor and nutritional value but also promote gut health, immunity, hydration, and overall wellness. Functional mango beverages are often fortified with proteins, vitamins, minerals, and natural antioxidants, making them highly popular among health-conscious consumers. In

addition, advanced processing and fermentation technologies improve shelf life, product stability, and nutrient retention, thereby creating new opportunities for value addition and commercialization in the mango processing industry.

4. Mango Pulp Processing Technologies

ICAR-CISH has established advanced mango pulp processing technologies aimed at large-scale value addition and industrial utilization of mango fruits. Mango pulp serves as the basic raw material for numerous processed products including beverages, ice creams, bakery products, yogurts, confectionery items, baby foods, and export products. Special emphasis has been given to improving pulp recovery, minimizing contamination, preserving natural flavor, and extending shelf life. ICAR-CISH also established pilot-scale and commercial-scale pulp processing facilities to demonstrate industrial processing technologies for entrepreneurs, startups, and FPOs.

The mango pulp processing technology developed by the institute supports preparation of:

1. Mango pulp
2. Mango nectar
3. Ready-to-serve beverages
4. Mango squash
5. Mango concentrate
6. Mango puree

A major innovation promoted by ICAR-CISH is the custom-hiring pulp processing model, where farmers and Farmer Producer Organizations can process their produce using institutional facilities without investing heavily in expensive machinery. This approach is particularly beneficial for small and marginal farmers.

The pulp processing technologies developed by ICAR-CISH are contributing significantly toward development of sustainable mango value chains and commercialization of mango-based industries in India.

6. Conclusion ICAR-CISH has emerged as a leading institution in mango innovation by developing advanced mango-based products, export technologies, dehydration methods, and sustainable waste utilization systems. Its innovations have contributed significantly toward reducing post-harvest losses, promoting entrepreneurship, improving export competitiveness, and creating sustainable value chains in the mango sector. Continued research and commercialization of these technologies can transform India's mango industry into a globally competitive and environmentally sustainable sector.



Next-Generation Mango Processing: Technologies Driving Value Addition

Sumaiya Fatima, Vivek Kumar*, Nabila Arshad and Saachi Chaurasia

Department of Food Technology, Harcourt Butler Technical University, Kanpur-208002, Uttar Pradesh, India

Email: viveksachan99@rediffmail.com (Corresponding author)



Vivek Kumar

1. Introduction

Mango (*Mangifera indica* L.) is one of the most commercially significant tropical fruits, valued for its rich flavour, vibrant colour, and high content of bioactive compounds such as carotenoids, polyphenols, and vitamins. India dominates global mango production with 44.52% of the global production as recorded in 2025 (APEDA 2026); however, a considerable proportion of the harvested fruit is lost due to rapid ripening, microbial spoilage, and inadequate post-harvest management. To mitigate these losses, conventional processing methods such as pulping, canning, and hot-air drying are widely employed. While these techniques extend shelf life, they present several limitations. Thermal processing often leads to degradation of heat-sensitive nutrients such as vitamin C and carotenoids, along with loss of natural flavour and colour. Enzymatic browning and non-enzymatic reactions like Maillard browning further compromise product quality during drying and storage. Additionally, traditional drying methods are time- and energy-intensive, resulting in uneven moisture removal and inferior texture. Conventional processing also offers limited scope for diversification, largely restricting products to pulp, juice, and basic dried forms, thereby reducing economic returns (Bhusari et al., 2025).

In contrast, next-generation processing technologies are enabling the development of high-quality, diverse mango products by addressing these limitations. For dried products, advanced techniques such as freeze drying, microwave-assisted drying, and hybrid drying systems ensure rapid and uniform moisture removal while preserving colour, flavour, and nutritional quality (López and Hincapié-Llanos, 2024). In the case of pulped and liquid products, non-thermal technologies including high pressure processing and pulsed electric field treatment help retain fresh-like characteristics and extend shelf life without significant nutrient loss. Simultaneously, innovative approaches for the valorization of mango by-products, including peel and seed, are creating new opportunities for the development of high-value products in the food and nutraceutical sectors. These technological advancements are transforming the traditional mango value chain into a more sustainable and economically viable system. The present article explores the emerging processing technologies that are driving value addition in mango and highlights their potential to enhance profitability, reduce post-harvest losses, and support the development of a circular bioeconomy.

2. Need for Mango Processing and Its Role in Economic Value Generation

The processing of mango is essential to address its high perishability and to ensure efficient utilization of seasonal surplus. Processing extends the shelf life of the fruit, stabilizes supply, and enables its availability beyond the harvest season, thereby improving market accessibility. Beyond preservation, mango processing plays a pivotal role in value addition and income generation. A broad range of minimally processed, dehydrated, and value-added formulations made from both ripe and green mangoes through meticulously regulated unit operations are included in mango-based product diversification. Fig. 1 shows the diverse range of products developed through mango processing. Fresh-cut mango is made from sliced ripe fruit that has undergone moderate pre-treatments (such as osmotic dips and anti-browning chemicals) and is placed in low-temperature storage. dehydrated products like dried mangos undergo pre-treatments (osmotic dehydration, sulfiting, or blanching) followed by drying using mechanical, solar, or sophisticated methods like refractive window or freeze drying. Further, pulp may be extracted from ripe mangos to produce juice, juice concentrates (by evaporation or membrane concentration), squash/nectar (with regulated TSS and acidity), probiotic-enriched beverages (by fermentation), and even mango wine (by alcoholic fermentation). Also, mango pulp can be converted into stable powders by spray drying, freeze drying, drum drying, or refractive window drying. Besides, green mangoes are mostly used for pickles and chutneys, or they can be dried to make mango leather. These products cater to different consumer segments and markets, including food and nutraceutical industries. Importantly, value-added processing significantly enhances economic returns compared to the sale of fresh fruit. For instance, products such as dried mango and mango-based beverages offer higher profit margins, while the utilization of by-products like peel and seed further contributes to additional revenue streams. Thus, processing not only reduces losses but also transforms mango into a high-value commodity, supporting livelihood generation and agro-industrial growth.

3. Limitations of Conventional Mango Processing Methods

Conventional mango processing techniques, despite its benefits, face several limitations that restrict product quality and value realization. Thermal processing methods, widely



Fig. 1. Major mango-based products prepared from mango processing

used in pulp and juice production, often lead to degradation of heat-sensitive nutrients such as vitamin C and carotenoids, along with loss of natural flavour and aroma compounds. These processes may also induce undesirable colour changes due to non-enzymatic browning reactions, affecting consumer acceptability. Similarly, traditional hot-air drying methods used for producing dried mango products are often inefficient and energy-intensive. They can result in uneven drying, case hardening, shrinkage, and poor rehydration properties, ultimately compromising product quality. In addition, conventional techniques provide limited scope for innovation and diversification into high-value or functional products. Another major drawback is the underutilization of processing by-products such as mango peel and seed, which are often discarded despite being rich in valuable bioactive compounds. These limitations underscore the need for adopting advanced processing technologies that can overcome quality losses, improve efficiency, and enable comprehensive utilization of mango resources (Bhusari et al., 2025).

4. Next generation mango processing

The novel processing technologies being adapted for processing different types of mango-based food products are summarized in Fig. 2.

4.1. Processing Technologies for Fresh-Cut Mango and Pickle Products

Conventional processing of fresh-cut mangoes and pickles primarily involves cutting, salt addition, and thermal or chemical preservation to extend shelf life. However, these methods often lead to quality deterioration, including enzymatic browning, nutrient loss, texture softening, and high microbial load, while traditional pickle processing relies heavily on high salt content, raising health concerns. To overcome these limitations, emerging technologies are being explored to improve product quality and safety. Cold plasma treatment has shown potential in preserving the nutritional and

sensory attributes of fresh-cut mangoes by reducing phenolic oxidation, inhibiting reactive oxygen species formation, and controlling microbial growth (Wang et al., 2022). Similarly, antimicrobial coatings and microencapsulation techniques are being developed to provide controlled release of antioxidant and antibacterial agents at the product surface, thereby enhancing shelf life. Edible coatings enriched with bioactive compounds such as vitamin C, dietary fibre, and antioxidants help reduce moisture loss and microbial contamination while maintaining freshness. The combined use of anti-browning agents and modified atmosphere packaging further minimizes discolouration and quality degradation during storage. In pickle production, controlled fermentation and the use of natural preservatives are emerging as effective alternatives to conventional high-salt methods, offering improved safety, extended shelf stability, and better alignment with consumer demand for healthier products (Alikhani, 2024).

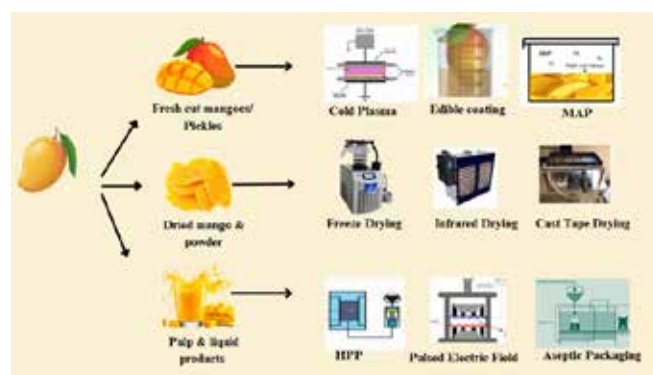


Fig. 2. Novel processing techniques used for mango processing

4.2. Processing Technologies for Dried Mango and Powder

Traditionally, green mango slices are preserved by sun drying, requiring long exposure times of 16 hours at 35–40°C, while cabinet drying reduces time but still affects product quality. These conventional methods are often associated with uneven drying, contamination risks, colour deterioration, and loss of heat-sensitive nutrients, even when pre-treatments such as blanching and sulfiting are applied. To address these limitations, advanced drying technologies have been developed to improve efficiency and product quality. Freeze drying, which operates under low temperature and vacuum conditions, effectively preserves colour, flavour, and nutritional components. Techniques such as foam mat drying and convection drying at controlled temperatures further reduce drying time while maintaining product integrity. Cast-tape drying offers an alternative approach for high-sugar materials, enabling the production of mango powder directly from pulp without the need for additives like maltodextrin. Additionally, hybrid drying methods that combine hot air with microwave or infrared energy enhance drying efficiency, resulting in uniform moisture removal and improved product characteristics. These next-generation techniques produce stable products with better solubility, higher retention of bioactive compounds, and superior overall quality compared to conventional drying methods (López and Hincapié-Llanos, 2024).

4.3. Processing Technologies for Pulp and Related Products

Conventionally, mango pulp is processed using thermal treatments such as pasteurization and canning, which, although effective for microbial safety, often result in degradation of heat-sensitive nutrients, loss of bioactive compounds, and undesirable changes in colour and flavour. These limitations reduce the overall quality and market value of pulp-based products. To overcome these challenges, advanced processing technologies are being adopted. High Pressure Processing (HPP), for instance, enables microbial inactivation with minimal heat exposure, resulting in better retention of ascorbic acid, phenolics, and flavonoids, along with improved colour stability compared to conventional methods. When combined with suitable packaging, such as retort pouches, HPP significantly extends shelf life while maintaining product quality (Kaushik et al., 2018). Similarly, Pulsed Electric Field (PEF) treatment enhances juice extraction efficiency by disrupting microbial cells and improves shelf stability without compromising nutritional attributes. In addition, advancements in aseptic processing and packaging allow for long-term storage of mango pulp with minimal quality deterioration (Salinas et al., 2017). These next-generation technologies ensure safety, preserve nutritional and sensory properties, and support large-scale production and distribution of high-quality mango pulp products.

4.4. Processing technology for mango By- Products

Mango processing generates substantial amounts of by-products, particularly peel and seed, which are often discarded or underutilized, leading to resource loss and environmental concerns. Traditional utilization methods, such as simple drying and grinding, are limited in efficiency and do not fully exploit the potential of these materials. To address this, advanced and sustainable valorisation approaches are being developed. Fig. 3 summarises various applications and properties of by-products of mango processing. Green extraction techniques such as supercritical fluid extraction, ultrasound-assisted extraction, microwave-assisted extraction, pulsed electric field extraction, and enzyme-assisted extraction enable efficient recovery of bioactive compounds with minimal environmental impact. These extracted compounds can be utilized in the development of value-added products including

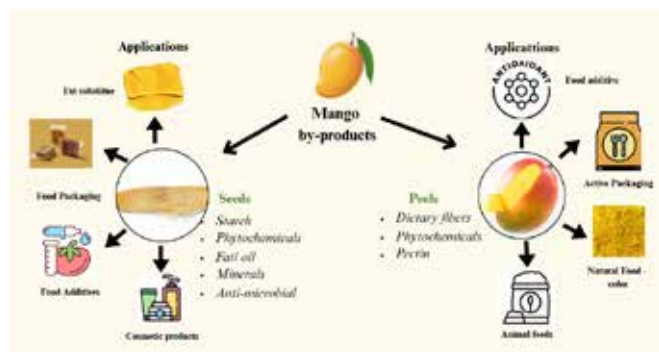


Fig. 3 Different by-products of mango and their key applications and properties

functional foods, active packaging materials, and cocoa butter substitutes. Additionally, innovative product formulations such as mango-based breakfast cereals, shrikhand, and dry chutney enhance the economic potential of by-products. Biotechnological approaches further expand utilization by converting mango waste into bio-products such as bioethanol, biogas, and single-cell protein. These advancements not only reduce waste but also contribute to sustainable and economically viable mango processing systems.

5. Economic and Sustainability Impact of Next-Generation Mango Processing

The adoption of next-generation processing technologies in the mango sector significantly enhances both economic returns and sustainability (Fig. 4). By improving product quality and enabling diversification, these technologies increase the market value of mango products, with items such as dried products, minimally processed juices, and mango seed-based ingredients fetching premium prices. This shift toward high-value products improves profit margins and reduces reliance on volatile fresh fruit markets. Additionally, advanced technologies enhance efficiency and minimize losses, while the utilization of by-products like peel and seed creates new revenue streams. Further, in terms of sustainability, next-generation processing promotes resource optimization and waste reduction. Technologies such as hybrid drying and energy-efficient systems reduce energy consumption, while valorization of by-products contributes to a circular economy by minimizing environmental burden. Collectively, these advancements not only improve the economic viability of mango processing but also support environmentally responsible production systems, making the sector more resilient and sustainable in the long term (Owino and Ambuko 2021).



Fig. 4. Circular economy in mango processing

6. Challenges and Future Scope of Next-Generation Mango Processing

Despite its advantages, the adoption of next-generation mango processing technologies remains constrained by high capital costs, the need for skilled manpower, and maintenance requirements, limiting accessibility for small-scale processors. Inadequate infrastructure, low technical awareness, and

limited access to finance further hinder widespread implementation, while market competition and low consumer awareness of premium products restrict commercialization. Future advancement depends on the development of cost-effective, scalable technologies suited to small and medium enterprises. Furthermore, strengthened collaboration among research institutions, industry, and policymakers will be critical to facilitate technology adoption and drive value addition in the mango sector.

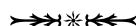
7. Conclusion

Next-generation mango processing technologies are playing a crucial role in transforming the traditional mango sector into a more efficient, value-driven, and sustainable system. By addressing the limitations of conventional methods, these advanced technologies enable the production of high-quality products with better retention of nutritional and sensory attributes. The categorization of processing approaches based on product types—fresh-cut and pickles, dried products, and pulp-based products—highlights the importance of selecting appropriate technologies to maximize quality and economic returns. Moreover, the integration of innovative techniques not only enhances product diversification but also opens new avenues for value addition through improved processing efficiency and extended shelf life. The effective utilization of mango resources, including by-products, further contributes to waste reduction and supports a circular economy approach. Despite existing challenges related to cost, infrastructure, and awareness, the future of mango processing remains promising.

With continued advancements, supportive policies, and increased industry adoption, next-generation technologies have the potential to significantly enhance profitability, reduce post-harvest losses, and strengthen the overall mango value chain.

References

- i. Alikhani, M. (2014). Enhancing safety and shelf life of fresh-cut mango by application of edible coatings and microencapsulation technique. *Food Science & Nutrition*, 2(3), 210-217.
- ii. APEDA (2026). <https://apeda.gov.in/>. [Accessed on 01-05-2026].
- iii. Bhusari, R. M., Patil, S. D., Gethe, A. S., & Patil, S. R. (2025). A Comprehensive Review on Processing Technologies and Value-added Products of Mango. *European Journal of Nutrition and Food Safety*, 17(9), 19-28.
- iv. López, L. C., & Hincapié-Llanos, G. A. (2024). Comparison of Mango (*Mangifera indica*) dehydration technologies: A systematic review. *Agriengineering*, 6(3), 2694-2717.
- v. Kaushik, N., Rao, P. S., & Mishra, H. N. (2018). Comparative analysis of thermal-assisted high pressure and thermally processed mango pulp: Influence of processing, packaging, and storage. *Food Science and Technology International*, 24(1), 15-34.
- vi. Owino, W. O., & Ambuko, J. L. (2021). Mango fruit processing: Options for small-scale processors in developing countries. *Agriculture*, 11(11), 1105.
- vii. Salinas-Roca, B., Elez-Martínez, P., Welte-Chanes, J., & Martín-Belloso, O. (2017). Quality changes in mango juice treated by high-intensity pulsed electric fields throughout the storage. *Food and Bioprocess Technology*, 10(11), 1970-1983.
- viii. Yi, F., Wang, J., Xiang, Y., Yun, Z., Pan, Y., Jiang, Y., & Zhang, Z. (2022). Physiological and quality changes in fresh-cut mango fruit as influenced by cold plasma. *Postharvest Biology and Technology*, 194, 112105.



Radioanalysis in Food, Water and Environment

Dr. Praveen Bansal

Director

AumNamah Radioanalytical Laboratory LLP

Plot No-1 First Floor, Community Centre, Lawrence Road, Delhi -110035

Phone no: 8920723168, 9717686925

aumnamah.ral@gmail.com

Radioactive substances (radionuclides) are materials that contain unstable atomic nuclei. These nuclei spontaneously break down (decay) and release energy in the form of radiation such as alpha, beta and gamma. Radioanalysis is the science of identifying and quantifying radioactive substances within various samples, crucial for protecting public health and the environment from the harmful effects of ionizing radiation. It ensures that food, drinking water, and the environment comply with safety standards, particularly by monitoring both natural and man-made contaminants. Fig 1 shows various pathways where in radionuclides in atmosphere reach to humans [1].

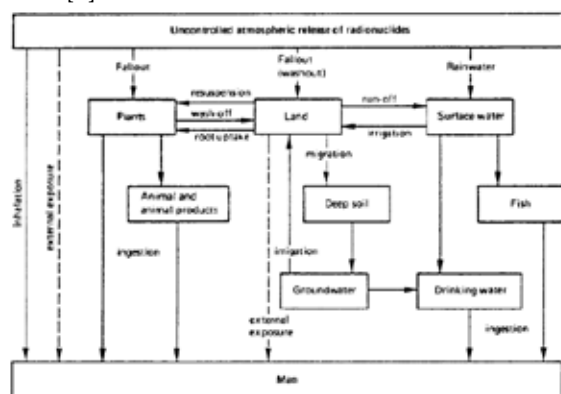


FIG. 1. Major pathways of radionuclides to man in the event of an uncontrolled release of radioactivity [1].

Fig. 1: Major pathways of radionuclides to man in the event of an uncontrolled release of radioactivity (Courtesy: IAEA Technical report series 295)

Importance of Radioanalysis in Water Safety

Common radioactive contaminants in water include Natural radionuclides: Uranium (U-238), Radium (Ra-226, Ra-228), Radon-222, and K-40 etc. and Artificial radionuclides: Cesium-137, Cesium-134, Strontium-90, Iodine-131 etc. These may enter water through geological formations, industrial discharge or nuclear incidents.

In India there are certain risk areas where there is potentially high chances of finding the radioactive contaminants for example groundwater in certain parts of Punjab, Rajasthan,

parts of Delhi NCR is found to be rich in specially uranium content causing larger incidences of cancer cases in local population. Recent data from the Central Ground Water Board (2025) indicates that uranium contamination in groundwater exceeds permissible limits in up to 62.5% of samples in Punjab, ~15% in Delhi NCR, and significant portions of Haryana and Rajasthan. These findings highlight a major geogenic radiological risk in northwestern India, with potential long-term health implications” [2].

Industrial regions associated with effluent discharge into surface and groundwater systems have been reported to exhibit elevated concentrations of radionuclides, often exceeding permissible limits. Such contamination may arise not only from direct radioactive sources but also from the mobilization of naturally occurring radioactive materials (NORM) due to industrial activities, including thermal power generation, phosphate fertilizer production, and metallurgical processing.

In addition, areas located in proximity to nuclear installations and uranium mining sites, particularly in regions such as Jharkhand notably Jaduguda are recognized as potential hotspots for radionuclide contamination in water resources. These regions are influenced by activities related to uranium extraction, ore processing, and tailings management, which can contribute to the release of radionuclides such as uranium, radium, and their decay products into surrounding groundwater and surface water systems.

Consequently, systematic radiological monitoring of water bodies in industrial and mining-influenced zones is essential to assess potential exposure risks and to ensure compliance with national and international safety standards. Radioanalysis is thus vital for evaluating the radiological quality of water sources for human consumption and agricultural use.

Importance of Radioanalysis in Food Safety

Radioanalysis detects artificial radionuclides (like Cesium-137, Strontium-90) and natural ones (like Lead-210, Polonium-210 etc) that can enter the food chain, especially near high risk areas or following nuclear incidents. It ensures

that food products do not exceed established radioisotope concentrations, which is critical for consumer safety and international trade.

Countries like the European Union, South Korea, Japan, and the Russian Federation often require certificates of non-contamination for imported food products. Exporters of fish and seafood, dairy products, and mushrooms are frequently required to provide radioactivity test reports to clear customs. In India, the Export Inspection Agency (EIA) verifies that food exports meet international radionuclide limits.

Importance in Radioanalysis in Environmental Monitoring



Radioanalysis is used to quantify radioactivity levels in environmental matrices such as air, soil, water, and sediments, enabling the evaluation of ecological risk and long-term contamination trends [3]. It plays a critical role in the rapid assessment and spatial mapping of radioactive contamination, particularly during and following nuclear or radiological emergencies. Environmental radioactivity mapping involves the systematic collection and georeferencing of samples, followed by laboratory or in-situ radiometric measurements. The resulting data are integrated into Geographic Information Systems (GIS) to generate contamination maps, dose distribution models, and risk zonation layers.

Economic Impact of Radiological Analysis in India

Radiological analysis of food and water is a critical component of national safety infrastructure. It ensures that radioactive contaminants are within permissible limits, safeguarding public health, supporting international trade, and fostering economic stability.

For countries like India, where agriculture, food processing, and water resources are key economic drivers, investment in radioanalytical capabilities yields long-term economic and strategic benefits. This report evaluates the economic implications of implementing robust radiological testing systems across food and water sectors.

1. Public Health and Healthcare Savings

Radiological testing helps in early detection of contaminants that can lead to diseases such as cancer. Thus leading to reduced burden on healthcare systems, lower long-term

treatment costs and improved labor productivity.

2. Trade and Export Competitiveness

Global agencies such as World Health Organization and Food and Agriculture Organization enforce strict safety limits. Stringent radiation testing at the exporting country ensures increased acceptance of Indian exports, reduction in shipment rejections while enhancing global credibility of Indian exports.

3. Industrial and Service Sector Growth

Radiological testing promotes the development of accredited laboratories, analytical instrumentation markets, and skilled scientific workforce. This enhances Job creation in high-skill sectors, growth of private analytical service providers, and increased foreign and domestic investment.

4. Environmental Risk Mitigation

Routine monitoring prevents large-scale contamination of water resources, agricultural land, fisheries etc. Early detection means better management. In the event of any contamination detection, costly remediation projects can be avoided, agricultural productivity can be protected and rural economy can be sustained.

5. Disaster Preparedness and Economic Stability

Radiological testing enables rapid response during nuclear or industrial emergencies by ensuring uninterrupted food supply chains, prevention of market panic, minimization of economic losses.

6. Regulatory Strengthening and Governance

Radiological data supports policy formulation and enforcement through improved compliance across industries, strengthened regulatory frameworks and increased investor confidence

Role of Radiological Testing in the Food & Water Industry



Water is the backbone of Food and water industry, Water is used in washing raw materials, food processing, beverage production, packaged drinking water etc. Contaminated water can directly transfer radionuclides into food products.

Radiological testing is essential to ensure compliance with



AumNamah Radioanalytical Laboratory LLP

(NABL Approved APEDA Recognized Testing Laboratory)



Ensure Safety Beyond Chemistry: Radiological Testing for Water & Food

" Why Radiological Testing Matters "

Water is the backbone of Food & Water Industry

Though chemical and microbiological testing are standard, Radiological contamination monitoring is equally important.

The Hidden Risk to Your Brand

- Undetected contamination → consumer health hazards
- Non-compliance with standards → regulatory penalties
- Export rejection due to non-alignment with World Health Organization / Food and Agriculture Organization norms
- Brand reputation damage → loss of consumer trust

Industries That Need Us

- Packaged Drinking water manufacturers
- Beverage companies (juices, soft drinks, breweries)
- Dairy & food processing industries
- Export-oriented food brands

Add Value to Your Product

- ✓ In a competitive market like India, safety is not just compliance—it's a selling point.
- ✓ Radiological certification can be your next trust badge.



Use Our Services For :

- ✓ Advanced instrumentation (Gamma Spectrometry, LED Fluorimeter, Alpha and Beta counters)
- ✓ Fast turnaround time
- ✓ Expert team of highly qualified & trained Radioanalysts
- ✓ NABL-aligned quality systems
- ✓ Customized testing packages

9717686925, 8920723168

aumnamah.testing@gmail.com, info@aumnamahrai.com

<https://aumnamahrai.com>

Plot No-1 First Floor, Community Center, Lawrence Road Industrial Area, Delhi -110035

Our Solution: Comprehensive Radiological Testing

Core Services

- Gross Alpha & Beta Screening in water samples
- Radionuclide-specific analysis (Cs-134, Cs-137, Co-60, K-40 etc.)
- Process water & finished product testing
- Testing in All Kind of Water Uranium Contain

Ancillary Services

- All parameters in food and water testing other than radiological parameters will be supported through our partner NABL accredited laboratory

Business Benefits for You

Regulatory Confidence

Meet BIS, FSSAI and global safety norms with confidence

Export Readiness

Avoid shipment rejections and gain global acceptance

Brand Protection

Strengthen your positioning as a safe & premium brand

Competitive Advantage

Differentiate with advanced safety assurance



safety standards set by bodies like World Health Organization and national regulators such as Bureau of Indian Standards. BIS Standards like IS 10500 (drinking water standard), IS 14543: 2024 (packaged drinking water standard) and international guidelines require stringent gross alpha and gross beta activity limits in water. As per these standards, typical screening limits for Gross Alpha Activity in Drinking Water ≤ 0.1 Bq/L and Gross Beta Activity ≤ 1.0 Bq/L [4,5]. If these limits are found to be exceeded then it demands for a detailed radionuclide analysis for the sample (e.g., Ra-226, Cs-137).

Countries importing food products require strict compliance with international safety standards established by the Food and Agriculture Organization and the World Health Organization, particularly those developed under the Codex Alimentarius framework, to ensure food safety and facilitate fair trade [6]. Radiological certification builds global trust and enhances export competitiveness. For exporting companies strict compliance to international norms would avoid costly recalls and brand damage thus helps in retaining export approvals.

Laboratory Framework for testing of Food, water and Environmental samples



Radiological testing facilities in India span government laboratories such as Bhabha Atomic Research Centre or National Accreditation Board for Testing and Calibration Laboratories (NABL) accredited private labs. M/s Aumnamah Radioanalytical Laboratory LLP is an advanced Radioanalytical Testing facility, NABL accredited Laboratory, at Lawrence Road Industrial Area, Delhi. The laboratory specializes in accurate, reliable, and compliant radiochemical and radiation analysis to support industries, research institutions, healthcare, environmental monitoring, and Regulatory bodies. The laboratory is authorized by AERB (Atomic Energy Regulatory Board, Department of Atomic Energy).

The laboratory is equipped with state-of-the-art instrumentation, including a gamma spectrometer, alpha and

low-beta scintillation counting systems, LED fluorimeter, and radiation contamination-cum-survey meters. These advanced analytical tools offer high accuracy, precision, and low detection limits, enabling reliable quantification of radionuclides across a wide range of matrices. Aumnamah Radioanalytical Laboratory has a team of highly qualified (Ph.D. and M.Tech in Nuclear Science and Technology, M. Pharma in Quality Assurance etc.) well trained radiation scientists. This diverse academic expertise ensures strong technical competence in radiological analysis, instrumentation, regulatory compliance, and quality management systems, enabling the laboratory to deliver accurate, reliable, and standards-compliant results. The laboratory boasts to prepare technically validated and reliable test reports while strictly adhering to quality protocols, safety and regulatory norms. The laboratory assures complete confidentiality of client information and timely turnaround of test results.

Conclusion

Radiological testing of water is a critical yet underutilized component of food safety. Radiological analysis of food and water is a strategic investment with direct and indirect economic benefits. It strengthens public health systems, boosts trade, supports industrial growth, and enhances national resilience.

For laboratories and policymakers, integrating radiological testing into mainstream food and water safety frameworks is not only essential for safety but also a catalyst for economic development. Radiological testing is expected to become a standard requirement in food safety systems in coming age.

References:

- i. International Atomic Energy Agency. (1989). Measurement of radionuclides in food and the environment (Technical Reports Series No. 295). Vienna: IAEA.
- ii. Central Ground Water Board (CGWB), Ministry of Jal Shakti. Annual Ground Water Quality Report, 2025, Government of India.
- iii. Standard Protocol For Evaluation Of Environmental Transfer Factors Around NPP Sites by A.G. Hegde, P.C. Verma and D.D. Rao, E-023, Health Safety & Environment Group, BARC, 2008.
- iv. Bureau of Indian Standards (2012) IS 10500: Drinking water—Specification (2nd revision) New Delhi, BIS.
- v. Codex Alimentarius Commission (2019) General standard for contaminants and toxins in food and feed (CODEX STAN 193-1995), Rome, FAO/WHO.
- vi. Codex Alimentarius Commission, General Principles of Food Hygiene (CXC 1-1969) and related standards, FAO/WHO.



Solar Refrigerated Evaporative Cooling (SREC) System: A Sustainable and Cost-Effective Solution for Safe Storage of Mangoes in India



Dr. R. Pandiselvam

Subrata Mandal¹, Sangeeta Chopra¹, R. Pandiselvam¹, Devinder Dhingra²

¹Division of Agricultural Engineering, ICAR-Indian Agricultural Research Institute, Pusa Campus, New Delhi – 110012, India

²Krishi Anusandhan Bhawan II, Indian Council of Agricultural Research, New Delhi- 110012.

E-mail:- anbupandi006@gmail.com

1. Introduction

Mango (*Mangifera indica* L.) is a widely consumed tropical fruit known for its delightful taste, rich aroma, high nutritional value (Mandal, Chopra, et al., 2025). India leads global mango production, contributing nearly 50% of the total production with around 26.3 million metric tons produced in 2023–2024 across 2.3 million hectares, with major producing states including Uttar Pradesh, Andhra Pradesh, Karnataka, Bihar, and Maharashtra (NHB, 2023–24). Mango not only contributes significantly to nutritional security but also plays a vital role in rural livelihoods, agro-processing industries, and export earnings. Despite this strong production base, the mango value chain in India faces a critical challenge in the form of post-harvest losses. post-harvest losses remain a major challenge, with up to 15% of mangoes wasted due to improper handling, transportation, and storage (NABCONS, 2022). The economic impact of such losses is substantial, with total post-harvest losses in horticultural produce estimated at nearly Rs. 92,000 crores annually, highlighting a major inefficiency in the supply chain (NABCONS, 2022). Mango, being a highly perishable climacteric fruit, is particularly vulnerable to rapid deterioration. After harvest, mangoes continue to respire and ripen, leading to increased ethylene production, softening, moisture loss, and susceptibility to microbial spoilage. In the absence of proper storage, mango shelf life is typically limited to 5–7 days at ambient conditions ($34\pm3^{\circ}\text{C}$ and $60\pm5\%$ RH), resulting in significant quantitative and qualitative losses (Mandal, Chopra, et al., 2025). The recommended storage conditions for mangoes are 10 to 13°C temperature and 85 to 95% RH (Mandal, Chopra, et al., 2025). The lack of accessible and affordable cold storage infrastructure further exacerbates the problem. Conventional cold storage systems require high capital investment, continuous electricity supply, and sophisticated maintenance, making them unsuitable for small and marginal farmers who constitute a major portion of India's agricultural community. As a result, farmers are often forced into distress selling immediately after harvest, leading to reduced income and market inefficiencies (Mandal et al., 2025).

In this context, there is an urgent need for decentralized, energy-efficient, and cost-effective storage technologies that

can be deployed at the farm level. The Solar Refrigerated Evaporative Cooling (SREC) system, developed as a hybrid solution combining solar-powered refrigeration and evaporative cooling, offers a promising approach to address these challenges (Chopra et al., 2022). By utilizing renewable energy and simple cooling principles, SREC systems can maintain favourable temperature and humidity conditions required for safe storage of mangoes, thereby reducing post-harvest losses and improving economic returns.

2. Concept and Working Principle of SREC System

The Solar Refrigerated Evaporative Cooling (SREC) system is a hybrid storage technology that integrates solar-powered refrigeration with evaporative cooling to provide an energy-efficient and sustainable solution for storing perishable commodities like mangoes, especially in rural and off-grid areas (Chopra et al., 2023). Solar photovoltaic panels generate electricity to operate the refrigeration unit and auxiliary components, enabling active cooling during the day without grid dependency. The refrigeration system lowers the storage temperature through a vapor compression cycle, ensuring rapid cooling of freshly harvested produce. Simultaneously, evaporative cooling maintains high relative humidity by circulating water over wetted surfaces, reducing moisture loss and preserving fruit freshness. Additionally, thermal storage in the form of a water battery stores excess cooling during the day and releases it at night, ensuring continuous operation (Chopra et al., 2022, 2023).

3. Pusa Farm Sun Fridge (PFSF): A Practical SREC Model

The Pusa Farm Sun Fridge (PFSF) (Fig. 1), developed at ICAR–Indian Agricultural Research Institute (IARI), New Delhi, represents a practical and field-scale implementation of the Solar Refrigerated Evaporative Cooling (SREC) system designed for decentralized, farm-level storage of perishable commodities (Ranjan et al., 2026). The structure, typically measuring 10 ft x 10 ft x 10 ft with a storage capacity of approximately 2 tonnes, is equipped with a 5 kW rooftop solar photovoltaic system that powers a refrigeration unit along with auxiliary components such as a submersible pump for water circulation (Chopra et al., 2023). The system integrates both active refrigeration and passive evaporative cooling, where wetted surfaces enhance cooling efficiency by reducing

air temperature and increasing relative humidity inside the chamber. Insulation using materials such as autoclaved aerated concrete (AAC) blocks and Styrofoam panels minimizes heat gain, while a water-based thermal storage system helps maintain lower temperatures during non-solar hours. The PFSF operates entirely off-grid, eliminating dependence on conventional electricity, and maintains temperatures of 6–15°C and relative humidity of 85–100% (Fig. 2) which are suitable for extending the shelf life of mangoes (Mandal, Chopra, et al., 2025). Its modular design, low maintenance requirements, and energy efficiency make it particularly suitable for small and marginal farmers, enabling on-farm storage, reducing post-harvest losses, and allowing farmers to make informed marketing decisions rather than engaging in distress selling.



Fig. 1. Pusa Farm Sun Fridge constructed at mela ground, IARI

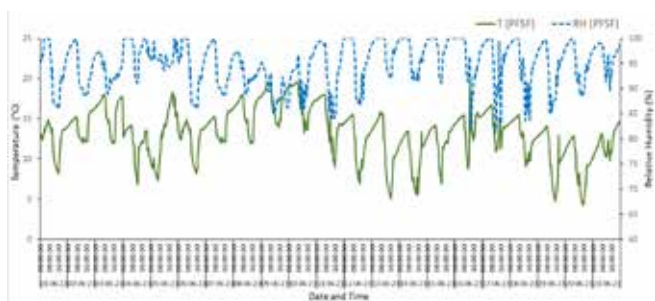


Fig. 2. Temperature (T) and RH profile inside PFSF during the storage study of mangoes

4. Performance Evaluation of SREC System for Mango Storage

The performance of the Solar Refrigerated Evaporative Cooling (SREC) system for mango storage has been validated through experimental observations under controlled and ambient conditions (Mandal, Chopra, et al., 2025). Mangoes stored in the SREC-based structure (Fig. 3), such as the Pusa Farm Sun Fridge, exhibited significantly improved quality retention compared to those stored under ambient conditions. The controlled environment of low temperature and high relative humidity effectively reduced physiological loss in weight (PLW), thereby minimizing moisture loss and maintaining fruit freshness. Additionally, the rate of softening was slower, indicating better firmness retention due to reduced metabolic activity. The color parameters (L^* , a^* , b^*) changed more gradually, reflecting delayed ripening and better visual quality. Furthermore, the increase in total soluble solids (TSS), which

is associated with ripening, was more controlled, suggesting a balanced biochemical transformation. The respiration rate of mangoes was also observed to be lower in the SREC system, indicating a slowdown in physiological processes. In contrast, mangoes stored under ambient conditions showed rapid deterioration, higher weight loss, faster softening, and reduced marketability. Overall, these findings confirm that the SREC system effectively maintains optimal storage conditions, thereby extending shelf life and preserving the physicochemical quality of mangoes (Mandal, Chopra, et al., 2025).



Fig. 3. Storage of mangoes inside PFSF in CFB boxes

5. Sustainability and Environmental Benefits

The Solar Refrigerated Evaporative Cooling (SREC) system offers significant sustainability and environmental benefits by utilizing renewable solar energy as its primary power source, thereby reducing dependence on conventional grid electricity and fossil fuels. This leads to a substantial decrease in greenhouse gas emissions and overall carbon footprint associated with cold storage operations. The integration of evaporative cooling further enhances energy efficiency by leveraging the natural process of water evaporation to lower temperature while maintaining high relative humidity, minimizing the need for energy-intensive refrigeration alone. Additionally, SREC systems contribute to resource conservation through low energy consumption and efficient water use. By extending the shelf life of mangoes and reducing post-harvest losses, these systems help prevent food wastage, which is a major contributor to environmental degradation. Overall, SREC technology supports sustainable agricultural practices, improves food security, and aligns with global and national goals for climate-resilient and energy-efficient food systems.

6. Analysis of Cost of Storage of mangoes inside Pusa Farm Sun Fridge

Assumptions

The assumptions for calculations are as under:

- The structure PFSF shall be constructed on farmers' land.
- Cost of PFSF: around Rs. 11,00,000 (Chopra et al., 2023)
- Average storage capacity of the unit: 2000 kg
- Annual repair and maintenance charges: 5 % of the cost of the structure.
- Depreciation: 10 % per annum
- Total number of days for which the structure is expected to be used in year: 300

Table 1. Annual Cost of PFSF under mango storage

Sr. No.	Item	Cost (in Rs)
a.	Depreciation (per annum)	Rs 110,000.00
b.	Repair and maintenance cost (5% of fixed capital per annum)	Rs 55,000.00
c.	Wages (Rs 783/day for 90 days in a year)	Rs 70,470.00
d.	Miscellaneous contingencies (per year)	Rs 8,000.00
	Total annual operating cost	Rs 2,43,470.00

Cost of storage of commodities in the structure =

$$\frac{\text{(Annual Cost (in Rs))}}{\text{Capacity (kgs) } \times \text{ No of days (pa)}}$$

Hence, cost of storage of mangoes in Pusa Farm Sun Fridge =
Rs 0.41 kg⁻¹ day⁻¹

In order to make comparison with cost of storage in commercial storage structures, rates were collected on phone from few operators and these are as under:

Table 2. Cost of storage – existing commercial cold storages in Delhi

Sr No	Cold Storage	Cost of Storage (Rs. kg ⁻¹ day ⁻¹)
1.	Cold Store A	Rs 0.50
2.	Cold Store B	Rs 0.30
3.	Cold Store C	Rs. 0.34
	Average cost	Rs 0.38

The cost of storing commodities in the PFSF structure was determined to be Rs. 0.41 per kilogram per day. Comparatively, the average cost of storing agricultural commodities in commercial cold storage facilities, available for rent, was approximately Rs. 0.38 per kilogram per day (Setiawan et al., 2021). This cost in commercial cold storage slightly undercuts the expense of storing perishables in the newly developed PFSF. It is worth noting that many commercial cold storage facilities in India have been established with government subsidies. However, in regions lacking access to such facilities or where they are situated at a considerable distance, transportation costs to these cold storage units can be significant. Additionally, the potential loss in market value due to deteriorating product quality could far exceed the cost of storage.

The PFSF storage structure is recommended for on-farm storage of agricultural commodities, particularly when considering government subsidies that further reduce storage costs. When used for high-value products, the cost of storage becomes even more affordable, leading to improved profitability. Another advantage of the PFSF is its ability to extend the on-farm storage duration, resulting in higher market prices. This storage facility can accommodate vegetables, fruits, and flowers for shorter durations, allowing small and marginal farmers to store their products briefly and sell them without quality deterioration, thus obtaining better market prices and reducing distress sales.

7. Policy Support and Government Initiatives

India's policy framework strongly supports the adoption of sustainable, decentralized cold storage technologies like the Solar Refrigerated Evaporative Cooling (SREC) system by aligning renewable energy promotion with post-harvest

infrastructure development. Schemes such as the Pradhan Mantri Kisan SAMPADA Yojana (PMKSY) and the Mission for Integrated Development of Horticulture (MIDH) provide financial assistance for establishing cold chain and on-farm storage facilities, where SREC structures like the Pusa Farm Sun Fridge can be effectively integrated as low-cost, farm-level pre-cooling and storage units. Simultaneously, the PM-KUSUM scheme and the National Solar Mission promote the use of solar energy in agriculture, directly supporting the solar-powered operation of SREC systems and reducing dependence on grid electricity. Initiatives like Operation Greens aim to stabilize supply chains and reduce post-harvest losses in perishable commodities, where SREC can play a critical role by enabling localized storage and reducing distress sales. Collectively, these policies create a conducive ecosystem for scaling SREC technology, as it fulfils national priorities of enhancing farmer income, minimizing food losses, promoting renewable energy adoption, and strengthening sustainable cold chain infrastructure in rural India.

8. Conclusion

The Solar Refrigerated Evaporative Cooling (SREC) system represents a transformative innovation in the field of post-harvest management of mangoes in India. By combining solar energy with evaporative cooling, it provides a sustainable, cost-effective, and decentralized storage solution. Experimental findings clearly demonstrate that SREC systems significantly improve mango shelf life, maintain quality attributes, and reduce post-harvest losses. With strong policy support and increasing focus on renewable energy, SREC systems have the potential to revolutionize India's cold chain infrastructure. Adoption of such technologies will not only enhance farmer income but also strengthen the vision of "India's Mango Magic – Sustenance through Policy and Innovation."

References

- National Horticulture Board (2023-24) Ministry of Agriculture & Farmer's Welfare. Retrieved on 5th April, 2026 https://nhb.gov.in/report_files/mango/mango.htm accreditation format.
- Nabard Consultancy Services, (NABCONS, 2022). Retrieved on 5th April, 2026 https://www.mofpi.gov.in/sites/default/files/study_report_of_post_harvest_losses.pdf
- Chopra, S., Mueller, N., Sharma, R., Dhingra, S., & Beaudry, R. M. (2022). Off-grid cooling and refrigeration: technical innovations for the storage of perishables for smallholder farmers. XXXI International Horticultural Congress (IHC2022): International Symposium on Postharvest Technologies to Reduce Food Losses 1364, 265–270.
- Chopra, S., Müller, N., Dhingra, D., Pillai, P., Kaushik, T., Kumar, A., & Beaudry, R. (2023). Design and performance of solar-refrigerated, evaporatively-cooled structure for off-grid storage of perishables. *Postharvest Biology and Technology*, 197, 112212.
- Mandal, S., Chopra, S., Rudra, S. G., Dhingra, D., Parray, R. A., Pandiselvam, R., Islam, M., Kinge, A., & Dharmender. (2025). Comparative assessment of mango quality under solar-refrigerated evaporative cooling, cold storage, and ambient conditions. *Journal of Food Measurement and Characterization*, 1–15.
- Mandal, S., Sharma, P. K., Chopra, S., & Dhingra, D. (2025). IoT-enabled solar powered temperature and relative humidity monitoring system for cold storages. *Journal of Agricultural Engineering*, 62(1), 95.
- Ranjan, J., Chopra, S., Mani, I., Dhingra, D., Parray, R. A., & Mandal, S. (2026). Effect of Circadian Variation of Temperature and Relative Humidity on Physicochemical Changes in Potatoes Stored in a Solar-Powered Battery-Less Cold Store. *Agricultural Research*, 1–12.
- Setiawan, E. A., Thalib, H., & Maarif, S. (2021). Techno-economic analysis of solar photovoltaic system for fishery cold storage based on ownership models and regulatory boundaries in Indonesia. *Processes*, 9(11), 1973.



Roadmap to help the mango industry dilute their present day crisis to a great extent

The global mango processing industry currently stands at a critical crossroads. Faced with the dual pressures of fluctuating crop yields and stagnating unsold stocks, processors are seeing their financial health strained to the limit.

The Problem: The "Pulp Trap"

Traditionally, the industry has relied almost exclusively on mango pulp production. However, this model carries inherent risks:

- **Limited Utilization:** Pulp is primarily restricted to the beverage sector, leaving processors vulnerable to shifts in consumer demand.
- **Waste Inefficiency:** During standard processing, up to 40% to 50% of the fruit (peels, seeds, and fibrous waste) is discarded.
- **Environmental & Financial Cost:** Discarding this biomass results in significant disposal costs and environmental degradation, representing a massive "lost" revenue stream.

The Solution: Embracing the Circular Economy

We believe the path to resilience lies in diversification through circularity. By transitioning from a "waste-generating" model to a "value-adding" one, the industry can dilute its market risk and unlock new profit centers.

1. Valorization of Processing By-Products

What was once considered waste is actually a treasure trove of high-value raw materials:

- **Peels:** Rich in pectin and polyphenols for the

pharmaceutical and food-thickening industries.

- **Seeds/Kernels:** A source of high-quality fats (mango butter) for cosmetics and specialized flour.
- **Fibrous Waste:** Can be converted into dietary fiber supplements or sustainable packaging materials.

2. Advanced Product Diversification: The Power of Powder

Instead of remaining tethered to liquid pulp—which is expensive to transport and has a shorter shelf life—converting mango pulp into High-Grade Mango Powder opens doors to:

- The bakery and confectionery industries.
- Nutraceuticals and health supplements.
- Instant beverage premixes with reduced logistics costs.

Partnering with T&I Global Ltd

Transforming a production line to support a circular economy requires more than just intent; it requires deep technical expertise.

T&I Global Ltd brings five decades of heritage and engineering excellence to the table. As a leading Indian entity, we offer end-to-end technology, systems, and solutions designed to:

- **Integrate** waste-processing modules into existing lines.
- **Optimize** yield through advanced extraction and drying technologies.
- **Future-proof** your facility against market volatility.

By turning environmental challenges into economic opportunities, T&I Global Ltd helps you ensure that every part of the mango contributes to your bottom line.

THINK DEHYDRATION. THINK T&I GLOBAL.



www.globaldryers.com



+91 9488168417



marketing@tiglobal.com



ENGINEERED DRYING SOLUTIONS FOR FOOD INDUSTRY

*Continuous Dehydration Plants for Vegetables,
Herbs, Spices, Granules, Soya, and more*

**DRYING & HEAT
GENERATION
SYSTEM**



**CUSTOMISED
DRYING
SOLUTIONS**



**PRODUCT
HANDLING**

**PREPARATORY
LINE**



ISO 9001:2015
Certified



Over 50 Years of
Industry Expertise



100+ Global
Installations



CE
Certified

THINK DEHYDRATION. THINK T&I GLOBAL.



www.globaldryers.com



+91 9488168417



marketing@tiglobal.com

Sustainable Management and Environmental Implications of Mango Processing Waste

Swosti S. Das and Ningthoujam Samarendra Singh

Division of Post-harvest Management, ICAR-Central Institute for Subtropical Horticulture,
Lucknow (UP) - 226101, India

The mango (*Mangifera indica* L.), often hailed as the "king of fruits," is a critical component of the global horticultural economy. However, the industry faces a significant sustainability challenge, approximately 40-50 % of the fruit's total weight is discarded as waste during industrial processing. However, mango processing generates large quantities of waste, including peels, seeds, kernels, and spoiled fruits. At the same time, mango by-products are rich in valuable bioactive compounds including polyphenols, carotenoids, dietary fiber, pectin, starch, and antioxidants. Sustainable utilization and valorization of mango waste through biotechnological and circular bioeconomy approaches can reduce environmental burden while generating value-added products for food, pharmaceutical, cosmetic, and packaging industries. This article discusses the environmental impact of mango waste, sustainability concerns, and modern approaches for mango waste management and valorisation.



Fig 1 *Mangifera indica* - Mango

1. Introduction

Mango is popularly known as the "King of Fruits" because of its excellent flavour, aroma, nutritional quality, and commercial importance. India is the largest producer of mango in the world, contributing more than 25 million tonnes annually. Mango is processed into several commercial products such as pulp, nectar, juice, puree, jam, chutney, dehydrated slices, canned products, and beverages. During these processing operations, enormous quantities of waste are generated in the form of peels, stones, kernels, damaged fruits, and fibrous residues. Mango processing waste constitutes approximately 45-50% of the total fruit mass.

Traditionally, these by-products are discarded as solid waste without proper treatment. Such disposal practices create major environmental and public health concerns.

However, scientific studies have revealed that mango waste contains several nutritionally and functionally important compounds such as phenolics, flavonoids, carotenoids, pectin, starch, and dietary fiber. Therefore, sustainable utilization of mango waste has become an important research area under the concepts of circular economy, zero waste processing, and sustainable food systems.

2. Environmental Impact of Mango Waste

2.1 Solid Waste Generation

The mango processing industry generates large amounts of organic waste during peeling, slicing, pulping, and seed removal operations. Mango peel alone contributes about 15-25% of the fruit weight, whereas mango stones account for 20-40% depending upon cultivar and processing method. During peak mango season, accumulation of untreated waste near processing facilities creates disposal challenges and environmental pollution.

2.2 Greenhouse Gas Emissions

When mango waste is dumped in open landfills, anaerobic decomposition occurs, resulting in methane (CH₄) and carbon dioxide (CO₂) emissions. Methane is a potent greenhouse gas with a global warming potential much higher than carbon dioxide. The uncontrolled degradation of fruit waste contributes significantly to climate change and atmospheric pollution.

2.3 Soil and Water Pollution

Leachates generated from decomposing mango waste contain high organic load, sugars, acids, and suspended solids. These substances may infiltrate into soil and groundwater systems, causing

nutrient imbalance, microbial contamination, and reduction in soil quality. Disposal of mango processing effluents into water bodies increases biological oxygen demand (BOD) and chemical oxygen demand (COD), adversely affecting aquatic ecosystems.

3. Sustainability Aspects of Mango Waste Utilization

Sustainable management of mango waste focuses on reducing waste generation, maximizing resource efficiency, and converting waste into economically valuable products. The concept aligns with Sustainable Development Goals (SDGs), particularly responsible consumption and production, climate action, and environmental sustainability.

3.1 Circular Bioeconomy Approach

The circular bioeconomy model emphasizes recycling and reutilization of agro-industrial residues into useful materials. Mango waste can be transformed into food ingredients, nutraceuticals, biodegradable packaging materials, biofuels, and animal feed. This approach minimizes environmental burden while generating additional income for industries and farmers.

3.2 Resource Recovery

Mango by-products are rich in recoverable compounds including:

1. Polyphenols and flavonoids
2. Dietary fiber
3. Pectin
4. Carotenoids
5. Mango kernel fat
6. Starch and bioactive compounds

These compounds possess antioxidant, antimicrobial, anti-inflammatory, and nutraceutical properties. Extraction and utilization of these compounds promote sustainable resource utilization and reduce dependence on synthetic additives.

3.3 Waste-to-Wealth Concept

The “waste-to-wealth” strategy converts low-value agro-waste into commercially important products. Mango peel powder, mango kernel flour, pectin, bioactive extracts, edible coatings, and biodegradable films are examples of value-added products developed from mango waste. Such innovations improve sustainability and profitability of mango processing industries.

Table 1: Nutritional composition of Mango Peel and Mango Seed Kernel

Composition	Mango Peel	Mango Seed kernel
<i>Proximate</i>		
Water (g)	72.5	9.1
Energy (kCal)	-	327
Protein (g)	3.6	6.61
Lipid (g)	2.2	9.4
Carbohydrate (by difference) (g)	28.2	18.2
Total dietary fiber (g)	40-72.5	70
Sugar (g)	25	2.8
<i>Vitamins</i>		
Vitamin C (total ascorbic acid) (mg)	18-257	17
Thiamin (mg)	-	0.08
Riboflavin (mg)	-	0.13
Niacin (mg)	-	0.19
Vitamin A, retinol activity equivalents (mg)	100	-
Vitamin E (α-tocopherol) (mg)	0.25–0.59	1.3
Vitamin K (phylloquinone) (mg)	-	59
<i>Minerals</i>		
Calcium (Ca) (mg)	150	450
Iron (Fe) (mg)	40.6	11.9
Magnesium (Mg) (mg)	100	100
Phosphorus (P) (mg)	-	140
Potassium (K) (mg)	75	365
Sodium (Na) (mg)	50	150

Source: Gupta et al. (2022)

4. Valorization and Utilization of Mango Waste

4.1 Mango Peel Utilization

Mango peel is rich in dietary fiber, pectin, polyphenols, carotenoids, and antioxidants. Mango peel powder has been successfully incorporated into bakery products, noodles, biscuits, cakes, and functional foods. It improves antioxidant activity and enhances nutritional quality.

Mango peel extracts are also used in active food packaging systems. Incorporation of peel extracts into biodegradable films improves antioxidant capacity and reduces permeability of packaging materials.

4.2 Mango Seed Kernel Utilization

Mango seed kernels contain starch, proteins, and fats rich in stearic and oleic acids. Mango kernel fat has physicochemical properties similar

to cocoa butter and can replace cocoa butter in chocolate industries. Mango kernel flour has also been utilized in bakery products to improve energy value and shelf life.



Fig 2. Raw mango with Visible Kernel

4.3 Pectin Extraction

Mango peel is an excellent source of pectin, a natural polysaccharide widely used as a gelling and stabilizing agent in food industries. Modern extraction methods such as ultrasound-assisted extraction, microwave-assisted extraction, and enzymatic extraction improve pectin recovery and sustainability.

4.4 Biofuel and Biogas Production

Organic mango waste can be converted into bioethanol, biogas, and compost through fermentation and anaerobic digestion. These renewable energy sources reduce dependence on fossil fuels and contribute to carbon-neutral waste management systems.

4.5 Biodegradable Packaging Materials

Starch extracted from mango seed kernels has potential for developing biodegradable films and edible coatings. These eco-friendly materials can replace petroleum-based plastics and reduce plastic pollution.

5. Biotechnological Approaches for Sustainable Mango Waste Management

5.1 Fermentation Technology

Fermentation using beneficial microorganisms enhances the nutritional and antioxidant properties of mango waste. Fermented mango residues can be used in animal feed, probiotic products, and organic fertilizers. Fermentation also improves release of bound phenolic compounds.



Fig 3 : Mango Fermentation

5.2 Enzymatic Treatment

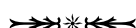
Enzymatic technologies involving pectinases, cellulases, and amylases improve extraction efficiency of valuable compounds from mango peels and kernels. These eco-friendly methods reduce chemical usage and energy consumption during processing.

5.3 Green Extraction Technologies

Sustainable extraction techniques such as ultrasound-assisted extraction, supercritical CO₂ extraction, and microwave-assisted extraction are gaining importance due to their lower solvent requirement, reduced processing time, and improved recovery of bioactive compounds.

6. Conclusion

Mango processing waste is both an environmental challenge and a valuable bioresource. Improper disposal of mango waste contributes to greenhouse gas emissions, soil and water pollution, and public health problems. However, mango peels and kernels are rich sources of bioactive compounds, dietary fiber, pectin, starch, and antioxidants with wide industrial applications. Sustainable valorization of mango waste through biotechnological and circular economy approaches can transform agro-industrial waste into valuable products such as functional foods, nutraceuticals, biodegradable packaging materials, and biofuels. Therefore, adoption of sustainable mango waste management strategies is essential for environmental protection, economic development, and long-term sustainability of the mango processing industry.



Rethinking Mango Waste: Sustainable Utilization of Mango Waste

Dr. Lekshmi S G^a, Dr. Shruti Sethi^b, Dr. Ajmal S^c, Dr. Anagha P K^a

^aDepartment of Agriculture and Farmers Welfare, Kerala

^bDivision of Food Science and Postharvest Technology, ICAR-Indian Agricultural Research Institute, New Delhi

^cICAR-Central Island Agricultural Research Institute, Andaman Nicobar

E-mail:- lekshmishaji1994@gmail.com



Dr. Lekshmi

Introduction

Mango (*Mangifera indica* L.) is celebrated globally for its flavor, nutritional richness, and economic importance in our country. As one of the most widely cultivated tropical fruits, mango plays a vital role in the livelihoods of farmers, food processors, and exporters across Asia, Africa, and Latin American countries. However, behind this success lies a less-discussed challenge, the generation of large quantities of mango waste. During processing into products such as pulp, juice, nectar, and dried slices, approximately 35–60% of the fruit is discarded as peel, seed, and kernel. Traditionally treated as waste, these byproducts often accumulate in large volumes, posing environmental and economic concerns. In recent years, however, a paradigm shift toward sustainability and circular economy principles has led to renewed interest in mango waste utilization as a valuable resource.

Mango Waste Generation: Scale and Nature of the Problem

The magnitude of mango waste generation is directly linked to global production and industrial processing. Countries like India, which is the largest producer of mango, generates approximately 20 million tons of mango waste annually from non-consumed fractions (peels and seeds), with processing industries generating roughly 40-50% solid waste per raw mango. Processing industries, particularly pulp and juice manufacturing units, are major contributors to this waste stream.

Mango waste is primarily composed of:

- Peel (epicarp): Constituting about 15–20% of the fruit weight
- Seed (endocarp + kernel): Making up around 20–25%
- Kernel (inside the seed): About 10–15%

These residues are highly perishable due to their high moisture content and rich organic composition. If left unmanaged, they rapidly undergo microbial degradation, leading to environmental issues.

Environmental Impact of Mango Waste

Improper disposal of mango waste poses several ecological challenges:

- **Solid waste accumulation:** Large volumes of organic residues from processing industries create disposal issues, especially during peak harvest seasons. Processing units, such as those in the Krishnagiri district, generate approximately 5,000-7,000 tonnes of solid waste annually.
- **Greenhouse gas emissions:** This agro-industrial waste has a high biological oxygen demand and pollutes the environment. Decomposition of mango waste under anaerobic conditions releases methane, a potent greenhouse gas.
- **Water and soil pollution:** Leachates from decomposing waste can contaminate soil and groundwater, affecting surrounding ecosystems.
- **Odor and pest issues:** Accumulated waste attracts insects and rodents, leading to sanitation concerns.

These impacts highlight the urgent need for sustainable waste management strategies in mango-producing regions.

Fruit Waste as an Untapped Resource

Citrus and mango wastes share a common challenge: despite being rich in valuable compounds, they are often underutilized due to lack of industrial-scale recycling and reuse. The inability or reluctance to process these residues leads to unnecessary waste generation, ultimately depleting environmental capital. This represents not only an ecological issue but also a missed economic opportunity. Mango waste, in particular, stands out due to its rich biochemical composition and versatility. Among its components, mango peel has been extensively studied and utilized as a source of high-value compounds.

Bioactive Compounds in Mango Waste

Mango peel is not merely a fibrous waste material, it is a reservoir of bioactive phytochemicals with significant health-promoting properties. Studies have identified a wide range of compounds in mango peel, mango kernels and mango seeds including polyphenols, flavonoids and these compounds are present in substantial quantities

Mango waste	Bioactive compounds identified
Mango peel	vanillic aldehyde, caffeic acid, oleanolic acid, chlorogenic acid, gallic acid, procyanidin B2

Mango kernal	iriflophenone glucoside, maclurin C-glucoside, maclurin digalloyl glucoside, mangiferin, 5-galloyl quinic acid, trigalloyl glucose, thiazole
Mango pulp residue	gallic acid, mangiferin, quercetin glycosides, gallotannins, ellagitannins

Mango Peel as a Source of Pectin

One of the most important applications of mango peel is its use as a source of pectin, a natural polysaccharide widely used as a gelling agent in food products such as jams, jellies, and confectionery. Scientific researches established mango peel as a viable and nutritious source of pectin. Mango peels are a sustainable source of pectin, with yields varying between 20-25%. Mango peel is a rich source of high methoxyl pectin (HMP), typically yielding pectin with a degree of esterification (DE) between 66-79%. Pectin derived from mango peel is particularly valuable because it is associated with dietary fiber benefits, including improved digestion, cholesterol reduction, and glycemic control. Compared to conventional sources like citrus peel, mango peel offers an alternative raw material that can be exploited, especially in mango-producing regions.

Antioxidant Properties of mango waste

Mango waste, particularly the peel and seed kernel, contains a wide range of antioxidants such as polyphenols, flavonoids, carotenoids, and vitamins. Among these, phenolic compounds play a crucial role in neutralizing free radicals and preventing oxidative stress. Key bioactive compounds identified in mango waste include mangiferin, quercetin, gallic acid, and various flavonol glycosides.

Mango peel is especially rich in flavonoids, gallotannins, xanthone C-glycosides, and benzophenone derivatives. These compounds exhibit strong radical scavenging activity by donating hydrogen atoms or electrons to unstable free radicals, thereby stabilizing them. The antioxidant capacity of mango waste is commonly evaluated using assays such as DPPH (1,1-diphenyl-2-picrylhydrazyl), ABTS, and FRAP, where mango peel extracts consistently show high scavenging efficiency.

Flavonoids present in mango waste are known to interact with reactive species such as peroxynitrite and hydroxyl radicals. Their ability to chelate metal ions and inhibit lipid peroxidation further enhances their antioxidant potential. Studies have shown that certain mango cultivars, such as Tommy Atkins, possess higher concentrations of polyphenolics in their peels, making them particularly suitable for antioxidant extraction.

The antioxidant properties of mango waste are significant not only for human health but also for food systems. Natural antioxidants derived from mango peel can be used to delay oxidation in food products, thereby extending shelf life and reducing the need for synthetic preservatives.

Antimicrobial Properties

In addition to antioxidant activity, mango waste exhibits notable antimicrobial properties against a range of pathogenic microorganisms. Extracts from mango peel and seed kernel have been shown to inhibit the growth of bacteria such as *Escherichia coli*, *Staphylococcus aureus*, and *Salmonella* species, as well as certain fungi.

The antimicrobial activity is primarily attributed to phenolic compounds, tannins, and flavonoids, which can disrupt microbial cell membranes, interfere with enzyme activity, and inhibit nucleic acid synthesis. Mangiferin, a bioactive compound found in mango peel and kernel, has been widely reported for its antimicrobial and anti-inflammatory effects.

Tannins present in mango waste contribute to antimicrobial action by forming complexes with proteins in microbial cell walls, leading to cell lysis. Similarly, flavonoids can alter membrane permeability and inhibit energy metabolism in microorganisms.

Recent studies have also demonstrated that mango waste extracts can act synergistically with conventional antimicrobial agents, enhancing their effectiveness. This opens up possibilities for developing natural preservatives and alternative therapeutic agents, especially in the face of rising antibiotic resistance.

Applications in Food, Industry, and Agriculture

1. Food Industry

Mango peel powder can be used to fortify bakery and functional foods with dietary fiber and antioxidants. Extracts rich in polyphenols can serve as natural preservatives.

2. Pharmaceutical and Nutraceutical Sector

Bioactive compounds such as mangiferin and flavonoids are being explored for their anti-inflammatory, antimicrobial, and anticancer properties.

3. Industrial Uses

Pectin extraction, biodegradable film production, and enzyme generation are key industrial applications of mango waste. Inclusion in products for skin protection and anti-aging

4. Agricultural Applications

Mango waste can be composted into organic fertilizers or processed into animal feed, contributing to sustainable farming practices.

Environmental Significance of Mango Waste Utilization

The improper disposal of mango waste has several environmental consequences, including methane emissions, soil and water contamination, and increased pest activity. By converting mango waste into useful products, these impacts can be significantly reduced.

Key Environmental Benefits:

- Reduction in landfill waste
- Lower greenhouse gas emissions
- Prevention of soil and water pollution
- Improved waste management efficiency

Utilization strategies such as composting, bioenergy production, and extraction of bioactive compounds contribute to a more sustainable agro-industrial system.

Challenges in Mango Waste Utilization

Despite its potential, several challenges hinder large-scale adoption:

1. Seasonal Availability

Mango waste is generated in large quantities over a short period, making continuous processing difficult.

2. High Moisture Content

The high moisture content of mango waste leads to rapid spoilage, requiring efficient preservation methods.

3. Technological Constraints

Advanced technologies for extraction, processing, and product development are not widely available or affordable in many regions.

4. Lack of Awareness

Limited awareness among farmers, processors, and policymakers restricts the adoption of waste utilization practices.

5. Regulatory Issues

Standardization, quality control, and regulatory approvals

are necessary for the commercialization of products derived from mango waste.

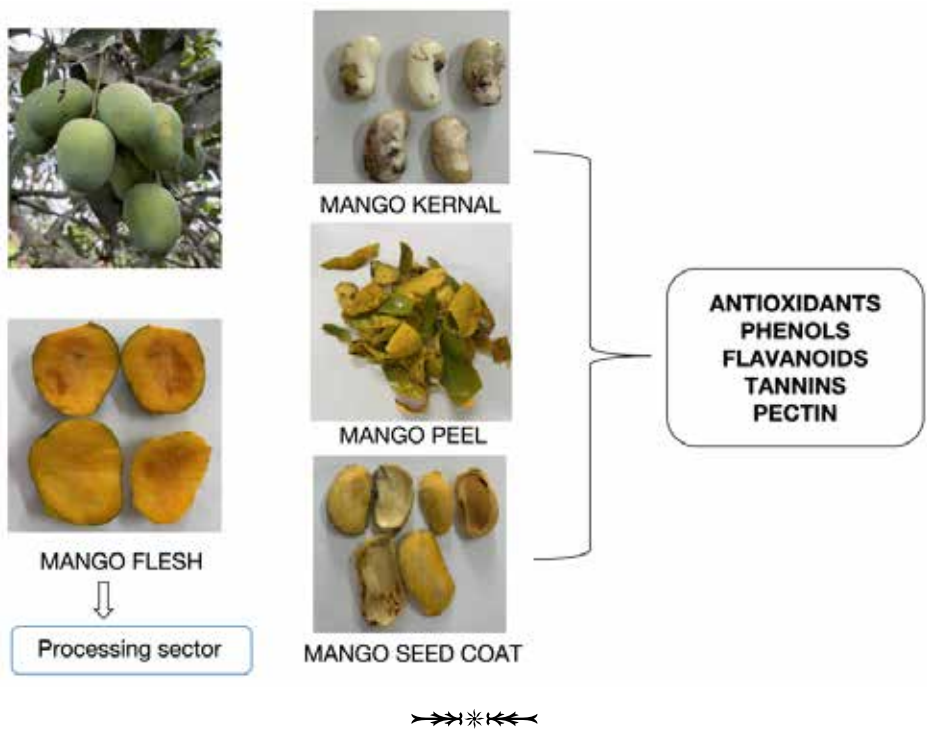
Future Prospects and Innovations

The future of mango waste utilization lies in innovation and integration

1. **Biorefinery approach:** Developing integrated systems that extract multiple products from mango waste
2. **Green extraction technologies:** Use of environmentally friendly methods such as supercritical fluid extraction and ultrasound-assisted extraction
3. **Nanotechnology:** Development of nano-based delivery systems for bioactive compounds
4. **Policy support:** Government initiatives and incentives to promote waste valorization
5. **Public-private partnerships:** Collaboration between research institutions and industries

Conclusion

Mango waste utilization represents a compelling opportunity to transform an environmental liability into a valuable resource. The presence of pectin, polyphenols, flavonoids, and other bioactive compounds in mango peel and seed underscores its potential in food, pharmaceutical, and industrial applications. When viewed through the lens of sustainability, mango waste is not merely a byproduct but a critical component of a circular bioeconomy. By adopting innovative technologies and promoting awareness, it is possible to reduce environmental impact while unlocking the economic value of mango waste. Such approaches will be essential for building a more sustainable and resource-efficient future in the agro-processing sector.



Valorization of Mango Processing Waste: Unlocking the Potential of Mango Kernel Fat for Sustainable Lipid Applications

Monika Chand^a and Rajni Chopra^b

^aAssistant Professor, Department of Food Science and Technology, National Institute of Food Technology Entrepreneurship and Management, Kundli, Haryana, India

^bProfessor and Head, Department of Food Science and Technology, National Institute of Food Technology Entrepreneurship and Management, Kundli, Haryana, India



Rajni Chopra

Mango seed kernel fat (MKF), derived from the cotyledons of *Mangifera indica* L., is a underutilized fat abundantly available as a byproduct of India's large-scale mango processing industry. MKF has high content of stearic acid (31-57%), oleic acid (38-53%), and major 1,3-distearoyl-2-oleoyl-glycerol (StOSt) and triacylglycerol content (40-55%). It has a beneficial minor lipid content, comprising phytosterols (1,500-4,000 mg/kg), tocopherols (150-300 mg/kg), and trace amounts of squalene. This review explores the physicochemical composition of MKF, critically evaluates extraction methodologies ranging from conventional solvent and mechanical pressing to emerging techniques such as supercritical CO₂, ultrasound-assisted, and enzymatic extraction, and discusses fractionation approaches for quality improvement. Industrial applications of MKF across the food, cosmetic, and nutraceutical sectors are also reported, along with its sustainability advantages as a waste-derived lipid resource. Despite its well-documented functional potential, the commercial valorization of MKF in India remains disproportionately low given the scale of available raw material. Addressing key gaps in varietal compositional data and process standardization, which are essential to establishing MKF as a strategically significant domestic specialty lipid.

1. Introduction

India is the largest producer of *Mangifera indica* (Mango) worldwide, with an annual output of (>20 million metric tonnes) cultivated across 2.5 million hectares (FAO, 2023). The rapid increment of the organized mango processing sector including jam, juice, jellies clarification, pulp extraction, nectar production, and pickle manufacturing led to generation of huge volume of agro-industrial waste. Among all these wastes, mango seeds are major underutilized fraction despite their considerable valorization potential. The mango seed kernel consists of 9 to 18% of the fresh fruit weight varying with different cultivar and agroclimatic conditions (Jahurul et al., 2015), a unique rich source of lipids. The mango kernel fat (MKF), obtained from the cotyledons of mango seeds, exhibits a specific fatty acid composition as compared to other conventional tropical vegetable fats. It has high proportion of stearic acid (31-57%), oleic acid (38-53%), and low polyunsaturated fatty acids (<5%), leading to higher oxidative stability (Nadeem et al., 2016; Solís-Fuentes et al., 2010).

The triacylglycerol (TAG) level of MKF is majorly composed of 1,3-distearoyl-2-oleoyl-glycerol (StOSt), that constitutes approximately 40 to 55% of the total TAG profile. This composition aligns with that of cocoa butter, qualifying MKF as Class V cocoa butter equivalent under Codex Alimentarius Commission standards (Lipp & Anklam, 1998; Garti & Widlak, 2012). EU Directive 2000/36/EC further allow MKF among six permitted vegetable fats for incorporating in chocolate formulations thereby establishing a clear regulatory framework for its application in international confectionery market. Other than its functional role in food systems, MKF contains various bioactive compounds proving its significant nutraceutical and cosmetic relevance. These

include phytosterols (1,500 to 4,000 mg/kg), tocopherols (150-300 mg/kg), and trace levels of squalene (Normen et al., 2000; Lutterodt et al., 2011; Makhlof et al., 2015). Phytosterol, majorly β -sitosterol, are correlated with reduced intestinal cholesterol absorption and lowering low-density lipoprotein (LDL) cholesterol (Kris-Etherton et al., 2005). Tocopherols contribute strongly to oxidative stability through chain-breaking antioxidant mechanisms, enhancing shelf life and functional quality (Waraho et al., 2011).

The global specialty lipids market that has cocoa butter equivalent functionality like shea butter, illipe butter, and kokum butter are well-known commercial commodities. However, MKF has received comparatively less industrial attention within India, despite irrespective of its superior physicochemical and functional properties. Current commercial utilization is majorly driven by international markets, specifically in Europe and Japan (Mwaurah et al., 2021). This review aims to explore MKF: its compositional attributes, fatty acid profile, extraction technologies, pretreatments, fractionation technologies, and industrial applications. Additionally, it explores the effective valorization of MKF within a sustainable, economically viable domestic value chain.

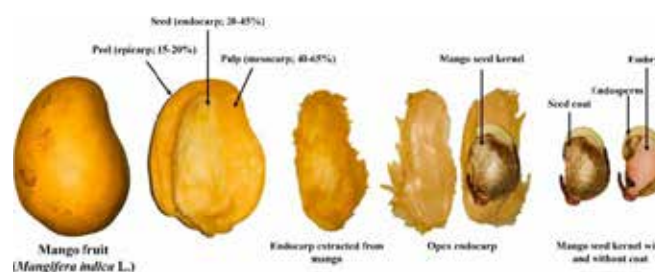


Fig1. Parts of Mango

2. Physicochemical Composition of Mango Seed Kernel Fat

2.1 Proximate Composition of Mango seed Kernel

Mango kernel also known as “gutli” or mango seed kernel: it is the inner edible embryonic cotyledon part surrounded by fibrous rich outer testa. MSK constitutes 20% of the whole mango fruit (fresh weight basis). The mango seed kernel (husk-removed cotyledon) has fat content 7-18% (varietal range; Alphonso, Totapuri, and Kesar varieties tend to be on the higher end), starch 40-50%, protein 5-8%, crude fibre 2-5%, and ash 1-3%. Mango seeds and peel are rich sources of proteins, fats, carbohydrates, and various specific bioactive compounds. The mango seed kernel contains approximately 53.34-76.81% carbohydrates, 5.20-10.48% proteins, 9.84-18.0% fat/oil, and 0.26-10.60% crude fiber. Table 1 represents proximate composition of mango seed kernel. Table 1 represents the Proximate composition of MSK based on different country region

Table 1: Proximate composition of mango seed kernel

Variety (country of origin)	Crude protein (%)	Fat/oil (%)	Ash (%)	Crude fiber (%)	Carbohydrates (%)	References
Nigeria	10.48	18.00	1.14	2.48	64.57	Ifesan (2017)
Kibangou Nigeria	6.36	13.00	3.20	2.02	32.24	Nzikou et al. (2010)
Egypt	7.53	11.0	2.77	2.45	69.20	Abd-Elaziz (2018)
Kenya	6.74-9.2	9.84-13.08	2.64-3.71	1.78-2.89	72.86-75.92	Mutua et al., 2017
India	6.8	10.5	2.68	2.00	76.81	Kaur et al., 2004

2.2 Fatty Acid Profile

The MSK consists of 7.8 to 13.6 5 oil on dry weight basis varying as per different variety. MKF has unusual fatty acid composition for a tropical vegetable fat: a very high content of saturated 44- 48% majorly stearic acid (C18:0) combined with unsaturated fatty aid 52-56% majorly monounsaturated oleic acid (C18:1) with less than 5% PUFA content. MSK oil has no trans fatty acids at room temperature and is therefore less prone to oxidation, leading to higher stability (Nadeem et al., 2016; Solís-Fuentes et al., 2010). This unique fatty acid profile give two major properties to MSK. First, MKF has been shown to maintain physicochemical stability for up to 24 months under ambient storage conditions without antioxidant treatment, due to the lack of PUFAs, significantly reducing sensitivity to oxidative rancidity (Nadeem et al., 2016; Waraho et al., 2011). Second, the balanced saturated/monounsaturated architecture gives it a quick, complete melting transition near

body temperature and a hard, brittle consistency at room temperature, which is similar to the quality of cocoa butter substitutes (Lipp & Anklam, 1998). The symmetric triglyceride 1,3-distearoyl-2-oleoyl-glycerol (StOSt), which normally accounts for 40-55% of the overall TAG fraction, dominates MKF at the molecular level (Jahurul et al., 2015; Solís-Fuentes et al., 2010). The asymmetric species 1-stearoyl-2,3-dioleoyl-glycerol (StOO) and trace amounts of 1,3-dioleoyl-2-stearoyl-glycerol (OStO).

Table 2: Comparative fatty acid profile of MKF against cocoa butter and palm oil.

Parameter	MKF	Cocoa Butter	Palm Oil	Shea Butter	Reference
Palmitic acid (C16:0)	3-9%	5-11%	24-30%	10-14%	Jahurul et al. (2015)
Stearic acid (C18:0)	31-57%	32-36%	2-5%	45-50%	Lipp & Anklam (1998)
Oleic acid (C18:1)	38-53%	52-57%	55-65%	35-40%	Okullo et al. (2010)
Linoleic acid (C18:2)	1-5%	3-6%	12-18%	6-9%	Makhlouf et al. (2015)
Linolenic acid (C18:3)	Trace	<1%	Trace	<1%	Solis-Fuentes et al. (2010)
Arachidic acid (C20:0)	0-2%	1-3%	Trace	<1%	Becker et al. (2021)
StOSt fraction (TAG)	40-55%	75-80%	N/A	55-65%	Kaleemullah et al. (2005)
Melting point (°C)	34-38	32-35	17 (liquid)	33-36	Nadeem et al. (2016)
Iodine value (g I ₂ /100g)	38-50	35-40	52-58	36-44	Codex 2010
Saponification value	185-198	192-196	196-205	185-195	FSSAI (2023)

2.3 Minor Lipid Constituents

Beyond these fatty acids various minor lipid constituents including stigmasterol, campestral, avaanastrol are present in MKF leading to its biological activity. Another important constituents are tocopherols (predominantly alpha- and gamma-tocopherol) at 150-300 mg/kg extensively contributing to the oxidative stability of MKF and its therapeutic properties.

Phytosterols are important minor fraction, having total phytosterol concentrations ranging from 1,500-4,000 mg/kg in MKF (Normen et al., 2000; Zheng et al., 2018). Among all three phytosterols present namely beta-sitosterol, stigmasterol, campesterol. β-Sitosterol is the majorly found species ranging from 800-2,000 mg/kg, followed by stigmasterol 300-900 mg/kg and campesterol (200-500 mg/kg). β-Sitosterol is strongly related to reducing cholesterol absorption, resulting in LDL chokesterol reduction 8 to 15% by 1.5 -3 g/ day intake (Kris-Etherton et al., 2005; Normen et al., 2000). Another important minor constituent is tocopherols. Tocopherols, basically α- and γ-tocopherol, are reported to be in the range of

150-300 mg/kg in unrefined MKF (Lutterodt et al., 2011; Kaleemullah et al., 2005). These tocopherols, also widely known as vitamin E homologs, significantly enhance the oxidative stability of MKF through chain-breaking antioxidant mechanisms, scavenging peroxy radicals and thus interrupting lipid autoxidation propagation (Waraho et al., 2011; Zhang et al., 2010). Conventional hexane extraction followed by alkali refining and deodorization substantially depletes the tocopherol fraction; cold-pressed and SC-CO₂-extracted MKF retains superior tocopherol profiles.

Squalene is found to be in trace amount (<50 mg/kg) in MKF (Makhlouf et al., 2015). Polyphenolic co-extractants such as mangiferin and gallic acid may co-extract with the lipid fraction under specific solvent systems (Masibo & Han, 2008; Soong & Barlow, 2004). Phospholipids (0.1-0.5%) serve as natural emulsifiers in crude MKF but are conventionally removed during degumming in refining operations (Bhalerao et al., 2003). The minor lipid composition of MKF is highly sensitive to extraction methodology and post-extraction refining. Cold-pressed and SC-CO₂-extracted MKF retains superior minor lipid profiles compared to solvent-extracted and subsequently deodorized products.

Table 3. Minor lipid constituents of mango kernel fat (MKF): typical concentrations, biological functions, and literature references.

Minor Constituent	Typical Concentration	Biological / Functional Role	References
Tocopherols (α + γ)	150-300 mg/kg	Antioxidant; radical scavenging; oxidative shelf-life extension	Lutterodt et al. (2011); Kaleemullah et al. (2005)
β-Sitosterol	800-2,000 mg/kg	LDL cholesterol reduction; competitive inhibition of cholesterol absorption in intestinal micelles	Normen et al. (2000)
Stigmasterol	300-900 mg/kg	Anti-inflammatory; oestrogen-modulating; hepatoprotective effects in animal models	Okullo et al. (2010); Patel & Shah (2017)
Campesterol	200-500 mg/kg	Synergistic LDL-lowering with β-sitosterol; reported PPAR-γ activation	Normen et al. (2000)

Squalene	Trace (<50 mg/kg)	Skin emollient; anti-tumour activity reported in murine models; preserved in cold-pressed MKF only	Makhlouf et al. (2015)
Mangiferin (co-extracted)	Variable; <100 mg/kg in lipid fraction	Antioxidant; anti-diabetic; anti-inflammatory; antiviral — partitions predominantly to aqueous phase	Masibo & Han (2008); Sogi et al. (2005)
Gallic acid (co-extracted)	Variable; trace in lipid fraction	Potent antioxidant; antimicrobial against Gram-positive bacteria	Masibo & Han (2008)
Phospholipids	0.1-0.5%	Natural emulsifiers; relevant for food texture; removed during degumming	Bhalerao et al. (2003)



Fig 2. Mango Butter

3. Extraction Methods

Extraction is an important step that determines fat yield, quality, application, regulatory compliance, and process economics. Unlike liquid vegetable oils, MKF's solid consistency at ambient temperature requires either elevated-temperature processing or solvent assistance. Mango kernels, contain approximately 8-15% fat with a dense cellular matrix consists of starch, proteins, and fibrous materials. Conventional extraction techniques, such as mechanical and solvent extraction. Mechanical extractions are ecofriendly, solvent free, simpler methods but results in lower oil yield (Gurjar and Raj, 2022). In contrast solvent extraction yield higher oil yield but have major challenge in solvent recovery and harmful environmental impact. Emerging extraction technologies like supercritical CO₂ extraction, enzyme-assisted extraction, ultrasound, and microwave in mango seed kernel (Mwaurah et al., 2021) have been explored (Table 3).



Advertise here !

To book this space
Contact :
Mr. Vishal Sarin
Mobile : +91-7838012842 Email : vishal@aifpa.net

Table 3: Extraction method with their process conditions, yield ranges, and limitations.

Method	Process Conditions	Oil Yield (w/w)	Advantages	Limitations	References
THERMAL METHODS					
Solvent extraction	n-Hexane; immersion or percolation; reflux 60-70°C; miscella stripping; meal desolventising	8 to 20%	Highest reported yield, Industrially proven Well-established infrastructure Low cost at scale	Hexane residue in crude oil Desolventising required Environmental/regulatory concern Not food-grade preferred	Yadav et al. (2017) ¹ Mwaurah et al. (2021) ² Awolu & Manohar (2022) ³
Expeller / hot screw press	Mechanical screw press; friction-generated heat 60-120°C; continuous operation; no solvent added	4 to10%	No solvent residue Continuous operation Simple, robust equipment Lower capital than	Heat degrades heat-labile bioactives Oxidation risk at high temp. Lower yield than solvent Press-cake retains residual oil	Awolu & Manohar (2022) ³ Kittiphoom & Sutasinee (2013) ⁴
Microwave-assisted extraction (MAE)	Microwave (2.45 GHz) with hexane or ethanol; rapid dielectric heating; 5-30 min at 40-80°C	14% higher oil yield as compare to Soxhlet	Rapid vs conventional Soxhlet Reduced solvent volume 20-60% Higher throughput per cycle Improved oil quality indices	Solvent still required Scale-up engineering complex Risk of thermal hot-spots Not solvent-free	Mwaurah et al. (2021) ²
Steam / hydrodistillation	Steam injected at ≥100°C; volatile fraction co-distilled and condensed; water-oil separation by density	Very low (volatile fraction only)	Captures volatile aroma compounds No organic solvent Food-safe distillate	Not suited for fixed oil recovery High energy demand Thermal degradation of some volatiles Negligible fixed-oil yield	Awolu & Manohar (2022) ³
Expeller / cold press	Mechanical screw press; temperature ≤40-60°C controlled; no solvent; batch or continuous	4 to 5%	No solvent residue Preserves minor lipids and bioactives Food-grade output Simple operation	Lowest yield among all methods Higher energy per unit oil vs hexane Residual oil in press-cake	Yadav et al. (2017) ¹ Kittiphoom & Sutasinee (2013) ⁴
Supercritical CO ₂ (SC-CO ₂ / SFE)	CO ₂ at 35-40 MPa; extractor 60°C / separator 50°C; pressure release precipitates oil; optional ethanol co-solvent	2.5 to 8.3% (dry weight basis)	Zero solvent residue Preserves heat-labile bioactives Superior oil quality (lower FFA) Tunable selectivity via pressure/temp	High capital and operating cost Batch-limited throughput Yield higher than hexane Soxhlet	Jahurul et al. (2014) ⁵ Cerón-Martínez et al. (2021) ⁶ Solís-Fuentes & Durán-de-Bazúa (2011) ⁷
Aqueous enzymatic extraction (AEE)	Cellulase + protease pretreatment; aqueous extraction at 35-50°C; pH 4-5; centrifugal demulsification	66% of theoretical (analogous fruit seed AEE) ^c	No organic solvent Green / sustainable process Protein-rich residual cake Mild operating conditions	Long processing time (hours) Enzyme procurement cost Emulsion demixing step required Scale-up remains challenging	Solís-Fuentes & Durán-de-Bazúa (2011) ⁷ Mwaurah et al. (2020) ⁸
Ultrasound-assisted extraction (UAE)	Ultrasonic probe 20-40 kHz in solvent or water; acoustic cavitation disrupts cell walls; 20-40°C; 15-60 min	Yield enhanced 10-55% vs control ^c	Rapid cell disruption Reduced solvent and processing time Combinable with enzymatic methods Near-ambient temperature	Organic solvent often still required Probe erosion and maintenance Scale-up acoustic uniformity difficult Limited standalone MKF yield data	Mwaurah et al. (2021) ² Ngo et al. (2024) ⁹
Pulsed electric field (PEF)	High-voltage pulses 1-50 kV/cm; electroporation of cell membranes; <40°C; pretreatment before aqueous/solvent extraction	Pretreatment Based on pulsed	Very low thermal impact (<40°C) Preserves thermolabile compounds Fast pulse cycles (<1 s) Reduced downstream solvent needed	High capital cost of PEF unit Primarily a pretreatment, not standalone	Barba et al. (2015) ¹⁰ Tiwari (2015) ¹¹

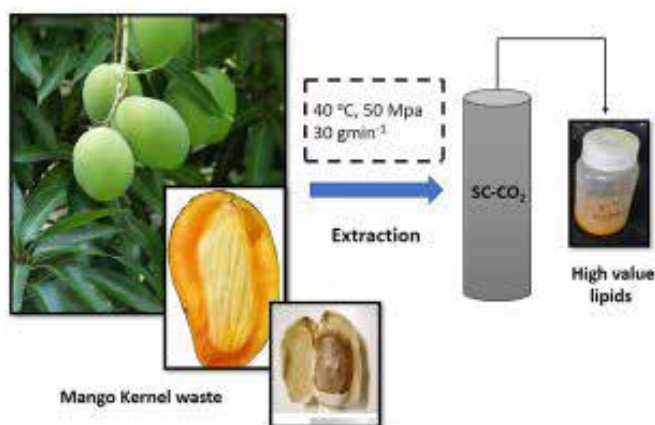


Fig 3.1 Mango Kernel Fat Extraction by Solvent Extraction method

3.1 Pretreatment

Pretreatment is an important step prior to oil extraction, which influences the final yield and product quality regardless of the extraction method used. Firstly, the fibrous outer husk must be separated from the cotyledon (manual splitting or mechanical hulling in medium-scale operations). This is then followed by drying the seed kernel to a moisture content of around 6-8%, which is a critical step for efficient mechanical pressing and a mold growth prevention process during storage. The seeds may also be flaked or micronized to reduce the particle size and improve cell disruption (this being an optional step). Before expelling, thermal preconditioning, i.e., cooking the seeds to 80-100°C for 15-20 minutes, increases oil yield by denaturing the kernel protein matrix and reducing the viscosity of the fat phase.

3.2 Fractionation of MKF

MKF has uniquely high amount of fat consisting of 1,3-distearoyl-2-oleoyl-glycerol (28.9-65.0%) conferring to its exceptional oxidative stability index (58.8-85.2 h at 110 °C). This TAG, along with sterols, tocopherols, and squalene, makes the MKF suitable as high-value-added food ingredients, such as cocoa butter alternatives, sources of natural antioxidants, and trans-free shortenings. Unfortunately, a major constraint on its industrial

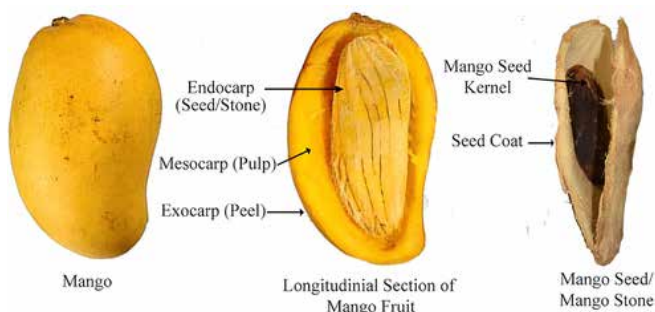


Fig 4. Pretreatment of Mango Seed Kernel



Fig 3.1 Cold Press Machine to extract fat from Mango Kernel

utilization is the prolong dehydration of mango kernels at high temperature resulting in hydrolysis of TAGs. The resulting high levels of free fatty acids and diacylglycerols limit the application of MKF in food sector. Moderate refining, degumming followed by physical deacidification, fractionation, and interesterification are major prerequisites to restore the physico-chemical and functional properties of MKF. The process of dry fractionation can be used to separate a higher-melting stearin fraction enriched in StOst, ideal for CBE applications from a lower-melting olein fraction :suitable as an emollient base for crude MKF. The fractionation process involves controlled crystallization from the melt, followed by membrane filtration, a well-established step in palm oil and shea butter processing, but with limited industrial adoption for MKF in India.

4. Industrial and Commercial Applications

The superior nutritional profile and functional properties of MKF lead to its applications in multiple industries, including the food, cosmetics, and nutraceuticals industries. A systematic overview of the principal application categories, classified by sector, is presented in Figure 1.

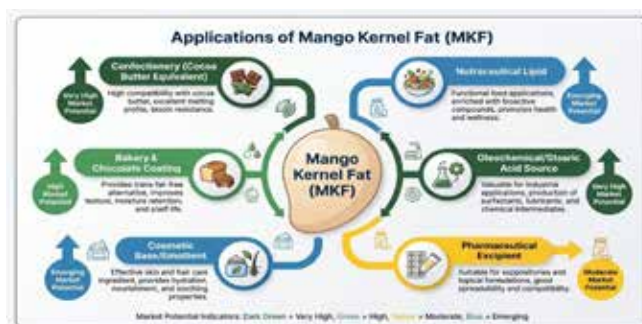


Fig 1: Application of Mango Kernel Fat in different industries

4.1 MKF as Cocoa Butter Equivalent

One of the major applications of MKF is its uses as a substitute for cocoa butter. For MKF to work

well as a cocoa butter replacement, three main requirements need to be met as follows: (i) TAG compatibility - To prevent eutectic depression or polymorphic incompatibility, SOS fraction in MKF needs to match those in cocoa butter; (ii) crystallization behaviour - MKF must crystallize in the beta (Form V) polymorph under conventional chocolate tempering protocols (Svanberg et al., 2011; Garti & Widlak, 2012) and (iii) sensory parity - gloss, snap, mouthfeel, and melt rate must not be distinguishable compared to a 100% cocoa butter chocolate at the replacement level used. Earlier research using several testing methods - Differential Scanning Calorimetry (DSC), Pulsed NMR solid fat content (SFC) determination, and X-ray diffraction (XRD) analysis has shown that blending MKF with cocoa butter at 10-20% (w/w) satisfies all three criteria (Jahurul et al., 2013; Solís-Fuentes et al., 2010; Nadeem et al., 2016). When tested, these blends behave like cocoa butter in the temperature range of 15 to 37°C, with a sharp melting point around 35-36°C that gives chocolate its desired texture and snap.

5. Sustainability and Environmental Impact Assessment

5.1 Waste Generation

India's mango processing industry generates around 400,000 -600,000 metric tonnes of mango seed (husk and kernel) annually from the organized sector alone. Recovered kernel from waste mango husk with fat content of 10-15% (dry weight) accounts for annual MKF resource equivalent to 25,000-45,000 metric tonnes, comparable to India's annual production of kokum butter or sal fat (Jahurul et al., 2015).

5.2 Life Cycle Considerations

MKF extracted from processing industry waste could serve as a potential alternative to commercial fat resources, helping lower the environmental carbon footprint. Several conventional fat resources are:

Cocoa butter: Cocoa cultivation is associated with significant water demand and also requires more land usage, resulting in deforestation in West African regions. On the contrary, MKF is a waste by-product, eliminating the need for high-water demand, land availability, and primary crops (Ntiamoah & Afrane, 2008).

Shea butter: Shea butter is also increasingly under pressure from land-use change and is predominantly wild-harvested in the Sahel belt. This also represents the challenge that raw material supply chains are geographically remote

from Indian processors (Fitzherbert et al., 2008).

Palm kernel oil: Palm kernel oil is well-documented for its biodiversity and deforestation concerns in Southeast Asia, despite RSPO certification schemes.

Utilizing MKF reduces the environmental burden at the processing facility level by diverting seed material from open dumping, eliminating issues such as methane and leachate generation, along with odour and pest pressure. This will also contribute to ISO 14001 environmental management system targets (Mwaurah et al., 2021).

5.3 Carbon Footprint Reduction Potential

A processor recovering and selling MKF instead of landfill disposal of seed waste can help avoid emissions from decomposition (0.3- 0.5 kg CO₂ equivalent per kg of organic waste) and also become eligible for carbon credits under voluntary carbon markets if waste-to-value projects are structured accordingly (IPCC, 2006). These sustainability practices are increasingly demanded by European and Japanese buyers purchasing Indian mango pulp and juice, who are extending their ESG supplier assessments to include waste management practices.

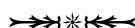
6. Conclusion

Mango seed kernel fat is a specialty lipid, characterized by a StOst-rich triacylglycerol profile, high oxidative stability, and bioactive compounds like phytosterols, tocopherols, and squalene. These attributes all together makes it as alternative to cocoa butter, shea butter, and sal fat across food, cosmetic, and nutraceutical applications. Extraction methodology remains a critical determinant of MKF quality, with supercritical CO₂ and cold-press techniques demonstrating superior retention of bioactive constituents relative to conventional solvent-based approaches. India is world's largest mango producer; the scale of available seed waste represents a largely underutilized domestic resource. Bridging this gap through improved processing technologies, varietal studies, and standardized processing could transform mango seed waste into a commercially valuable and sustainable specialty lipid, benefiting both the Indian processing industry and the broader global food market.

REFERENCES

- i. Awolu, O.O. & Manohar, B. (2022). Solvent extraction of oil from three cultivars of Nigerian mango seed kernel: Process modeling, GA-optimization, nonlinear kinetics and comparative characterization. *Scientific African*, 18, e01391. DOI: 10.1016/j.sciaf.2022.e01391
- ii. Barba, F.J., Parniakov, O., Pereira, S.A. et al. (2015). Current applications and new opportunities for the use of pulsed electric fields in food science and industry. *Food Research International*, 77, 773-798. DOI: 10.1016/j.foodres.2015.06.037
- iii. 3. Cerón-Martínez, L.J., Hurtado-Benavides, A.M., Ayala-Aponte,

- A., Serna-Cock, L. & Tirado, D.F. (2021). A pilot-scale supercritical carbon dioxide extraction to valorize Colombian mango seed kernel. *Molecules*, 26(8), 2279. DOI: 10.3390/molecules26082279
- iv. FAO (2023). FAOSTAT Crop Production Database. Food and Agriculture Organization of the United Nations.
 - v. Fitzherbert, E. B., et al. (2008). How will oil palm expansion affect biodiversity? *Trends in Ecology & Evolution*, 23(10), 538-545.
 - vi. Garti, N., & Widlak, N. (2012). *Cocoa Butter and Related Compounds*. AOCS Press.
 - vii. Gurjar, H. R., & Raj, J. D. (2022). Extraction of mango kernel (*Mangifera indica*) oil.
 - viii. Ifesan, B. O. T. (2017). Chemical properties of mango kernel and seed and production of biscuit from wheat-mango kernel flour blends. *International Journal of Food and Nutrition Research*, 1-5
 - ix. IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 5: Waste. Intergovernmental Panel on Climate Change, Geneva.
 - x. Jahurul, M. H. A., et al. (2015). Mango (*Mangifera indica* L.) by-products and their valuable components. *Food Chemistry*, 183, 173-180.
 - xi. Jahurul, M.H.A., Zaidul, I.S.M., Norulaini, N.N.A., Sahena, F., Jaffri, J.M. & Omar, A.K.M. (2014). Supercritical carbon dioxide extraction and studies of mango seed kernel for cocoa butter analogy fats. *CyTA - Journal of Food*, 12(1), 97-103. DOI: 10.1080/19476337.2013.801038
 - xii. Kaur, M., Singh, N., Sandhu, K. S., & Guraya, H. S. (2004). Physicochemical, morphological, thermal and rheological properties of starches separated from kernels of some Indian mango cultivars (*Mangifera indica* L.). *Food Chemistry*, 85(1), 131-140. <https://doi.org/10.1016/j.foodchem.2003.06.013>
 - xiii. Kittiphoom, S. & Sutasinee, S. (2013). Mango seed kernel oil and its physicochemical properties. *International Food Research Journal*, 20(3), 1145-1149. ISSN: 1985-4668
 - xiv. Kris-Etherton, P. M., et al. (2005). Plant sterols and stanols. *Arteriosclerosis, Thrombosis, and Vascular Biology*, 25(6), 1159-1164.
 - xv. Lipp, M., & Anklam, E. (1998). Review of cocoa butter and alternative fats. *Food Chemistry*, 62(1), 73-97.
 - xvi. Lutterodt, H., et al. (2011). Tocopherol composition of mango kernel fat. *Journal of the American Oil Chemists' Society*, 88, 1001-1008.
 - xvii. Makhoul, H., et al. (2015). Minor lipid constituents of mango kernel fat. *Industrial Crops and Products*, 74, 156-163.
 - xviii. Mutua, J. K., Imathiu, S., & Owino, W. (2017). Evaluation of the proximate composition, antioxidant potential, and antimicrobial activity of mango seed kernel extracts. *Food Science & Nutrition*, 5(2), 349-357. <https://doi.org/10.1002/fsn3.399>
 - xix. Mwaurah, P.W., Kumar, S., Kumar, N., Panghal, A., Attkan, A.K., Singh, V.K. & Garg, M.K. (2021). Ultrasound and microwave-assisted solvent extraction of mango kernel oil and evaluation of physicochemical properties and fatty acid profile. *Journal of Food Processing and Preservation*, 45(2). DOI: 10.1111/jfpp.16090
 - xx. Mwaurah, P.W., Kumar, S., Kumar, N., Panghal, A., Attkan, A.K., Singh, V.K. & Garg, M.K. (2020). Physicochemical characteristics, bioactive compounds and industrial applications of mango kernel and its products: A review. *Comprehensive Reviews in Food Science and Food Safety*, 19(5), 2421-2446. DOI: 10.1111/1541-4337.12598
 - xxi. Nadeem, M., et al. (2016). Oxidative stability, characterization and potential food applications of mango (*Mangifera indica* L.) kernel fat. *Journal of Food Science and Technology*, 53(4), 1917-1925.
 - xxii. Ngo, T.V. et al. (2024). Ultrasound-assisted extraction of mango seed kernel. *Food Science & Nutrition*. DOI: 10.1002/fsn3.4036
 - xxiii. Normen, L., et al. (2000). Phytosterol content of mango seed fat. *European Journal of Nutrition*, 39, 84-91.
 - xxiv. Ntiamoah, A., & Afrane, G. (2008). Environmental impacts of cocoa production and processing in Ghana. *Journal of Cleaner Production*, 16(16), 1735-1740.
 - xxv. Nzikou, J. M., K, A., M, L., Loum, B., Pambou, N. P. G., Kimbonguila, A., et al. (2010). Extraction and characteristics of seed kernel oil from mango (*Mangifera indica* L.). *Research Journal of Environmental and Earth Sciences*, 2(1), 31-35.
 - xxvi. Solís-Fuentes, J.A. & Durán-de-Bazúa, M.C. (2011). Mango (*Mangifera indica* L.) seed and its fats. In: *Nuts and Seeds in Health and Disease Prevention*, pp. 741-748. Academic Press/Elsevier. DOI: 10.1016/B978-0-12-375688-6.10088-X
 - xxvii. Svanberg, L., et al. (2011). Effect of pre-crystallization process and solid particle addition on cocoa butter crystallization and final chocolate properties. *Journal of Food Engineering*, 104(1), 70-78.
 - xxviii. Tiwari, B.K. (2015). Ultrasound: A clean, green extraction technology. *TrAC Trends in Analytical Chemistry*, 71, 100-109. DOI: 10.1016/j.trac.2015.04.013
 - xxix. Waraho, T., et al. (2011). Mechanisms of lipid oxidation in food dispersions. *Trends in Food Science & Technology*, 22(1), 3-13
 - xxx. Yadav, K.K., Garg, N., Verma, A., Kumar, S. & Trivedi, M. (2017). Optimization and extraction of oil from mango seed kernel (*Mangifera indica*). *Indian Journal of Agricultural Sciences*, 87(7), 943-946. DOI: 10.56093/ijas.v87i7.71925



**Your Brand. Their Attention.
Get Featured Today!**

To book this space
Contact :
Mr. Vishal Sarin
Mobile : +91-7838012842 Email : vishal@aifpa.net

Cold Chain Readiness and Infrastructure for Mango Supply

Karma Beer¹, Akath Singh² and T. Damodaran³

Division of Post -Harvest Management
ICAR- Central Institute for Sub-Tropical Horticulture, Rehmanikhera,
Post- Kakori, Lucknow- 226 101

Introduction

Mango, often called the “king of fruits,” is a perishable fruit if not handled properly from harvesting to reach of consumer. Temperature and humidity plays important role to maintain its freshness and quality. Ensuring that to maintain its freshness and quality, cold chain system is essential to reduce post-harvest losses, maintain quality, and expand domestic as well as export markets. Mango supply chains require an efficient cold chain system to maintain fruit quality, reduce post-harvest losses, and ensure market competitiveness, particularly for international as well as domestic market. Cold chain readiness refers to the availability of infrastructure, technologies, and management practices that maintain optimal temperature and humidity to maintain the mango fruit from harvest to consumer. Since mango is a climacteric fruit with high respiration and ripening rates, inadequate temperature management can lead to rapid deterioration, weight loss, fruit shrivelling, post-harvest disease incidence and reduced shelf life. The cold chain for mango begins immediately after harvest with proper handling on field level and exclusion of field heat (pre-cooling). Pre-cooling is an essential step because it removes field heat and slows physiological activity.

Packhouses play a critical role in cold chain readiness. Modern packhouses equipped with washing units, fungicidal treatment facilities, grading lines, waxing systems, and packaging sections that help to maintain quality standards and extension of shelf life of mangoes. Packaging materials must provide ventilation (8-10%), cushioning (foam net), and humidity control to reduce mechanical injury during transportation. Suitable corrugated fiberboard boxes having 5 ply for international market and 3 ply for domestic market with ventilation holes/mesh are commonly used for mango transport. Cold storage infrastructure is a major requirement in mango supply chains. Mature mangoes are generally stored at 12°C with relative humidity of 85–95% (protocol developed by ICAR- CISH, Lucknow). Temperatures below the 12°C may induce chilling injury, while higher temperatures accelerate ripening. Cold rooms should include temperature monitoring, humidity control, airflow management, and backup power systems to ensure uninterrupted operation. Refrigerated transport, including reefer trucks and refrigerated containers, is essential for maintaining the cold chain during transit. For long-distance export shipments, controlled atmosphere

storage and transport systems are increasingly adopted. These systems regulate oxygen and carbon dioxide concentrations, helping delay ripening and preserve fruit firmness. Controlled atmosphere containers (CA) are particularly useful for shipping mangoes over travel periods exceeding 15 days. Infrastructure readiness also depends on logistics integration, trained manpower, and digital monitoring systems. Temperature loggers, remote sensors, and Internet of Things (IoT)-based monitoring can track environmental conditions throughout the supply chain. In many developing regions, inadequate cold storage facilities, unreliable electricity, fragmented logistics, and high operational costs limit cold chain efficiency. Strengthening infrastructure through government support, private investment, and farmer cooperatives can significantly improve post-harvest management.

Importance of Cold Chain in Mango Supply

The importance of cold chain management in the Mango supply system is critical because mango is a highly perishable climacteric fruit that continues to respire and ripen after harvest. Once harvested, physiological and biochemical activities remain active, leading to rapid softening, moisture loss, color changes, and deterioration in quality. Without proper temperature regulation, mangoes may undergo uneven ripening, microbial spoilage, and significant post-harvest losses during storage. A well-established cold chain ensures that mangoes are maintained under controlled temperature and optimum humidity conditions throughout the supply chain, beginning from harvesting and pre-cooling to storage, transportation, distribution, and retail display. Maintaining suitable storage conditions slows respiration rate, delays ethylene production, and reduces metabolic activity, thereby preserving freshness and extending shelf life.

In the absence of cold chain infrastructure, shelf life can reduce drastically, often leading to deterioration in texture, flavor, aroma, and visual appearance. Post-harvest losses in mango may reach 25–30% due to inadequate handling and temperature mismanagement. Such losses negatively affect farmers, traders, exporters, and consumers by reducing product availability and profitability. Cold chain systems also contribute to improved market access and better price

¹Scientist, Division of Post -Harvest Management,

²Principal Scientist & Head, Division of Post -Harvest Management,

³Director, ICAR- Central Institute for Sub-Tropical Horticulture, Lucknow

realization. High-quality mangoes maintained under cold storage fetch premium prices in domestic and international markets. Technologies such as refrigerated transport, cold rooms, pre-cooling units, and controlled atmosphere storage help retain fruit firmness and reduce post-harvest diseases.

Key Components of Mango Cold Chain Infrastructure

a. Pre-cooling Facilities

Pre-cooling facilities play a crucial role in the post-harvest management of mango by rapidly removing field heat immediately after harvest. Mangoes are harvested under warm environmental conditions where temperature typically ranges from 35-42 °C, and the retained field heat accelerates respiration, physiological weight loss, accelerate ripening, and microbial activity. If this heat is not removed promptly, fruit quality deteriorates rapidly, resulting in reduced shelf life and increased post-harvest losses. Therefore, pre-cooling is considered the first and one of the most important steps in establishing an effective cold chain system for mango supply. By decreasing the respiration rate, pre-cooling preserves firmness, maintains skin color, minimizes weight loss, and reduces the incidence of fungal decay and physiological disorders.

Several pre-cooling methods are commonly used in mango handling systems. Forced-air cooling is widely adopted because it allows chilled air to circulate uniformly around packed fruits, ensuring rapid and consistent cooling. Room cooling is another option but is generally slower compared with forced-air systems. The ideal pre-cooling temperature for mangoes generally ranges between 14–15°C (protocol developed by ICAR- CISH, Lucknow). Overall, pre-cooling facilities significantly improve storage life, maintain fruit quality, support long-distance transportation, and reduce economic losses. They form a vital link between harvesting and refrigerated storage, ensuring that mangoes enter the cold chain under optimal physiological conditions.

b. Packhouses and Grading/ Sorting Units

Packhouses and grading units are essential components of the post-harvest management system for Mango, helping maintain fruit quality and reduce handling losses. Modern packhouses provide facilities for cleaning, washing, sorting, grading, and packaging of harvested mangoes before storage or transportation. Cleaning and washing remove field dust, latex stains, and microbial contaminants, improving fruit appearance and hygiene. Sorting helps separate damaged,



diseased, or immature fruits, ensuring only marketable produce enters the supply chain.

Grading is particularly important because it classifies mangoes according to size, weight, color, maturity, and quality standards. Uniform grading promotes consistent ripening and enhances consumer acceptance. Properly graded fruits are easier to market and often receive better prices in domestic and export markets.

Process flow chart in packhouse



c. Cold Storage Units

Cold storage units are an important part of the post-harvest handling system for Mango, particularly for short-term holding before transportation, marketing, or export. After harvesting and pre-cooling, mangoes are stored under controlled environmental conditions to slow physiological activity and maintain fruit quality. Cold storage reduces respiration rate, delays ripening, minimizes moisture loss, and suppresses microbial growth, thereby extending shelf life.

For mango storage, an optimal temperature range of 12–13°C with relative humidity of 85–90% is generally recommended. These conditions help preserve firmness, color, flavor, and nutritional quality while reducing shrinkage and weight loss. Proper humidity is essential to prevent physiological weight loss and fruit shrinkage. However, temperatures below 10°C should be avoided because mangoes are highly sensitive to chilling injury.



d. Reefer Transport (Reefer Logistics)

It is a vital component of the cold chain system for mango, ensuring that fruits maintain their quality during transit

from production areas to markets. Since mangoes are highly perishable and continue to respire after harvest, maintaining a controlled temperature during transportation is essential to slow ripening and reduce spoilage. Temperature fluctuations during transit can accelerate softening, moisture loss, and microbial decay, leading to significant post-harvest losses.

Refrigerated trucks and reefer containers are commonly used to transport mangoes under controlled environmental conditions. These systems maintain a stable temperature (12–13°C) with relative humidity of 85–90%, typically within the recommended range for mango storage, helping preserve texture, flavor, appearance, and nutritional quality. Proper airflow circulation within refrigerated units ensures uniform cooling and prevents hot spots that may cause uneven ripening.

e. Ripening chambers

Ripening chambers are an essential post-harvest facility for mango that enable safe, uniform, and scientifically controlled ripening. Traditional ripening practices often involve the use of harmful chemicals such as calcium carbide, which may pose health risks and lead to uneven fruit quality. Modern ripening chambers provide a controlled environment where temperature, humidity, airflow, and ethylene concentration are carefully regulated to achieve consistent ripening. Ethylene gas is commonly used in ripening chambers because it is a natural plant hormone responsible for initiating fruit ripening. Under controlled conditions, ethylene promotes uniform color development, softening, aroma formation, and improved sweetness. Scientific ripening ensures better appearance, enhanced taste, and longer shelf life compared with uncontrolled methods.

f. Distribution and retail cold storage facilities

Distribution and retail cold storage facilities play a crucial role in maintaining the quality of mango after transportation and before final sale. Cold rooms at wholesale markets, distribution centers, and retail outlets help maintain appropriate storage temperatures, slowing ripening and preserving freshness. These facilities are particularly important during peak supply periods when fruits may remain in storage for extended durations. Proper cold storage at retail points prevents excessive softening, moisture loss, and microbial spoilage. It

also reduces wastage caused by over-ripening and improves shelf presentation. Maintaining mangoes under controlled conditions enhances consumer satisfaction by ensuring fruits retain their texture, flavor, and visual quality until purchase.

Major Challenges in India

Despite being one of the world's largest producers of mango, India faces several challenges in developing an efficient cold chain system for mango supply. One of the major constraints is inadequate cold storage and pre-cooling infrastructure in rural and production regions, where most mango cultivation occurs. Limited access to packhouses, refrigerated transport, and cold rooms often results in high post-harvest losses. High capital investment and operational costs further restrict cold chain adoption, particularly for small and marginal farmers. Establishing refrigerated storage, ripening chambers, and reefer transport requires substantial financial resources and technical expertise.

Farmer awareness regarding post-harvest management and cold chain benefits is often limited, affecting adoption of improved practices. Power supply inconsistencies and inadequate backup systems also disrupt cold storage operations, especially in rural areas. These challenges collectively hinder quality preservation, reduce shelf life, and limit export competitiveness.

Role of Government and Private Agencies

Government and private sector participation plays a crucial role in strengthening the cold chain infrastructure for mango supply. Government initiatives provide subsidies and financial assistance for establishing cold storage units, packhouses, refrigerated transport, and ripening chambers. Programs supporting infrastructure development encourage farmers and agribusinesses to adopt modern post-harvest technologies. Promotion of Farmer Producer Organizations (FPOs) helps small farmers collectively access storage, grading, and marketing facilities. Public-private partnerships enhance investment, technology transfer, and efficient management of cold chain operations. Additionally, the development of integrated cold chain networks improves coordination among production, storage, transportation, and retail sectors, reducing post-harvest losses and improving market access for mango growers.





Make Your Brand The Next Big Talk !

Promote . Connect . Grow

To book this space
Contact :
Mr. Vishal Sarin
Mobile : +91-7838012842 Email : vishal@aiipa.net

Misbranded Food: An Avoidable Violation

K. M. Nath

Former Joint Director, MoFPI/FSSAI

E-mail:- Kmnath51@gmail.com



Introduction and regulatory evolution

In the early years under the Prevention of Food Adulteration Act, the regulatory approach was rigid and narrow. Adulteration was the central offence. Even minor labelling defects were often treated at par with adulteration and led to prosecution. The law did not clearly differentiate between risk to health and procedural non-compliance. This resulted in fear driven enforcement rather than risk-based compliance.

With the enactment of the Food Safety and Standards Act, 2006, India moved to a modern and scientific regulatory framework. The Act clearly classified violations and linked them with proportionate penalties and punishments. Under the Act the violations are broadly categorised into three types:

- Misbranded food
- Substandard food
- Unsafe food

Misbranding relates to information given to the consumer through packages or labels. Substandard food relates to quality parameters not meeting prescribed standards. Unsafe food relates to direct risk to human health. Depending on the category of violation, the penalty and punishment are decided. This distinction is critical for Food Business Operators and Regulators alike.

Misbranded food

As per Section 3(1)(zf) of the Food Safety and Standards Act 2006, misbranded food means an article of food if:

- it is represented or offered or promoted for sale with false or misleading or deceptive claims through labels of the package or through advertisement
- the article is an imitation of or substitute for or resembles in a manner likely to deceive another article of food under the name of which it is sold, not labelled indicating its true nature
- the label bears any statement, design, picture, or device regarding ingredients or substances which is false, misleading, or deceptive with respect to its content
- it contains artificial colour, flavour, preservative, or added nutrients like vitamins or minerals without proper declaration or information
- the product is not labelled in accordance with the Act or the Regulations made thereunder

Penalty for misbranded food

Section 23 of the Act mandates that all packaged foods shall be marked and labelled as specified in the Regulations. Unlike the PFA era, most misbranding cases are now handled through monetary penalties instead of prosecution.

The penalties prescribed under the Act are as given below:

- For any deficiency in declarations on the label of a package, a penalty up to three lakh rupees under Section 52 may be imposed
- For false or misleading claims through labels or advertisements, the penalty can go up to ten lakh rupees under Section 53

Why misbranding is common

Many legal samples collected by Food Safety Officers and analysed by Food Analysts are declared misbranded, basically due to labelling defects. These occur because:

- The provisions under the Regulations are not read carefully
- Label development is treated as a marketing activity
- Font size, placement, and manner of declaration are ignored
- Claims are made without regulatory validation

It is important to understand that compliance is not only about making declarations, but the manner of declaration is also equally important.

Mandatory provisions that must be complied with

Name of the product:

The name of the product must be as per its essential composition or as specified under product standards. It should clearly reflect the true nature of the food contained in the package. Many Food Business Operators display only fancy or trade names prominently and either miss or make the actual common name. Just adding 1-2% fruit or nut or any such ingredient or only with natural/artificial flavour of that ingredient does not qualify the name of the product after that ingredient. This violation is often observed.

List of ingredients:

All ingredients must be declared in descending order of their composition by weight or volume at the time of manufacture. The product sold as mixtures of ingredients

or that is highlighted through names, words or pictures must also carry percentage declaration of such ingredients. This is one of the most frequently violated requirements.

Nutritional information:

Nutritional values specified under the regulations must be declared per 100 g or 100 ml or per single consumption pack. Values should be based on laboratory analysis or validated data. Per Serve percentage contribution to RDA must be calculated as prescribed. Values must remain within permissible tolerance limits.

Veg or non- veg logo:

The vegetarian or non-vegetarian symbol must be displayed near the product name or brand name. It should not be placed arbitrarily anywhere on the pack. The size, colour, and shape of the circle or triangle must comply with regulations.

Food additives:

The functional class of the additive must be declared along with either the common name or the INS number. In case of artificial flavouring substances, the class name and common name must be declared. A common error is declaring only the INS number or common name without the functional class.

Name and address:

The name and complete address of the brand owner must be declared as Marketed by or Manufactured and Marketed by, as the case may be, The FSSAI logo and licence number are mandatory. If there are multiple manufacturers or packers, a suitable identification code for each must be provided. This requirement is often ignored.

Date marking:

Date of manufacture and expiry or use by date must be declared together and not separately. The format should be MM/ YY if the shelf life is more than three months, and DD/ MM/ YY if the shelf life is less than three months.

Batch or lot number:

Batch, code, or lot number must be declared clearly to ensure traceability.

Instructions for use & storage:

Instructions such as dilution, reconstitution, thawing, keeping in refrigerator after opening must be provided wherever applicable. If a dish, image, or picture is shown along with other ingredients as a serving suggestion, a clear statement "For Serving Suggestion Only" must be given.

Allergen declarations:

Allergen warning/advice for the ingredients such as wheat, milk, nuts, soy, fish, eggs, sulphites must be declared prominently in the manner specified. If the product does not contain allergen ingredients but there is a possibility of

cross contamination, precautionary statements such as "May contain traces of (name of the allergen)." or "Manufactured in a facility that also processes ((name of the allergen)" must be given.

Imported food:

For imported food, the country of origin must be declared prominently and comply with other requirements as per Food Safety and Standards Import Regulations, 2017.

Some specific provisions to be complied with

Certain food categories have additional labelling requirements prescribed under the Food Products Standards Regulations, Allied Regulations, and Schedule II of the Labelling and Display Regulations, 2020. These declarations must be displayed prominently, in capital letters where prescribed, and in the minimum font size specified.

Some examples of mandatory declarations/statements for the products or the ingredients contained in the products are:

Caffeinated beverages:

"Contains Caffeine. Not to consume more than 500 ml per day"

Pan Masala/ Supari:

"Chewing of Pan Masala / Supari is injurious to health"

Products containing Artificial Sweeteners such as Aspartame or Acesulfame Potassium:

"Not recommended for children, pregnant women and lactating mothers"

Products containing Monosodium Glutamate:

"Not recommended for infants below 12 months and pregnant women"

Admixture of edible Oils:

"MULTI-SOURCE EDIBLE OIL"

Plus name, nature, and percentage of each edible oil used in the blend must be clearly declared.

These declarations are product specific and cannot be diluted or modified for marketing convenience.

Legal Metrology requirements

Apart from Food Safety and Standards Act & Regulations, packaged foods must also comply with the Legal Metrology Act, 2009 and the Legal Metrology (Packaged Commodities) Rules, 2011.

The three mandatory declarations applicable to food products are:

- Net Quantity in standard units of mass for solid, volume for liquid and number for length
- Maximum Retail Price inclusive of all taxes in rupees in two decimal place and Unit Sale Price rounded off to the

nearest two decimal places

- Consumer complaint contacts details-name, address, contact number, email

Font size and manner of declaration must be complied with as specified under the LM(PC) Rules. The font size for the declarations of Expiry or Use by date, Net Quantity, MRP, and Consumer contact details must comply with minimum height of letters and numerals prescribed under LM(PC) Rules, which is higher than that prescribed under FSS Labelling & Display Regulations. Non-compliance can attract separate penal action under the Legal Metrology Act, even if it complies with FSS Act.

Many products are declared misbranded not because information is missing, but because the font size is smaller than that prescribed or the declaration is placed incorrectly.

Many misleading claims often made in violation of the FSS Advertisement and Claims Regulations, 2018.

Following provisions of the Regulations must be complied with:

- All claims must be factual, truthful, and not misleading in any manner
- Any health benefit claim must be supported by scientific evidence
- No Addition claim such as no added colour, preservative can be made only when such additives are permitted in the product and not used
- No added sugar claims must be followed by the statement "Contains naturally occurring sugars".
- Claims such as zero cholesterol, gluten free, high omega 3 fatty acid, high protein, high dietary fibre etc. can be made meeting the conditions laid down under Schedule I

- Claims for fortification or enrichment of vitamins & minerals must comply with the conditions prescribed under Schedule IV
- Use of adjectives such as Fresh, Natural, Original, Pure, Real, Authentic, Genuine, Traditional is permitted only when conditions under Schedule V are fulfilled.

Conclusion

The food sector today is fast moving, competitive, and marketing driven. Labels are developed quickly, and products are launched rapidly. At the same time, enforcement has become stricter, laboratories are vigilant, and consumers are well informed.

Misbranding is no longer a minor or technical lapse. Under the Food Safety and Standards Act, 2006, it is a clearly defined violation linked to consumer rights, transparency, and right to informed choice. A shift from the PFA era to the FSSAI framework has brought clarity, proportionality, and scientific reasoning into enforcement. However, this also places greater responsibility on Food Business Operators.

Most misbranding cases arise not from intent to deceive, but from lack of awareness, poor label review, or over emphasis on marketing considerations. These mistakes are avoidable. A careful reading of regulations, proper regulatory vetting of labels, and attention to font size, placement, and product specific declarations can prevent penalties, recalls, and brand damage.

In today's environment of strict enforcement and informed consumers, compliance is no longer optional. Correct labelling is not merely a legal requirement; it is a commitment to consumer trust and ethical business practice. For Food Business Operators, treating labelling as a core compliance function rather than a marketing tool, is the only sustainable way forward.



Make Your Brand The Next Big Talk !
Promote . Connect . Grow

To book this space
Contact :
Mr. Vishal Sarin
Mobile : +91-7838012842 Email : vishal@aifpa.net

Workshop on “Mango Processing, Product Diversification & Waste Management” held on 24th April 2026 at Tirupati

Dr. Raghav Jadli, President, All India Food Processors' Association (AIFPA), delivered a keynote address at the Tirupati Seminar, attended by senior government officials, scientists, academia, and leading representatives from the juice, beverage, and mango pulp industry. The Session was graced by distinguished dignitaries including, Sh. Chiranjiv Choudhary, IFS (Retd.), Principal Secretary to Government, (Food Processing), Industries & Commerce Deptt. and Dr. Sekhar Babu Geddam, IFS, CEO, AP Food Processing Society.

In his address, Dr. Jadli emphasized the critical importance of the mango sector as a pillar of India's agri-economy, processing over 10 lakh metric tonnes annually and supporting more than 180 lakh livelihoods across the value chain. He noted that mango continues to be a pride product for India with a strong global identity. Highlighting the ongoing crisis, he pointed out that over the past five years the industry has faced severe disruptions due to the COVID pandemic, Russia-Ukraine conflict, Middle East instability, Suez Canal logistics challenges, and rising import duties in global markets. As a result, nearly '1.5 lakh metric tonnes of mango pulp remain unexported', posing a serious threat to farmer incomes, particularly in key regions like Chittoor & Krishnagiri.

Dr. Jadli also outlined AIFPA's proactive initiatives, including the successful 'International Fruit Juice Conference 2024', organized in collaboration with the International Fruit and Vegetable Juice Association (IFU), bringing together over 70 global participants to strengthen India's engagement with international buyers. To address demand-side challenges, he stressed the need to boost domestic consumption of mango pulp through institutional channels such as defence forces, police, and school nutrition programs, along with expanding export markets and promoting innovative value-added mango products.

A key policy recommendation highlighted was the urgent need for GST rationalization on 'carbonated fruit Juices, currently placed under the "sin category" with 40% GST. AIFPA has proposed reducing GST to 5%-12% with a minimum 10-12% fruit pulp content, positioning these products as healthier alternatives for consumers, particularly Gen Z, and significantly boosting demand for mango pulp.

Dr. Jadli further emphasized diversification opportunities including development of value-added products like mango sauces, direct pulp consumption, and customized applications for the HORECA sector, along with increased participation in domestic and international trade exhibitions to enhance market visibility. He also acknowledged the efforts of the *Chittoor District Fruit Processor Federation* for their continuous work in representing the interests of local processors during challenging times.

On the policy front, Dr. Jadli urged the Government to extend immediate support measures, including:

- * Inclusion under 'ECLGS-style credit guarantee schemes'
- * Participation in export facilitation initiatives such as the 'RELIEF scheme'
- * Enhanced MSME liquidity support and access to the 'SME Growth Fund'
- * Provision of '3-6 month loan moratoriums'
- * Recognition of 'force majeure conditions' for exporters

The address was widely appreciated by senior officials, scientists, academia, and industry leaders, reinforcing the need for coordinated efforts between Government, industry, and research institutions to ensure sustainable growth and resilience of the mango processing sector.



Roadmap for Revitalising India's Mango Pulp Industry

A delegation of the AIFPA Mango Pulp Industry Forum, led by Dr Raghav Jadli, President, AIFPA, met the Economic Advisor (Joint Secretary), Ministry of Commerce, New Delhi, to discuss key issues concerning the mango pulp industry. The meeting focused on the sector's challenges, potential solutions, and the way forward to strengthen industry growth and exports. The delegation emphasised that the Indian mango sector should not be viewed merely as an agricultural commodity segment, but as a matter of India's agricultural heritage, national identity, farmer sustainability, and global soft power.

The delegation highlighted that:

- India possesses a unique natural advantage in mango cultivation with unmatched diversity, quality, taste profiles, climatic suitability, and production scale, which are difficult to replicate globally.
- Indian Mango pulp represents a strategic agri-resource comparable to the manner in which India has successfully positioned and protected Basmati rice in international markets.
- While substantial policy, branding, and promotional support have been extended to sectors such as Basmati, spices, tea, coffee, and millets, a similar integrated national mission for "Mango Pulp & its value-added Products " promotion is yet to emerge.
- The delegation expressed concern that, despite India being the world's largest mango producer, domestic and international consumption potential remains significantly underdeveloped due to inadequate branding, structured promotion, product diversification, and institutional market development.
- It was further emphasised that sustaining mango cultivation is not only critical for industry and exports, but also for protecting farmer livelihoods, biodiversity, traditional varieties, and India's global agricultural identity.
- The delegation urged the Government to consider a long-term "Indian Mango Pulp & its Value-added Products Promotion & Consumption Mission" involving: global branding campaigns for Indian mango and mango pulp; domestic consumption promotion initiatives; institutional support for value-added mango products; international market access facilitation; integration with tourism, food diplomacy, and cultural promotion initiatives.



Moreover, it was stated that India must proactively protect and promote mango as a national agri-heritage asset before losing long-term global competitive advantage. The Department appreciated the broader perspective presented by the delegation and advised that these aspects may also be formally submitted to APEDA for further deliberation and possible policy support.

Driving Growth in India's Mango Pulp Industry: AMPIF-APEDA Initiatives

A delegation of AIFPA Mango Pulp Industry Forum (AMPIF), led by Dr Raghav Jadli, President AIFPA, met Chairman APEDA & senior officials of APEDA on 22nd May, 2026, to discuss global promotion, export business growth and opportunities & issues in the mango pulp industry.

APEDA agreed to (a) AIFPA's suggestion to promote Mango Pulp and RTD / RTE products made thereof, in all the "mango promotion" events to be organised by APEDA abroad (b) to include wet sampling of items made using mango pulp at APEDA Pavilion in International exhibition abroad and (c) indicated willingness to accept our proposal of providing concession to mango pulp exporters for participation in APEDA pavilion in International exhibition(s). Further, Chairman, APEDA suggested Mango pulp industry participation in APEDA pavilion in internal exhibitions being participated by APEDA at the U.S.A, Australia & Russia during FY 2026-27.

APEDA official from New Delhi will visit Mango Pulp cluster in South India along with AIFPA, to scale up the export of mango pulp and other mango products. It is understood that APEDA is likely to invite global buyers during World Food India (WFI) regional editions, towards export promotion of food products

The matter of RoDTEP arrears of mango pulp exporters stuck in Chennai Port Customs was raised with concern. APEDA agreed to look into and take up the matter with the authorities again.

The meeting was successful, and APEDA is now expected to include mango pulp export promotion consistently.

About AMPIF: AIFPA Mango Pulp Industry Forum (AMPIF) is committed to promoting sustainable growth, enhancing exports, addressing industry challenges, and supporting policy advocacy for the sector. With a vision to establish India as a global leader in mango pulp and processed mango products, AMPIF strives to foster collaboration, innovation, and strategic partnerships to unlock new opportunities for the sector. Through collective efforts and constructive industry-government engagement, the Forum seeks to build a stronger, more resilient, and globally recognised Indian mango pulp industry for the future.





Hardcastle Restaurants Pvt. Ltd.

About Westlife:

Westlife Foodworld Limited (NSE: WESTLIFE, BSE: 505533), formerly known as Westlife Development Ltd (WDL), focuses on setting up and operating Quick Service Restaurants (QSR) in India through its subsidiary Hardcastle Restaurants Pvt. Ltd. (HRPL). The Company operates a chain of McDonald's restaurants in West and South India having a master franchisee relationship with McDonald's Corporation USA, through the latter's subsidiary.

About Hardcastle Restaurants Pvt. Ltd.:

HRPL is a McDonald's franchisee with rights to own and operate McDonald's restaurants in India's West and South markets. HRPL has been a franchisee in the region since its

inception in 1996. HRPL serves over 200 million customers, annually, at its 458 (as of December 31, 2025) McDonald's restaurants across 73 cities in the states of Telangana, Gujarat, Karnataka, Maharashtra, Tamil Nadu, Kerala, Chhattisgarh, Andhra Pradesh, Goa along with parts of Madhya Pradesh and Union Territory of Puducherry and provides direct employment to over 10,000 employees. McDonald's operates through various formats and brand extensions including standalone restaurants, drive-thrus, McCafé, 24x7, McDelivery, McBreakfast and dessert kiosks. The menu features Burgers, Wraps, Hot and Cold Beverages besides a wide range of desserts. Majority of the McDonald's restaurants feature an in-house McCafé. The pillars of the McDonald's system – Quality, Service, Cleanliness and Value – are evident at each of the restaurants that HRPL operates.





Collaboration with



100%



NEW
Protein
PLUS

SLICE



SAME FAVOURITES.
EXTRA PROTEIN.

14-25g
Protein

McEgg® Burger
with Protein Plus

McAloo Tikki Burger®
with 2 Protein Plus slices

25-40g Protein

McCheese Burger® Chicken
with Protein Plus

40+g Protein

McSpicy™ Premium
Chicken Burger
with 2 Protein Plus slices

+
ADD ON
GOODNESS
TO YOUR EVERY ORDER

NEW
PROTEIN PLUS
SLICE:



ADDS
5 GM PROTEIN



MADE WITH
VEG SOYA &
PEA PROTEIN



NO ONION
NO GARLIC



NO ARTIFICIAL
FLAVOURS & COLOURS



Ministry of Agriculture & Farmers Welfare



INDIA POST, MUMBAI REGION LAUNCHES INITIATIVE FOR DOORSTEP DELIVERY OF GI-TAGGED RATNAGIRI ALPHONSO MANGOES

प्रतिष्ठित तिथि: 13 APR 2026 6:13PM by PIB Mumbai

Mumbai, 13 April 2026

Reinforcing its commitment to inclusive growth and rural empowerment, India Post, Mumbai Region, has launched a special initiative to deliver GI-tagged Ratnagiri Alphonso mangoes directly to households across Mumbai, establishing a robust market linkage between farmers and urban consumers.

The initiative is designed not merely as a seasonal offering but as a sustainable support system for mango growers and small business entrepreneurs in Ratnagiri. By leveraging its extensive and trusted logistics network, India Post aims to eliminate intermediaries, ensuring that farmers receive a greater share of the final sale price, while enabling small-scale sellers and agro-entrepreneurs to expand their reach into metropolitan markets.

To enhance accessibility and convenience, the initiative offers seamless ordering options through QR codes and WhatsApp, along with the facility of Cash on Delivery (CoD). This approach empowers even the smallest producers to participate in organized retail without requiring significant investment in infrastructure or marketing.

Freshly sourced and carefully delivered, the premium Alphonso mangoes reflect both quality and the collective effort to strengthen rural livelihoods and promote local enterprise.

Speaking on the occasion, Ms Kaiya Arora, Postmaster General, India Post, Mumbai Region, stated, This initiative aims at leveraging the vast network of India Post to establish a connect between local mango farmers of Ratnagiri with the city dwellers. The doorstep delivery of GI-tagged mangoes supports rural economies and ensures farmers get a higher percentage of the final sale price, while Mumbaiers receive fresh fruits at competitive prices with the Cash on Delivery feature.

By transforming its logistics capabilities into a platform for economic inclusion, India Post continues to play a pivotal role in supporting farmers, nurturing small businesses, and delivering prosperity from rural India to urban homes.

PIB Mumbai | Sriyanka Chatterjee/Robin Singh/Darshana Rane

सोशल मिडियवर आदारा फॉलो करा: @PIBMumbai | /PIBMumbai | /pibmumbai | pibmumbai[at]gmail[dot]com | P

IBMumbai | /pibmumbai

(Release ID: 2251624) Visitor Counter : 180
Read this release in: Marathi

Chittoor's mango economy in crisis amid poor yields and West Asia tensions



According to the stakeholders, the yields are likely to be around 40% from the acreage of about 1 lakh hectares, coupled with the phenomenon of early flowering and early summer. The mango economy in Chittoor district, one of the country's largest pulp manufacturers, is reportedly heading for a major crisis this season, with only a month left for the procurement of stocks from farmers. The stakeholders maintain that this year, the yields are likely to be around 40% from the acreage of about 1 lakh hectares, coupled with the phenomenon of early flowering and early summer. Above all, the ongoing West Asia crisis has played a detrimental role, as the pulp industry remains tight-lipped about fresh procurement of stocks, claiming that "the fate of the already shipped pulp stocks stranded in the Gulf remains uncertain."

The vast acreage mainly contributes to the popular "Totapuri" variety, which feeds about two dozen active pulp industries in the region, with an annual production of more than 10 lakh tonnes, and supports thousands of farmers, remaining a key driver of the local economy. Octogenarian and doyen of the mango trade in Chittoor, N.C. Ramachandra Reddy, expressed serious concern at the present season, saying: "Though the mango gardens witnessed a very healthy and early flowering from January onwards, it was not uniform. This year's summer set in from early March, which caused the flowering to wither, affecting the prospects of yields. In several mandals, the baby mangoes started falling to the ground from mid-March. Our clear expectation is that the yield will not cross 40% at any point this year," he said.

All India Food Processors Association (AIFA) chairperson (South Zone) Kattamanchi Govardhan Bobby also presented a very grim picture regarding the prospects of trading this year. "As per the information available to the association, the pulp stocks worth over ₹1,500 crore from various factories in Chittoor and parts of Tamil Nadu were shipped a couple of weeks prior to the West Asia crisis. Some exporters from our region had recently paid a hefty charge running into 4 million USD at the Strait of Hormuz. It's a very scary situation for the exporters. The money flow has turned zero due to the West Asia tensions. As of now, we will not be able to say anything about the fresh procurement of mangoes for the current season, but only around mid-May," he said. Meanwhile, Mr. Kothur Babu, representative of the Chittoor District Mango Producers Association, said: "It is unfortunate that some pulp industries in Chittoor district are yet to clear the 2025 pending dues to the farmers, and even the district authorities are biased towards them. Despite adverse climatic conditions this year, the farmers are hopeful of a reasonable yield, and the government should intervene to start procurement as per the original schedule, which is from May. Talk about the West Asia crisis and stranded shipments is merely a lame excuse.

Source: - The Hindu, 22nd April 2026



'West Asia crisis has caused a rise in logistics costs, hitting profit margins'

Continued from page 3

Generally, 6 to 7 lakh tonnes of mangoes are procured from farmers during the season, producing about 3 to 4 lakh tonnes of pulp, mainly from Totapuri variety. About 70% of the total pulp is exported to the Gulf and European countries. The remaining 30% of pulp is sold in the domestic market. Currently, pulp is being exported in 215 kg six-layer aseptic bags instead of the older OTS tins to meet international quality standards.

Speaking to TNIE, Kattamanchi Govardhan Bobby, chairman of All India Food Processors Association (South Zone), said, "India is the world leading producer and exporter of mango pulp with nearly 70% of exports originating from Chittoor



alone. The West Asia crisis has caused a sharp increase in logistics costs, severely affecting profit margins of exporters."

He also pointed out that the mango pulp exports to the European countries have also been hit due to Houthi attacks

on commercial vessels in the Red Sea. The financial impact of the West Asia conflict on the mango pulp industry is estimated at ₹750 crore. With export payments from foreign buyers being delayed, processing units are struggling to maintain cash flow.

This has directly affected their ability to procure mangoes from farmers during the current season, creating a ripple effect across the supply chain. The association has written to the government seeking immediate intervention and support to help the mango pulp industry overcome the crisis, he said.

Farmers, who depend heavily on the mango pulp industry, are now facing uncertainty as the demand for fruit from processing units has declined.

Bikano partners with Each One Feed One



Bikano has partnered with Each One Feed One NGO to support underprivileged women through a special CSR initiative marking National Safe Motherhood Day. As part of the initiative, the company distributed 1,500 gifts to women across Delhi, celebrating maternal strength while spreading joy within communities. The distribution drive, held in the national capital, reached women from low-income backgrounds, bringing moments of warmth and celebration into their day. Each hamper included a selection of its popular namkeens, biscuits, and sweets, offering not just food but a gesture of care and inclusion. The initiative reflects its continued commitment to uplifting communities and supporting women through meaningful interventions. Mr. Kush Aggarwal, CMO, Bikano, said, women are the backbone of our families and communities, often displaying quiet strength and resilience every single day. Through our partnership with Each One Feed One, we wanted to do something meaningful to acknowledge and support this spirit. This initiative is a small step towards bringing smiles, sharing comfort, and letting these women know that they are seen and valued. At Bikano, we believe that true impact lies in creating moments of connection and care that go beyond business.

Lulu Group Signs MoU to Export 1,000+ MT Ladakhi Apricots, Boosting Global Market Access for Farmers



LuLu Group International has signed a Memorandum of Understanding (MoU) with the Union Territory administration of Ladakh to procure and export over 1,000 metric tonnes of the region's famed apricots to international markets, marking a major push to globalize Ladakhi horticulture. The initiative represents a sharp scale-up for the region, which exported just 1,500 kg of apricots over the past two years. The exports will primarily include the GI-tagged "Raktsey Karpo" and the region-specific "Halman" varieties, known for their premium quality. The agreement has been facilitated in collaboration with the Agricultural and Processed Food Products Export Development Authority and the Ministry of Food Processing Industries. Under the MoU, LuLu Group will handle end-to-end operations—from procurement and harvesting to processing, packaging, transportation, and international marketing—while preserving Ladakh's regional identity. The move is designed to ease the burden on farmers, who traditionally manage labor-intensive processes such as sorting, cleaning, and packaging, often leading to delays and post-harvest losses. By transferring these responsibilities to the exporter at the harvest stage, the initiative aims to improve efficiency and ensure better price realization for growers.

Mr. Vinai Kumar Saxena, who has been actively promoting the branding and export of Ladakhi produce, described the MoU as a "historic" step toward unlocking the region's untapped agricultural potential. He noted that apricots are among Ladakh's most valuable horticultural assets and that the agreement will create direct economic benefits by opening global market channels. The administration is also working on expanding the model to other produce, including apples, while planning to launch a dedicated e-commerce platform for Ladakh. The proposed portal, envisioned along the lines of Amazon and Flipkart, aims to provide farmers with direct market access and reduce reliance on intermediaries. Additionally, the government is focusing on building local processing infrastructure to minimise wastage and enhance value addition, further strengthening Ladakh's position in domestic and global agri-markets.

Maggi Turns 50: Nestlé Balances Nostalgia with Changing Consumer Tastes

As Maggi completes 50 years in India, parent company Nestlé India is navigating a delicate balance leveraging the brand's deep-rooted emotional connect while adapting to rapidly evolving consumer preferences. Widely credited with creating India's instant noodles category, Maggi has grown into a household staple, with over six billion servings consumed in FY2023–24, making India its largest global market. Today, the brand has expanded beyond noodles into spice mixes, pasta, and sauces, forming part of a ₹6,311-crore prepared dishes and cooking aids portfolio that contributes 31.4% to Nestlé India's FY25 revenue.

Maggi's journey in India began in 1975 with stock cubes, followed by the launch of its iconic 2-Minute Noodles in 1983—an innovation that transformed quick meals in Indian



kitchens. Globally, the brand traces its origins to 19th-century Switzerland, when it was founded by Julius Maggi and later acquired by Nestlé in 1947. Despite its scale and legacy, the brand's challenge today lies not in market reach but in staying relevant amid shifting consumption patterns. Nestlé executives say Maggi's core identity—including its iconic yellow packaging—will remain unchanged, even as the brand introduces contemporary elements such as Korean-inspired flavours and limited-edition innovations. Maggi continues to dominate India's instant noodles segment with an estimated 60% market share, although competition has intensified with the presence of brands like Sunfeast YiPPee!, Wai Wai, Top Ramen, Ching's Secret, and Patanjali Atta Noodles.

However, the company maintains that its primary competition is not other brands but changing consumer behaviour. To address this, Maggi has diversified its portfolio with perceived healthier options such as atta, millet, and oat-based noodles, while continuing to cater to its core audience—Indian households. The brand has also embraced regionalization, tailoring flavours and products to local tastes. While chicken variants perform better in North India, products like Masala-ae-Magic see stronger demand in southern markets. Urban centres, meanwhile, are driving adoption of health-focused offerings. Maggi's marketing strategy has evolved from traditional on-ground activations to a digital-first, omnichannel approach, focusing on emotional engagement and cultural relevance. Recent initiatives include limited-edition launches like the Signature Maggi Bowl and the expansion of "Maggi Hotspots"—small-format kiosks in high-footfall locations such as colleges and tourist destinations. The brand's recent foray into Korean-inspired flavours reflects its effort to tap into global cultural trends while retaining its core identity. At 50, Maggi remains a dominant force in India's food landscape. Yet, as consumer preferences shift toward health, variety, and experimentation, Nestlé's strategy appears clear: evolve with the times, but protect the familiarity that made Maggi a national comfort food.



Tetra Pak & Sterilgarda Alimenti launch industry-first 1-litre aseptic carton with paper-based barrier



Tetra Pak in collaboration with Sterilgarda Alimenti, has unveiled the first-ever 1-litre package featuring a paper-based barrier, marking another significant milestone in the carton packaging industry's transition towards low-carbon, renewable materials. Introducing a paper-based barrier into the Tetra Brik Aseptic 1000 Edge carton package - one of the company's most popular formats with its distinctive slanted top panel - increases the renewable content to 90% when combined with plant-based polymers.

Designed for ambient distribution, the carton supports shelf life and performance comparable to traditional aseptic packaging with an aluminium foil barrier, while reducing the carbon footprint by up to 50%, as verified by the Carbon Trust. Mr. Giampaolo Rossi, Plant Director at Sterilgarda Alimenti, said, we at Sterilgarda Alimenti stand for tradition and innovation. Through our continuous improvements in technology and product quality, we aim to provide the best possible answers to evolving consumer needs. Conscious consumption is becoming increasingly important, which is why adopting pioneering technologies is key across all our activities. Tetra Pak's new paper-based barrier is a groundbreaking innovation in beverage packaging and represents a significant contribution to our journey towards greater sustainability. Ms. Tatiana Liceti, Executive Vice President, Packaging Solutions at Tetra Pak, said, as environmental sustainability continues to shape the food and beverage industry, this innovation is an important step forward. It introduces a new barrier material that helps producers move closer to their environmental goals. For me, this is about harnessing the power of paper to make beverage cartons more renewable. This is the result of working hand-in-hand with our customers, supporting them as they offer responsible choices that consumers increasingly expect.'

PRODUCT LAUNCHES

Third Wave Coffee launches 'Beach Packers'



Third Wave Coffee has launched a limited-time 'Beach Packers' range. The collection of four frappes built around the season's flavours of Alphonso mango, Japanese matcha, coconut, and lime. Launched for the summer, the range includes the Tropical Alphonso Frappe, Alphonso Matcha Frappe, Brazilian Lemonade Frappe, and Vietnamese Coconut Frappe. Each drink blends summer favourite ingredients with smooth textures and perfectly captures the spirit of sunlit afternoons and tropical escapes, said the communication note. Mr. Rajat Luthra, CEO, Third Wave Coffee, said, we continuously experiment at our innovation lab, guided by evolving consumer preferences to create distinctive beverage experiences for every season. Each range is shaped by a thoughtful exploration of flavours, formats, and textures, balancing familiarity with a sense of newness. With the Beach Packers range, we set out to celebrate classic summer flavours like mango, coconut, and lime that evoke nostalgia, while reimagining them in refreshing formats for our patrons.

ITC Sunfeast Farmlite launches sugar free cookies for mindful snacking



ITC Sunfeast Farmlite, the better-for-you biscuits range is expanding its portfolio with the launch of an all-new sugar free cookies range. This launch aligns with the larger vision of ITC 'Help India Eat Better'. The new range is crafted for consumers who are actively rethinking sugar in their everyday snacking, without wanting to compromise on taste or indulgence. The range debuts in two indulgent variants, Choco Nut Cookies and Hazelnut & Oats Cookies, both of which are a source of protein and contain no trans-fat, while the Hazelnut & Oats variant is also lactose free. Mr. Suraj Kathuria, Vice President (Marketing), Biscuits, Foods Division ITC, said, we believe mindful snacking should never come at the cost of indulgence. With the launch of our Sugar Free Cookies range, we are catering to the growing need for guilt-free snacking while continuing to deliver a rich, satisfying cookie experience.

Dr Shantanu Das, Vice President & Head of Food Sciences, Foods Division ITC, said, as a trusted food brand, our responsibility is to deeply understand the expectations of today's consumers and translate them into meaningful solutions through our research and development expertise. With Sunfeast Farmlite Sugar Free Cookies, we have applied robust food science and formulation capabilities to develop cookies that are sugar free while preserving the taste and texture consumers love. By offering thoughtfully developed variants that cater to different health preferences, we aim to make mindful snacking enjoyable, credible, and a seamless part of everyday life.

Giani's launches premium ice cream bars



Giani's is all set to delight ice cream lovers with the launch of its all-new premium ice cream bars range. Known for its rich heritage, indulgent creations, and timeless flavours, its latest launch brings together luxurious ingredients, exciting textures, and contemporary taste preferences in a format that is both convenient and irresistible. The new range features four indulgent variants, each crafted to offer a distinct and elevated dessert experience. The Belgian Chocolate Bar is a treat for chocolate lovers, made with rich premium Belgian chocolate ice cream, coated in a thick chocolate shell and finished with crunchy nut bits. With deep cocoa notes and a satisfying crunch in every bite, it promises pure indulgence. The Classic Almond Bar brings together smooth and creamy vanilla ice cream dipped in rich chocolate and generously coated with roasted almonds, creating a timeless combination of creaminess and nutty texture. For those seeking mindful

indulgence, it has launched the Chocolate Noir (No Added Sugar), a bar specially crafted for the health-conscious consumer. With an intense dark chocolate flavour and no added sugar, it offers a guilt-free yet decadent experience without compromising on taste or texture. Adding a refreshing twist to the lineup is the Mix Berry White Chocolate Bar, a delicious fusion of tangy mixed berries and creamy ice cream, coated in smooth white chocolate and crunchy bits, making it the perfect blend of fruity freshness and sweet indulgence.

Mr. Gurpreet Singh, Managing Director at Giani's, said, at Giani's, we have always believed in creating desserts that bring together nostalgia, quality, and innovation. With our new Premium Ice Cream Bars, we wanted to introduce a range that feels indulgent, contemporary, and exciting for today's consumers while staying true to the taste and trust that Giani's has been known for over the decades. Each bar has been thoughtfully crafted to offer a unique experience, and we are confident that this new launch will resonate strongly with both loyal patrons and a younger generation of dessert lovers.

Klaw Snacks launches Super Sprout Sticks



KLAW Snacks announced the launch of its latest innovation — KLAW Super Sprout Sticks, expanding its portfolio of bold, flavour-packed, and better-for-you snacks. Launched after the success of its Supergrains Puffs, the new product is designed for modern consumers who believe that winding down is just as important as powering up. It is made with 47% sprouted moong & jowar, baked, and high in protein, delivering a smart snacking experience that combines intelligence with indulgence. Crafted with mindful nutrition and bold flavours, the new sticks offer a wholesome way to snack without compromise.

The newly launched products are available in five bold and exciting flavours, Masala Madness, Mango Mania, Achari Nimbu, Pudina Punch, and Korean Kick-bringing together familiar favourites and globally inspired tastes for adventurous snack lovers. The products are now available on Zepto, Instamart, Blinkit, and nearby retail stores across Delhi, Mumbai, Bangalore, and other major cities, making mindful snacking easily accessible to consumers.

Ms. Nikki Arora Singh, Founder & Director, said, with KLAW Super Sprout Sticks, we wanted to create a snack that supports

mindful winding down while delivering on taste and nutrition. Made with 47% sprouted moong and jowar, these baked, high-protein sticks are designed for those who believe that winding down is just as important as powering up. With bold flavours and wholesome ingredients, we're building snacks that support relaxation, stress relief, and better focus- making conscious snacking exciting, accessible, and intelligent.

LT Foods introduces Daawat Saffron Basmati



LT Foods, a leading billion-dollar global FMCG company in the consumer food space, has launched Saffron Basmati Rice, a limited edition basmati offering for connoisseurs who value the finest culinary experience. Infused with the finest saffron, sourced exclusively from Jammu & Kashmir (J&K), this is a first-of-its-kind innovation in the basmati category, reinforcing LT Foods' continued commitment to delivering an exceptional food experience. The rice is cultivated on farms fed by the Chenab River, where farmers nurture each grain using time honoured, pesticide free methods passed down through generations. Every grain is hand-selected with meticulous care, preserving the natural aroma, purity, and authenticity of traditional basmati cultivation.

Daawat Saffron Basmati Rice is a rare and collectable culinary offering. Each limited-edition pack is designed as a luxury experience, containing vacuum sealed saffron-basmati rice, a handcrafted jute storage bag, a personalized note by farmer with a certified pack number and curated cooking instructions-transforming the product into a keepsake as much as a culinary indulgence. Mr. Ritesh Arora, Chief Executive Officer- India Business & Far East, LT Foods, said, the introduction of Saffron Basmati Rice is a strategic step in strengthening our premium portfolio and enhancing value creation across the premium basmati ecosystem. As custodians of a legacy built over seven decades, we continue to invest in innovations that strengthen the premium end of the portfolio while deepening the value we create for consumers and stakeholders. This limited edition offering is an embodiment of our long-term vision-where authenticity, product excellence and exclusivity come together to reinforce leadership and set new category benchmarks. With this launch, we are not only celebrating heritage but also shaping the future trajectory of premium basmati worldwide.



NEWS ROUND UP

New jars help pasta sauce brands reduce costs and improve performance



Amcor, a global leader in developing and producing responsible packaging solutions, announced the launch of the next generation of its Apex family of premium packaging. The jars now include Elevate technology, which increases their ability to handle hot-fill process on the production line. The Apex line of clear, lightweight polyethylene terephthalate (PET) jars with metal lug closures offers pasta sauce brands a simpler, more consistent experience than traditional glass jars that can offer lower overall costs when factoring in freight, manufacturing downtime and breakage. The greater stability of Elevate technology enables food manufacturers to fill Apex jars at up to 205 degrees Fahrenheit and offers stronger resistance to deformation as temperatures on production lines fluctuate. Additionally, Apex costs less to ship than glass jars. With about seven unfilled Apex jars equal in weight to one unfilled glass jar, manufacturers could ship up to 36% more product per truckload. Apex is compatible with most manufacturers' existing lines and equipment, further reducing the cost to switch. Amcor's Apex jars are also customizable, allowing both consumer packaged goods brands and private-label manufacturers to create a signature look. At Amcor, we are elevating what our customers and consumers can expect from a PET jar. From greater consistency under changing conditions on production line to the unmistakable pop when a consumer opens the jar, Apex with Elevate offers certainty, said Mr. Dan Rardin, Director of Product Line Management for Amcor Rigid Packaging Solutions North America. Add in lower total cost of ownership and the switch just makes sense.

Amazon Strikes \$30 Million Carbon Credit Deal with Indian Rice Farmers in Landmark Climate Push

Amazon has signed a \$30 million (approximately ₹280 crore) agreement to purchase carbon credits generated by Indian rice farmers, marking one of the largest agricultural carbon credit deals globally and the first of its scale in India. The

deal has been inked with the Good Rice Alliance, an entity backed by Bayer, GenZero, and Shell. The agreement will enable Amazon to offset a portion of its global emissions as part of its broader net-zero commitments. The initial phase of the project is expected to generate more than 685,000 metric tonnes of carbon dioxide-equivalent credits.



The initiative works with over 13,000 smallholder farmers across 35,000 hectares in India, promoting sustainable rice cultivation practices aimed at reducing methane emissions—one of the most potent greenhouse gases. Traditional rice farming methods, which involve flooding paddy fields, contribute an estimated 8–10% of global methane emissions, making it a major source of agricultural pollution. India, the world's largest rice producer, is also among the top methane emitters globally. The programme encourages farmers to adopt improved water management techniques such as alternate wetting and drying and direct-seeded rice, significantly lowering methane output while maintaining productivity. Methane has a global warming potential more than 27 times that of carbon dioxide, making its reduction critical for near-term climate action.

Beyond emissions reduction, the initiative provides farmers with training, field-level support, and financial incentives to transition toward sustainable practices. The model aims to create a dual benefit—enhancing farmer incomes while contributing to global climate goals. The deal underscores increasing participation by large technology companies in carbon markets. Firms with significant carbon footprints are turning to carbon credits to offset emissions as they work toward sustainability targets. Recent examples include Microsoft signing a soil carbon credit agreement valued between \$171 million and \$228 million, and Meta entering a forestry credit deal worth up to \$16 million. Shell remains one of the largest buyers of carbon credits globally. The development comes as India prepares to roll out its first formal carbon trading framework, with current activity largely limited to voluntary markets. Amazon's agreement signals a growing convergence between agriculture, sustainability, and global carbon markets, positioning Indian farmers as key contributors to climate solutions while opening new income streams in the process.

Plant-based meat market expected to reach \$33.3bn by 2031, at a CAGR of 20.5%



The global plant-based meat market reached \$5.3 billion in 2021, and is projected to reach \$33.3 billion by 2031, registering a compound annual growth rate (CAGR) of 20.5% from 2022 to 2031, according to the report by Allied Market Research. The global plant-based meat market is undergoing a significant structural change, fuelled by the fast convergence of health consciousness, environmental awareness and rapidly developing technology in food science. The plant-based meat market size which had a value of \$5.3 billion in 2021 is projected to rise more than sixfold across the forecast decade, a trajectory that highlights the ongoing reorientation of consumer choices away from traditional animal proteins.

The impressive expansion of the plant-based meat market is supported by a number of macroeconomic and sociocultural tailwinds. A growing global population of vegetarians and flexitarians increases the addressable consumer market well beyond committed vegans. A growing body of scientific knowledge about the health consequences of overconsuming red and processed meat including cardiovascular disease, obesity, type 2 diabetes and some cancers is spurring consumers to more actively seek plant-based protein substitutes. And the documented environmental price of industrial livestock production including greenhouse gas emissions, deforestation and water depletion is shifting buying habits across demographic groups. The decisive catalyst has been the ability of modern plant-based meat products to closely mimic animal-based meat in terms of taste, texture, colour and other sensory qualities. Berkeley: To meet such demand, "foodservice and retail channels across the globe have become a lot more accessible and adoption has accelerated." Tofu, tempeh, seitan, or legume-derived products are now widely available. Continually, health-minded consumers are realising that plant-based meat items have a useful nutritional profile they're high in protein and dietary fibre, low in saturated fats and bad cholesterol, and strengthened and fortified with important vitamins: trace minerals including iron; zinc; calcium; vitamin B12. These nutritional benefits, paired with the ethical appeal of cruelty-free food sourcing, is broadening the consumer addressable market by age group, income bucket and geography.

Australian Avocados bring premium culinary experience to Mumbai



Australian Avocados marked their presence in India with an exclusive culinary showcase in Mumbai, in collaboration with the School of European Pastry (SEP). The experience featured a specially curated five-course menu by chef, Parvinder Singh Bali, Director, School of European Pastry and Culinary, designed to showcase the versatility of Australian Avocados across formats, textures, and cuisines. From fresh and light combinations such as mango, apple and nori crisp, to more indulgent creations like avocado mousse with compressed watermelon, avocado tempura chaat, and avocado-based desserts, each course highlighted the distinct applications of the ingredient across every day and contemporary formats. The avocados are grown across Queensland, Western Australia and other climatically diverse regions of Australia, ensuring premium quality produce is available year-round. Rich in good fats, including monounsaturated fats, the same beneficial fats found in olives, along with fibre, folate, potassium, and B vitamins, they avocados are well suited to the demands of modern, health-conscious diets. India represents a priority market for Australian avocados, with the focus firmly on building long-term trade partnerships across the value chain, from importers and distributors to retail and foodservice. This trade-led approach is supported by growing consumer awareness, driven by increasing exposure to global food culture and a stronger focus on health and nutrition. According to recent trade data, India imported over 20,358 tonnes of avocados in the 12 months to December 2025, with imports between July and December 2025 rising 127 per cent compared to the same period the previous year, reflecting strong and accelerating market demand.

Mr. Duncan Wells, Western Australia Director for Australian Avocados, said, India is one of the most exciting growth markets for Australian avocados. The scale of demand growth over the past year alone speaks to how rapidly consumer appetite is shifting here. Building a meaningful, long-term presence in this market means working closely with the right partners to ensure the product reaches consumers consistently and at the quality they expect. Engagements like this plays an important role in that journey, bringing the product to life for the industry professionals who shape how and where it is

adopted. Chef Bali said, Australian avocados are a genuinely exciting ingredient to work with professionally. The quality is consistent and the texture holds remarkably well across different applications, whether you are compressing it, blending it into a mousse, or using it as a base for a frozen dessert. What I wanted to show through this menu is that this is not a one-trick ingredient. It belongs across the entire meal, from the first course to the last and it fits naturally into the kind of contemporary, technique-driven cooking that chefs and food professionals in India are increasingly excited about.

India & NZ sign FTA with focus on balanced exchange of agri and food products



India and New Zealand officially signed a Free Trade Agreement and while the pact covers various sectors, the primary focus is on a balanced exchange of agricultural and food products designed to benefit both economies without disrupting local markets. The FTA was signed by Indian Union Minister for Commerce and Industry, Sh. Piyush Goyal, and Mr. Todd McClay, Minister for Trade and Investment, New Zealand. Under the terms of the agreement, New Zealand will eliminate tariffs on a wide array of Indian food exports. This includes immediate duty-free access for Indian rice, spices, tea, and processed organic foods.

Experts suggest this will allow Indian farmers and food processors to establish a stronger foothold in the premium New Zealand market, where demand for organic and ethnic Indian produce is rising. Conversely, India has granted New Zealand improved access for high-value agricultural items. Tariffs will be significantly reduced on New Zealand kiwifruit, apples, and blueberries. These reductions are expected to make fresh, seasonal produce more affordable for Indian consumers during the off-season. A notable feature of the deal is the preservation of India's dairy sector.

To protect the livelihoods of millions of domestic small-scale dairy farmers, India has excluded milk, cheese, and butter from tariff concessions. Instead, both nations have agreed to a Technical Cooperation Framework. This initiative focuses on sharing New Zealand's advanced cold-chain technology and livestock management practices to help Indian farmers modernise their operations and reduce post-harvest losses. This agreement is expected to double the bilateral trade in food

and agricultural products within the next four years, fostering a more resilient supply chain between the two nations.

Organic F&B market to reach \$594.6 bn by 2033



The global organic food and beverages market size reached \$247.9 billion in 2024, reflecting a robust demand trajectory. The market is experiencing a steady compound annual growth rate (CAGR) of 10.2% from 2025 to 2033. By the end of this forecast period, the global organic food and beverages market is projected to attain \$594.6 billion, according to Growth Market Report. This remarkable growth is primarily driven by the increasing consumer awareness regarding health and wellness, coupled with a growing preference for chemical-free, environmentally sustainable food and beverage options.

India's food landscape is undergoing a quiet yet powerful transformation. As consumers become more aware of what goes into their meals, the demand for organic food and beverages is accelerating. No longer confined to niche urban stores, organic products are steadily entering mainstream retail, digital platforms, and even local markets. This shift is not just about health-it reflects a broader change in lifestyle, environmental awareness, and trust in food systems. Organic food and beverages are produced without synthetic fertilisers, pesticides, genetically modified organisms (GMOs), or artificial additives. In India, certified organic products follow strict guidelines under regulatory frameworks such as the National Programme for Organic Production (NPOP).

Key Characteristics: Chemical-free cultivation; non-GMO ingredients; Sustainable farming practices; Ethical sourcing and animal welfare (for dairy and meat). The organic food and beverages market in India is experiencing steady growth, driven by urbanisation, rising incomes, and increased health awareness. While still a small fraction of the total food market, it is one of the fastest-growing segments. **Growth Drivers:** Increasing lifestyle diseases such as diabetes and obesity; Rising awareness about pesticide residues in conventional food; Expansion of e-commerce platforms offering organic products; Government support for organic farming initiatives. Indian consumer is more informed and selective. Organic products are no longer seen as luxury items but as investments in long-term health.



FOOD SAFETY & REGULATORY ISSUES

Eateries in Maharashtra told to label dishes if using cheese analogues



The Maharashtra Food and Drug Administration (FDA) has officially cracked down on the deceptive use of 'cheese analogues', ordering all food service establishments to provide clear disclosure to diners from May 1, 2026.

The new directive targets the growing industry practice of using vegetable-fat-based substitutes in place of traditional dairy without informing the consumer.

The directive requires restaurants, cafes, and street-food vendors to explicitly label dishes that utilise dairy analogues in the 'menu'. These substitutes, often crafted from edible vegetable oils and starches, mimic the texture of cheese or paneer but do not share the same nutritional profile as milk-based products.

Establishments have been granted a period until April 30, to revise their branding and 'menu' descriptions.

As per reports, from May 1, the FDA will launch state-wide inspections to ensure compliance.

Authorities warned that any establishment failing to update its menus by the deadline will face strict legal action, including the potential suspension of operating licences.

Thailand FDA approval of Balchem's OptiMSM expands innovation opportunities



Balchem announced that OptiMSM, its methylsulfonylmethane

(MSM) ingredient for skin and joint health, has been granted approval from the Thailand Food and Drug Administration (FDA) for supplement use, with an approved maximum daily intake of 1 gram per day. Secured in collaboration with leading food and pharmaceutical ingredients distributor Rama Production Co., this milestone enables manufacturers in Thailand to leverage this premium, science-backed MSM ingredient to develop new product formulations targeting beauty-from-within, joint health, muscle recovery and active aging.

The Thai population is growing older, with people aged 60 and over now outnumbering those under 15 by 4.1 million. This demographic shift is driving increased demand for solutions that support key aspects of active aging such as mobility, vitality and overall well-being. As consumers strive to feel and look their best for longer, beauty has also emerged as a key priority. 1 in 4 consumers in Asia-Pacific reported purchasing food or beverage products to improve physical appearance between 2024 and 2025.

OptiMSM can help brands meet this growing demand, offering a source of sulphur, a key building block of joints and connective tissues, that helps maintain joint function and support healthy skin, hair and nails. The nutrient also supports the production of molecules such as glutathione, which plays a key role in the body's antioxidant response. Together, these clinically researched benefits position OptiMSM as a trusted ingredient for active and healthy aging, reinforced by its exceptional purity, quality and efficacy. With approval in Thailand, OptiMSM provides formulators with greater flexibility to develop effective joint health, beauty-from-within, and active aging solutions aligned with local regulatory requirements.

Ms. Jessica Arnaly, Senior Marketing & Business Development Manager, Balchem Human Nutrition & Health, said, APAC represents a promising growth market and we're excited to help our partners unlock new regional opportunities with confidence, leveraging our premium, science-backed, FDA-approved ingredient, OptiMSM. We're proud to partner with Rama Production in Thailand to help local brands innovate and deliver products that meet consumers' evolving health and wellness needs.

Mr. Prakaithip Ekwongsupasarn, Assistant Sales Director at Rama Production, said, as an OptiMSM distributor in Thailand, we are proud to bring a globally recognised ingredient to support joint health and beauty-from-within solutions for the Thai market.

SC reserves order on PIL seeking Task Force to oversee FSSAI activities

The Supreme Court of India has reserved order on a Public Interest Litigation (PIL) that sought the establishment of a specialised National Task Force to oversee the Food Safety and Standards Authority of India (FSSAI) enforcement activities.

As per reports, the petitioner had argued that current

enforcement mechanisms are insufficient to tackle the rising instances of food adulteration and 'pesticide-heavy' produce across the country.



The SC observed that while the concerns regarding public health are 'serious', the creation of a parallel administrative body falls squarely within the domain of executive policy.

The SC noted that the FSSAI already operates under a comprehensive legislative framework, the Food Safety and Standards Act of 2006, which provides for scientific panels and expert committees and the petition lacks evidentiary material.

The reports added that during the proceedings, the bench expressed reservations about the 'generalised nature' of the petition.

The petition sought SC direction for setting up of the National Task Force under the apex court's supervision to ensure effective enforcement of food safety laws and a time bound audit & inspection drive across the country.

FSSAI Cracks Down on Ashwagandha Use, Bans Leaves in Food and Nutraceutical Products

India's food safety regulator, Food Safety and Standards Authority of India, has directed food manufacturers to use only Ashwagandha roots and their extracts in products, explicitly prohibiting the use of leaves in any form.

The clarification comes after the regulator observed that some food business operators were incorporating Ashwagandha leaves and leaf extracts in health and nutraceutical products, which is not permitted under existing regulations.



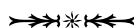
Ashwagandha, a widely used herb in wellness formulations, is allowed in categories such as health supplements, nutraceuticals, and foods for special dietary and medical purposes. However, the regulator has now reiterated that only the root and its extracts can be used in these products.

"It is clarified that the use of Ashwagandha leaves in crude, extract, or any other form is not permitted under the regulations," FSSAI said in its latest advisory.

The directive aligns with similar instructions issued by the Ministry of Ayush, which has also asked manufacturers of Ayurvedic and herbal products to restrict usage to Ashwagandha roots and their derivatives.

FSSAI has instructed state food safety commissioners and enforcement officials to maintain strict vigilance and ensure compliance across the industry.

The move signals tighter scrutiny of ingredient integrity in India's fast-growing nutraceutical and functional foods segment, where botanical ingredients are increasingly being used in diverse formats.



Advertise here !

To book this space
Contact :
Mr. Vishal Sarin
Mobile : +91-7838012842 Email : vishal@aifpa.net

UPCOMING EVENTS

Event : **Food Confluence-2026**
 Dates : 01-07-2026 to 03-07-2026
 Place : CODISSIA, Coimbatore
 For information, contact :
Synergy Exposures & Events India Pvt. Ltd.
 M:- 044-22780776
 E-mail:- info@synergyexposures.com



Event : **World Food India South-2026**
 Dates : 09-07-2026 to 10-07-2026
 Place : AU Engineering College Grounds, Visakhapatnam
 For information, contact :
Mr. Rameshwar Maurya
 M:- 8744088116
 E-mail:- maurya@aifpa.net



Event : **World Mithai Namkeen Convention**
 Dates : 16-07-2026 to 18-07-2026
 Place : Hitex Exhibition Centre, Hyderabad
 For information, contact :
Advance Info Media & Events
 M:- 8291918484
 E-mail:- wmnc@advanceinfomedia.com



Event : **Bharat Food Expo-2026**
 Dates : 23-07-2026 to 25-07-2026
 Place : JECC, Jaipur
 For information, contact :
Reenvision Events Pvt. Ltd.
 T:- 022-27812093
 E-mail:- info@reenvision.in



Event : **India International Hospitality Expo**
 Dates : 05-08-2026 to 08-08-2026
 Place : India Expo Centre & Mart, Greater Noida
 For information, contact :
India Exposition Mart Ltd.
 M:- 9289733490
 E-mail:- exhibition2@indiaexpocentre.com



Event : **Fi India-2026**
 Dates : 26-08-2026 to 28-08-2026
 Place : Bombay Exhibition Centre, Mumbai
 For information, contact :
Mr. Mahesh Verma
 E-mail:- Mahesh.verma@informa.com



Kejriwal
GROUP

Honey at it's best.

Kejriwal Honey is the pioneer of honey exports from the country of India, offering top notch honey to the quality conscious consumers across the world. The group started exporting honey in the year 1996 and within a short span of time it has established itself as a premier Honey exporter to many countries of Europe, USA and the Middle East.

Equipped with ultra modern facilities, with a capacity to handle about 100 MT daily, the company also maintains complete traceability for all honey from the farm level to match its quality with the trust of the customers. The packaging is done under the most stringent hygienic conditions, thus, retaining the natural flavours and vital enzymes of the product. All operations are fully automated, housing a well designed closed circuit honey processing line with no manual interface, ensuring quality and hygiene of the international standards.

We are the first honey exporters from India who are certified for ISO 22000-2005 for Food Safety Management System and the first **True Source Certified** honey exporters in the World. In short, if you are looking for quality honey from India, Kejriwal is your answer.



Talk to us for quality products from India

Regd. Office: W-42, Greater Kailash - II, New Delhi - 110048

Tel: +91-11-29219677/78/79 Fax: +91-11-29210985

Email: mail@kejriwalgroup.co.in

Certified for ISO: 9001-2000, 22000-2005 for food safety management system.



From bakery to beverage, dairy to nutrition bar, USA Blueberry Ingredients deliver:

- Superior Color & Natural Appeal
- Consistent Sweet-Tart Flavor Profile
- Excellent Bake, Freeze & Process Stability
- High Food Safety & Quality Standards
- Year-Round Reliable Supply



Available Formats for Industrial Use

Innovating Products with
USA Blueberries



IQF Blueberries

Dried Blueberries

Freeze Dried Blueberries

Micro Dried Blueberries

Purées & Concentrates

**Why Leading Indian Manufacturers
Choose USA Blueberries**

- Grown under strict U.S. quality systems
- Strong traceability & compliance
- Clean-label friendly
- Consumer-recognized premium origin
- Excellent performance in high-temperature processing

Connect with us for Samples & Technical Support

U.S. Highbush Blueberry Council (USHBC) – India

✉ rajkapoor@usblueberry.in

📍 www.usblueberry.in

**Blueberries
GO BIG**